

Electrochemistry

Question 1

The pH and conductance of a weak acid (HX) was found to be 5 and 4×10^{-5} S, respectively. The conductance was measured under standard condition using a cell where the electrode plates having a surface area of 1 cm^2 were at a distance of 15 cm apart. The value of the limiting molar conductivity is _____ $\text{Sm}^2 \text{ mol}^{-1}$. (nearest integer)

(Given : degree of dissociation of the weak acid (α) $\ll 1$)

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Answer: 6

Solution:

Step 1: Find $[H^+]$ from pH

The pH is given as 5.

$$\text{So, } [H^+] = 10^{-5} \text{ mol L}^{-1}$$

Step 2: Write the relation for weak acid dissociation

For a weak acid HX:

$$[H^+] = [HX] \cdot \alpha$$

Where $[HX]$ is the initial concentration and α is the degree of dissociation.

Step 3: Link α and molar conductivity

Degree of dissociation is also given by

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$$

Where Λ_m is the molar conductivity at given concentration, and Λ_m^∞ is the molar conductivity at infinite dilution (limiting molar conductivity).

Step 4: Find molar conductivity Λ_m

Molar conductivity is given by:

$$\Lambda_m = \frac{k \times 1000}{[HX]}$$

k is the specific conductance (conductivity).

Step 5: Calculate cell constant and specific conductance (k)

Given conductance (G) = 4×10^{-5} S

$$\text{Cell constant} = \frac{\text{distance between plates}}{\text{area of plates}} = \frac{15 \text{ cm}}{1 \text{ cm}^2} = 15 \text{ cm}^{-1}$$

So, $k = G \times \text{cell constant}$

$$\rightarrow k = 4 \times 10^{-5} \times 15 = 6 \times 10^{-4} \text{ S cm}^{-1}$$

Step 6: Combine equations to solve for Λ_m^∞

From above, $[H^+] = [HX] \cdot \alpha$

But, $\alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$, so:

$$[H^+] = [HX] \cdot \frac{\Lambda_m}{\Lambda_m^\infty}$$

$$\text{Recall } \Lambda_m = \frac{k \times 1000}{[HX]}$$

Substitute in:

$$[\text{H}^+] = [HX] \cdot \frac{\frac{k \times 1000}{[HX]}}{\Lambda_m^\infty}$$

$$= \frac{k \times 1000}{\Lambda_m^\infty}$$

$$\text{So, } \Lambda_m^\infty = \frac{k \times 1000}{[\text{H}^+]}$$

Step 7: Substitute values and solve

$$k = 6 \times 10^{-4} \text{ S cm}^{-1}$$

$$[\text{H}^+] = 10^{-5} \text{ mol L}^{-1}$$

So,

$$\Lambda_m^\infty = \frac{6 \times 10^{-4} \times 1000}{10^{-5}} = \frac{6 \times 10^{-1}}{10^{-5}} = 6 \times 10^4 \text{ S cm}^2 \text{ mol}^{-1}$$

Step 8: Convert units to $\text{S m}^2 \text{ mol}^{-1}$

$$1 \text{ S cm}^2 \text{ mol}^{-1} = 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$$

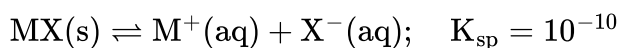
$$\text{So, } 6 \times 10^4 \times 10^{-4} = 6 \text{ S m}^2 \text{ mol}^{-1}$$

Final Answer:

Limiting molar conductivity $\Lambda_m^\infty = 6 \text{ S m}^2 \text{ mol}^{-1}$.

Question2

MX is a sparingly soluble salt that follows the given solubility equilibrium at 298 K.



If the standard reduction potential for $\text{M}^+(\text{aq}) + \text{e}^- \rightarrow \text{M(s)}$ is

$(E_{\text{M}^+/\text{M}}^\ominus) = 0.79 \text{ V}$, then the value of the standard reduction potential for the metal/metal insoluble salt electrode $E_{\text{X}^-/\text{MX(s)}/\text{M}}^\ominus$ is _____ mV. (nearest integer)

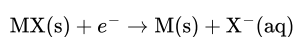
[Given: $\frac{2.303RT}{F} = 0.059 \text{ V}$]

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Answer: 200

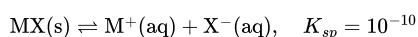
Solution:

For the metal/metal insoluble salt electrode, the reduction half-reaction is

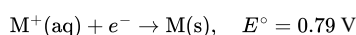


We can obtain this by adding the following two steps:

1) Solubility equilibrium:



2) Reduction of M^+ :



Adding (1) and (2), M^+ cancels and we get the required electrode reaction.

Now use ΔG° relation (NCERT):

- For equilibrium: $\Delta G^\circ = -RT \ln K$
- For a half-cell: $\Delta G^\circ = -nFE^\circ$

So,

For dissolution:

$$\Delta G_{\text{diss}}^\circ = -RT \ln(10^{-10}) = -RT(-10 \ln 10) = 10RT \ln 10$$

For reduction of M^+ :

$$\Delta G_{M^+/M}^\circ = -F(0.79)$$

Total:

$$\Delta G_{\text{total}}^\circ = 10RT \ln 10 - F(0.79)$$

For the overall electrode reaction ($n = 1$):

$$E_{X^-/MX(s)/M}^\circ = -\frac{\Delta G_{\text{total}}^\circ}{F} = -\left(\frac{10RT \ln 10}{F} - 0.79\right)$$

$$\text{Given } \frac{2.303RT}{F} = 0.059 \text{ V and } \ln 10 = 2.303,$$

$$\frac{RT \ln 10}{F} = \frac{2.303RT}{F} = 0.059 \text{ V}$$

$$\frac{10RT \ln 10}{F} = 10 \times 0.059 = 0.59 \text{ V}$$

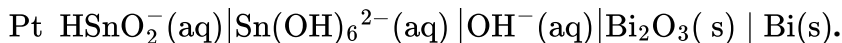
So,

$$E^\circ = -(0.59 - 0.79) = 0.20 \text{ V} = 200 \text{ mV}$$

Answer: 200 mV

Question 3

Consider the following electrochemical cell at 298 K



If the reaction quotient at a given time is 10^6 , then the cell EMF (E_{cell}) is
_____ $\times 10^{-1}$ V (Nearest integer).

Given the standard half-cell reduction potential as

$$E_{\text{Bi}_2\text{O}_3/\text{Bi}, \text{OH}^-}^\circ = -0.44 \text{ V and } E_{\text{Sn}(\text{OH})_6^{2-}/\text{HSnO}_2^-, \text{OH}^-}^\circ = -0.90 \text{ V}$$

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Answer: 4

Solution:

$$\begin{aligned} E_{\text{cell}}^\circ &= -0.44 - (-0.90) \\ &= +0.46 \text{ V} \end{aligned}$$

Applying Nernst equation :-

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.06}{n} \log Q$$

First, find n (number of electrons transferred). In this cell, $n = 6$.

Given $Q = 10^6$, so $\log 10^6 = 6$.

$$\Rightarrow E_{\text{cell}} = 0.46 - \frac{0.06}{6} \log 10^6$$

$$\Rightarrow E_{\text{cell}} = 0.46 - \frac{0.06}{6} \times 6$$

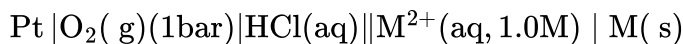
$$\Rightarrow E_{\text{cell}} = 0.46 - 0.06$$

$$\Rightarrow E_{\text{cell}} = 0.40 \text{ V} = 4 \times 10^{-1} \text{ V}$$

$$\Rightarrow x = 4$$

Question4

Consider the following electrochemical cell :



The pH above which, oxygen gas would start to evolve at anode is ____ (nearest integer).

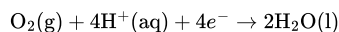
$$\left[\begin{array}{l} \text{Given : } E_{\text{M}^{2+}/\text{M}}^{\circ} = 0.994 \text{ V} \\ E_{\text{O}_2/\text{H}_2\text{O}}^{\circ} = 1.23 \text{ V} \end{array} \right\} \text{ standard reduction potential} \quad \text{and } \frac{RT}{F} (2.303) = 0.059 \text{ V at the given condi}$$

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Answer: 4

Solution:

For the oxygen electrode (reduction reaction):



Nernst equation:

$$E_{\text{O}_2/\text{H}_2\text{O}} = E_{\text{O}_2/\text{H}_2\text{O}}^{\circ} - \frac{0.059}{4} \log \left(\frac{1}{p_{\text{O}_2} [\text{H}^+]^4} \right)$$

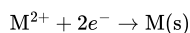
Given $p_{\text{O}_2} = 1 \text{ bar}$, so $\log p_{\text{O}_2} = 0$:

$$E_{\text{O}_2/\text{H}_2\text{O}} = 1.23 + 0.059 \log [\text{H}^+]$$

Since $\log [\text{H}^+] = -\text{pH}$:

$$E_{\text{O}_2/\text{H}_2\text{O}} = 1.23 - 0.059 \text{ pH}$$

For the metal electrode:



Given $[\text{M}^{2+}] = 1.0 \text{ M}$, hence

$$E_{\text{M}^{2+}/\text{M}} = E_{\text{M}^{2+}/\text{M}}^{\circ} = 0.994 \text{ V}$$

Oxygen will **start evolving at the anode** when the oxygen electrode becomes the **anode**, i.e. when its **reduction potential becomes less than** that of M^{2+}/M . The threshold is when both are equal:

$$1.23 - 0.059 \text{ pH} = 0.994$$

$$0.059 \text{ pH} = 1.23 - 0.994 = 0.236$$

$$\text{pH} = \frac{0.236}{0.059} = 4$$

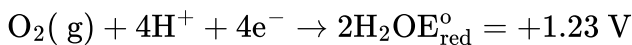
So, oxygen gas starts evolving at the anode for $\text{pH} > 4$.

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Question5

Electricity is passed through an acidic solution of Cu^{2+} till all the Cu^{2+} was exhausted, leading to the deposition of 300 mg of Cu metal. However, a current of 600 mA was continued to pass through the same solution for another 28 minutes by keeping the total volume of the solution fixed at 200 mL. The total volume of oxygen evolved at STP during the entire process is ____ mL. (Nearest integer)

[Given:



Molar mass of Cu = 63.54 g mol^{-1}

Molar mass of O_2 = 32 g mol^{-1}

Faraday Constant = 96500 C mol^{-1}

Molar volume at STP = 22.4 L]

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Answer: 111

Solution:

Eq of Cu = Eq of O_2

In electrolysis, the number of equivalents deposited and liberated are equal for the same quantity of electricity.

For copper deposition: $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$, so n -factor = 2.

For oxygen evolution (reverse of given reduction): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^{+} + 4\text{e}^{-}$, so n -factor = 4.

So, equivalents of Cu deposited = equivalents of O_2 evolved during the first part.

$$\frac{300 \times 10^{-3} \times 2}{63.54} = n_{\text{O}_2} \times 4$$

$$2.36 \times 10^{-3} = n_{\text{O}_2}$$

When current is further passed

After all Cu^{2+} is exhausted, the current now causes only water oxidation, producing O_2 at the anode.

Charge passed in 28 minutes at 600 mA:

$$n_{\text{O}_2} \times 4 = \frac{600 \times 28 \times 60}{96500 \times 1000}$$

$$n_{\text{O}_2} = 2.611 \times 10^{-3}$$

Total O_2 released

$$\text{Total moles of oxygen} = (2.36 + 2.611) \times 10^{-3}.$$

Volume at STP = $n \times 22400 \text{ mL}$.

$$= [10^{-3} \times (2.36 + 2.611)] \times 22400 \text{ mL}$$

$$= 111.35 \text{ mL}$$

Question 6

Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be 1/30 of the molar conductivity of another weak acid HZ with concentration of 0.02 M. If $\lambda^{\circ} \text{Q}^{-}$ happened to be equal with $\lambda^{\circ} \text{Z}^{-}$, then the difference of the pK_a values of the two weak acids ($\text{pK}_a(\text{HQ}) - \text{pK}_a(\text{HZ})$) is ____ (Nearest integer).

[Given: degree of dissociation (α) $\ll 1$ for both weak acids, λ° : limiting molar conductivity of ions]

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Answer: 2

Solution:

$$K_a(\text{HQ}) = C_1 \alpha_1^2 \quad \alpha_1 = \frac{\lambda_m(\text{HQ})}{\lambda_m^\circ(\text{HQ})}$$

$$K_a(\text{HZ}) = C_2 \alpha_2^2 \quad \alpha_2 = \frac{\lambda_m(\text{HZ})}{\lambda_m^\circ(\text{HZ})}$$

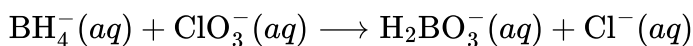
$$\frac{K_a(\text{HQ})}{K_a(\text{HZ})} = \frac{C_1}{C_2} \cdot \left(\frac{\alpha_1}{\alpha_2} \right)^2 = \frac{0.18}{0.02} \cdot \left[\frac{\lambda_m(\text{HQ})}{\lambda_m(\text{HZ})} \right]^2$$

$$\frac{K_a(\text{HQ})}{K_a(\text{HZ})} = 9 \times \left(\frac{1}{30} \right)^2 = \frac{1}{100}$$

$$\text{p}K_a(\text{HQ}) - \text{p}K_a(\text{HZ}) = 2$$

Question 7

Consider the following redox reaction taking place in acidic medium



If the Nernst equation for the above balanced reaction is

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q,$$

then the value of n is _____. (Nearest integer)

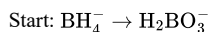
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Answer: 24

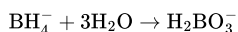
Solution:

Balance by ion–electron method in acidic medium.

Oxidation half-reaction (borohydride to dihydrogen borate):

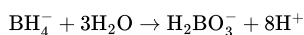


Balance O by adding water:



Balance H by adding H^+ :

Left H = 4 + 6 = 10, right H = 2, so add 8H^+ on RHS:



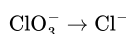
Balance charge by adding electrons (oxidation $\Rightarrow e^-$ on RHS):

LHS charge = -1 , RHS charge = $-1 + 8 = +7$, so add $8e^-$ to RHS:

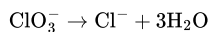


So, electrons lost = 8.

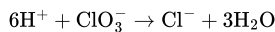
Reduction half-reaction (chlorate to chloride):



Balance O by adding water:

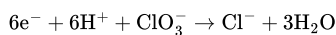


Balance H by adding H^+ on LHS:



Balance charge by adding electrons (reduction $\Rightarrow e^-$ on LHS):

LHS charge = $6 - 1 = +5$, RHS charge = -1 , so add $6e^-$ to LHS:



So, electrons gained = 6.

Make electrons equal (LCM of 8 and 6 is 24):

Multiply oxidation half by 3 and reduction half by 4:

Oxidation $\times 3$ gives $24e^-$, reduction $\times 4$ consumes $24e^-$.

After adding and canceling common species, the balanced overall reaction becomes:



Hence, the number of electrons transferred in the balanced reaction is

$$n = 24.$$

So, $n = 24$ (nearest integer).

Question 8

For strong electrolyte Λ_m increases slowly with dilution and can be represented by the equation

$$\Lambda_m = \Lambda_m^\circ - A c^{1/2}$$

Molar conductivity values of the solutions of strong electrolyte AB at 18°C are given below :

$c [\text{mol L}^{-1}]$	0.04	0.09	0.16	0.25
$\Lambda_m [\text{S cm}^2 \text{mol}^{-1}]$	96.1	95.7	95.3	94.9

The value of constant A based on the above data [$\text{in S cm}^2 \text{mol}^{-1}/(\text{mol/L})^{1/2}$] unit is _____.

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Answer: 4

Solution:

We use the given relation for a strong electrolyte:

$$\text{Using equation : } \Lambda_m = \Lambda_m^\circ - A\sqrt{c}$$

Now substitute the first set of data: $c = 0.04$ and $\Lambda_m = 96.1$.

$$96.1 = \Lambda_m^\circ - A\sqrt{0.04}$$

$$96.1 = \Lambda_m^\circ - A \times 0.2 \quad \dots (i)$$

$$95.7 = \Lambda_m^\circ - A \times \sqrt{0.09}$$

$$95.7 = \Lambda_m^\circ - A \times 0.3 \quad \dots (ii)$$

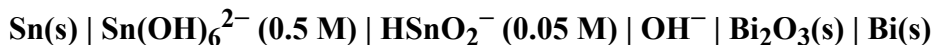
Now subtract equation (ii) from equation (i) so that Λ_m° cancels out:

From eq. (1) and eq. (2)

$$\begin{aligned}
 (96.1 - 95.7) &= (\Lambda_m^0 - A \times 0.2) - (\Lambda_m^0 - A \times 0.3) \\
 0.4 &= -0.2A + 0.3A \\
 0.4 &= 0.1A \\
 A &= 4
 \end{aligned}$$

Question9

A volume of x mL of 5 M NaHCO₃ solution was mixed with 10 mL of 2 M H₂CO₃ solution to make an electrolytic buffer. If the same buffer was used in the following electrochemical cell to record a cell potential of 235.3 mV, then the value of x = _____ mL (nearest integer).



Consider up to one place of decimal for intermediate calculations

$$\left[\begin{array}{l} \text{Given: } E^\circ_{\text{Sn(OH)}_6^{2-} \mid \text{HSnO}_2^-} = -0.90 \text{ V} \\ E^\circ_{\text{Bi}_2\text{O}_3 \mid \text{Bi}} = -0.44 \text{ V} \\ \text{pK}_a(\text{H}_2\text{CO}_3) = 6.11 \\ \frac{2.303RT}{F} = 0.059 \text{ V} \\ \text{Antilog}(1.29) = 19.5 \end{array} \right]$$

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Answer: 78

Solution:

For the half-cells (25 °C):

(1) Sn(IV)/Sn(II) couple (given $E^\circ = -0.90 \text{ V}$)

Balanced reduction in basic medium:



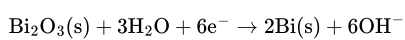
$$E_{\text{Sn}} = E_{\text{Sn}}^\circ - \frac{0.059}{2} \log \left(\frac{[\text{HSnO}_2^-][\text{OH}^-]^3}{[\text{Sn(OH)}_6^{2-}]} \right)$$

Given $[\text{Sn(OH)}_6^{2-}] = 0.5$, $[\text{HSnO}_2^-] = 0.05$:

$$\frac{[\text{HSnO}_2^-]}{[\text{Sn(OH)}_6^{2-}]} = \frac{0.05}{0.5} = 0.1$$

(2) Bi₂O₃/Bi couple (given $E^\circ = -0.44 \text{ V}$)

Reduction:



$$E_{\text{Bi}} = E_{\text{Bi}}^\circ - \frac{0.059}{6} \log([\text{OH}^-]^6) = E_{\text{Bi}}^\circ - 0.059 \log[\text{OH}^-]$$

Bi has higher E° , so it is the **cathode**. Thus

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} = E_{\text{Bi}} - E_{\text{Sn}}$$

Substitute and simplify:

$$E_{\text{cell}} = (-0.44 - 0.059 \log[\text{OH}^-]) - \left(-0.90 - \frac{0.059}{2} \log(0.1[\text{OH}^-]^3) \right)$$

$$E_{\text{cell}} = 0.46 - 0.059 \log[\text{OH}^-] + \frac{0.059}{2} (\log 0.1 + 3 \log[\text{OH}^-])$$

Since $\log 0.1 = -1$:

$$E_{\text{cell}} = 0.46 + \frac{0.059}{2} (-1) + \left(-0.059 + \frac{3 \cdot 0.059}{2} \right) \log[\text{OH}^-]$$

$$E_{\text{cell}} = 0.4305 + 0.0295 \log[\text{OH}^-]$$

Given $E_{\text{cell}} = 0.2353 \text{ V}$:

$$0.2353 = 0.4305 + 0.0295 \log[OH^-] \Rightarrow \log[OH^-] = \frac{-0.1952}{0.0295} \approx -6.6$$

$$\text{So } pOH \approx 6.6 \Rightarrow pH \approx 14 - 6.6 = 7.4.$$

Buffer calculation (H_2CO_3/HCO_3^-)

Henderson–Hasselbalch:

$$pH = pK_a + \log \frac{[HCO_3^-]}{[H_2CO_3]}$$

$$7.4 = 6.11 + \log \frac{[HCO_3^-]}{[H_2CO_3]} \Rightarrow \log \frac{[HCO_3^-]}{[H_2CO_3]} = 1.29 \Rightarrow \frac{[HCO_3^-]}{[H_2CO_3]} = 19.5$$

Moles mixed:

$$\bullet \quad n(H_2CO_3) = 2 \text{ M} \times 10 \text{ mL} = 2 \times 0.010 = 0.020 \text{ mol}$$

$$\bullet \quad n(HCO_3^-) = 5 \text{ M} \times x \text{ mL} = 5 \times \frac{x}{1000} = 0.005x \text{ mol}$$

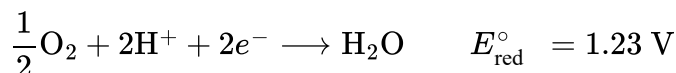
Ratio = 19.5:

$$\frac{0.005x}{0.020} = 19.5 \Rightarrow 0.005x = 0.39 \Rightarrow x = 78$$

Answer: 78 mL

Question 10

Consider the following two half-cell reactions along with the standard reduction potential given :



A fuel cell was set up using the above two reactions such that the cell operates under the standard condition of 1 bar pressure and 298 K temperature. The fuel cell works with 80% efficiency. If the work derived from the cell using 1 mol of CH_3OH is used to compress an ideal gas isothermally against a constant pressure of 1 kPa, then the change in the volume of the gas,

$$\Delta V = \text{_____ m}^3. \text{ (nearest integer)}$$

$$\text{Given : } F = 96500 \text{ C mol}^{-1}$$

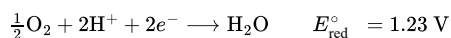
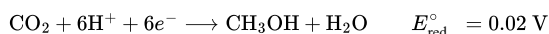
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Answer: 560

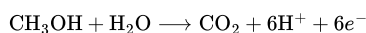
Solution:

For the fuel cell, methanol will be oxidised and oxygen will be reduced.

Given reduction half-cells:



Since methanol is the fuel, it will undergo oxidation. So we reverse the first reaction:



Its oxidation potential is:

$$E_{\text{ox}}^\circ = -0.02 \text{ V}$$

Now calculate standard cell potential:

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ + E_{\text{anode(ox)}}^\circ$$

$$E_{\text{cell}}^{\circ} = 1.23 + (-0.02) = 1.21 \text{ V}$$

Number of electrons transferred:

$$n = 6$$

So maximum electrical work from 1 mol of CH_3OH is

$$w_{\text{max}} = nFE_{\text{cell}}^{\circ}$$

$$w_{\text{max}} = 6 \times 96500 \times 1.21$$

$$w_{\text{max}} = 700590 \text{ J}$$

Fuel cell works with 80% efficiency, so actual work obtained is

$$w = 0.8 \times 700590$$

$$w = 560472 \text{ J}$$

This work is used to compress an ideal gas isothermally against constant pressure $P = 1 \text{ kPa} = 1000 \text{ Pa}$.

For compression against constant external pressure,

$$w = P\Delta V$$

So,

$$\Delta V = \frac{w}{P}$$

$$\Delta V = \frac{560472}{1000} = 560.472 \text{ m}^3$$

Nearest integer:

$$\Delta V \approx 560 \text{ m}^3$$

Since it is compression, volume decreases. So strictly,

$$\Delta V = -560 \text{ m}^3$$

If only magnitude is asked, then

$$\boxed{560}$$

Question 11

An electrochemical cell, consist of the following two redox couples, $\text{M}^{x+}(\text{aq})/\text{M}(\text{s})$ $\left[E_{\text{red}}^{\ominus} = +0.15 \text{ V} \right]$ and $\text{Fe}^{3+}(\text{aq})/\text{Fe}(\text{s})$ $\left[E_{\text{red}}^{\ominus} = -0.036 \text{ V} \right]$. The cell EMF (E_{cell}) is recorded to be 0.2057 V . If the reaction quotient of the electrochemical reaction is found to be 10^{-2} , then the value of x is _____. (Nearest integer)

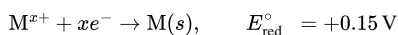
[Given : M is a p-block metal and $\frac{2.303RT}{F} = 0.059 \text{ V}$]

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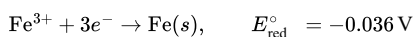
Answer: 2

Solution:

For the cell, the two reduction potentials are:



and



Since the reduction potential of M^{x+}/M is greater, it will act as the cathode.

So,

- Cathode: $M^{x+} + xe^{-} \rightarrow M(s)$
- Anode: $Fe(s) \rightarrow Fe^{3+} + 3e^{-}$

Hence,

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = 0.15 - (-0.036) = 0.186 \text{ V}$$

Now use the Nernst equation:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

Given:

$$E_{\text{cell}} = 0.2057 \text{ V}, \quad Q = 10^{-2}$$

Since $\log 10^{-2} = -2$,

$$0.2057 = 0.186 - \frac{0.059}{n}(-2)$$

$$0.2057 = 0.186 + \frac{0.118}{n}$$

$$0.2057 - 0.186 = \frac{0.118}{n}$$

$$0.0197 = \frac{0.118}{n}$$

$$n = \frac{0.118}{0.0197} \approx 5.99 \approx 6$$

So the number of electrons transferred in the balanced cell reaction is

$$n = 6$$

Now, for balancing electrons:

- Oxidation of iron gives 3 electrons
- Reduction of M^{x+} consumes x electrons

Therefore, total electrons transferred in the balanced equation will be $\text{LCM}(3, x)$.

Given this is 6, we get:

$$\text{LCM}(3, x) = 6$$

Possible values of x are 2 or 6.

Now it is given that M is a p-block metal. Common p-block metals generally form ions like 2^{+} , 3^{+} , 4^{+} , etc. Among the possible values, the most suitable is

$$x = 2$$

So, the nearest integer value of x is

$$\boxed{2}$$

Question 12

At 298 K, the molar conductivity of $x\%$ (w/w) MX solution (aqueous) is $123.5 \text{ S cm}^2 \text{ mol}^{-1}$. The conductance of same solution is $1.9 \times 10^{-3} \text{ S}$. The value of x is $____ \times 10^{-2}$.

(Given: cell constant = 1.3 cm^{-1} ; molar mass of MX is 75 g mol^{-1} , density of aqueous solution of MX at 298 K is 1.0 g mL^{-1})

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Answer: 15

Solution:

Conductivity (κ) is related to the conductance (G) and the cell constant (G^*) by the following relationship:

$$\kappa = G \times G^*$$

Substituting the given values:

$$\kappa = (1.9 \times 10^{-3} \text{ S}) \times (1.3 \text{ cm}^{-1})$$

$$\kappa = 2.47 \times 10^{-3} \text{ S cm}^{-1}$$

The molar conductivity (Λ_m) is related to the conductivity (κ) and the molarity (C in mol L^{-1}) by the formula:

$$\Lambda_m = \frac{\kappa \times 1000}{C}$$

Rearranging the formula to solve for molarity (C):

$$C = \frac{\kappa \times 1000}{\Lambda_m}$$

Substituting the values into the equation:

$$C = \frac{(2.47 \times 10^{-3} \text{ S cm}^{-1}) \times 1000 \text{ cm}^3 \text{ L}^{-1}}{123.5 \text{ S cm}^2 \text{ mol}^{-1}}$$

$$C = \frac{2.47}{123.5} \text{ mol L}^{-1}$$

$$C = 0.02 \text{ mol L}^{-1}$$

An $x\%$ (w/w) aqueous solution means there are x g of the solute (MX) in 100 g of the solution.

Using the density of the solution, we can find the volume of 100 g of the solution:

$$\text{Volume of solution} = \frac{\text{Mass of solution}}{\text{Density}}$$

$$\text{Volume} = \frac{100 \text{ g}}{1.0 \text{ g mL}^{-1}} = 100 \text{ mL}$$

Molarity is defined as the number of moles of solute per 1000 mL (1 liter) of the solution.

Since 100 mL of the solution contains x g of MX, 1000 mL of the solution will contain $10x$ g of MX.

Now, calculate the number of moles of MX in 1000 mL, which is exactly our molarity (C):

$$\text{Moles in 1 L} = \frac{\text{Mass of solute in 1 L}}{\text{Molar mass of solute}}$$

$$C = \frac{10x}{75}$$

We already calculated the molarity to be 0.02 mol L^{-1} . Equating the two values:

$$\frac{10x}{75} = 0.02$$

$$10x = 0.02 \times 75$$

$$10x = 1.5$$

$$x = 0.15$$

The problem requires x to be expressed in the form of $\text{____} \times 10^{-2}$.

$$x = 0.15 = 15 \times 10^{-2}$$

The value of x is 15.

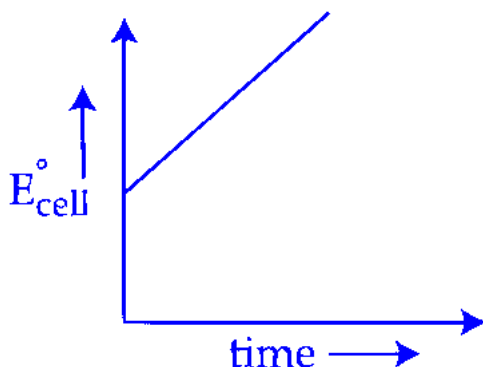
Question 13

For a closed circuit Daniell cell, which of the following plots is the accurate one at a given temperature?

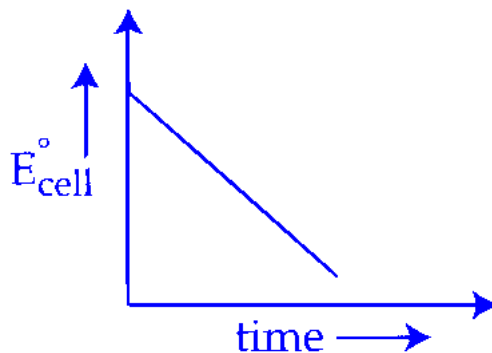
JEE Main 2026 (Online) 21st January Evening Shift

Options:

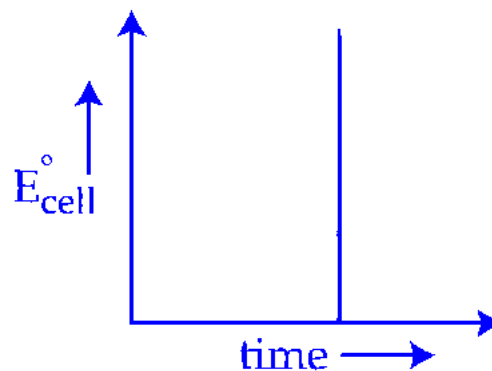
A.



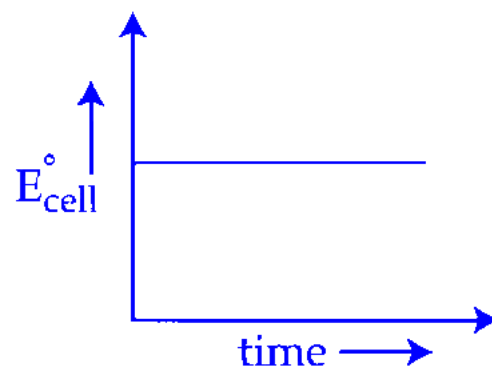
B.



C.



D.



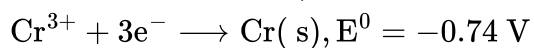
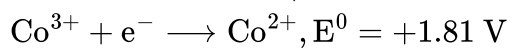
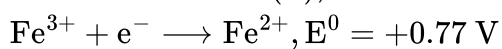
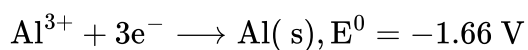
Answer: D

Solution:

Standard cell potential (E_{cell}^0) does not change as time passes.

Question14

Consider the following reduction processes :

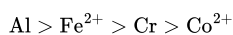


The tendency to act as reducing agent decreases in the order :

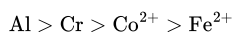
JEE Main 2026 (Online) 22nd January Evening Shift

Options:

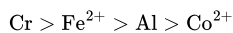
A.



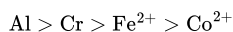
B.



C.



D.



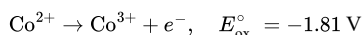
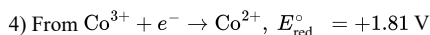
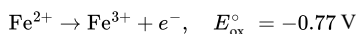
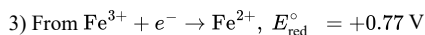
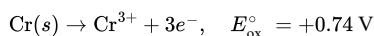
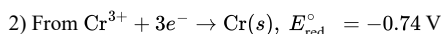
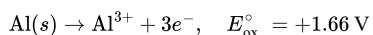
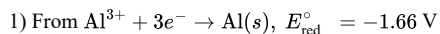
Answer: D

Solution:

Reducing agent is the species that **gets oxidised** (donates electrons). In NCERT terms:

- **More negative E_{red}°** for a couple $\Rightarrow$ the **reduced form** is a **stronger reducing agent** (it will be more easily oxidised).
- Equivalently, **higher $E_{\text{ox}}^{\circ} \Rightarrow$ stronger reducing agent.**

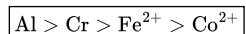
Now write oxidation potentials for the given reducing species:



Now arrange reducing strength (decreasing E_{ox}°):

$$+1.66 > +0.74 > -0.77 > -1.81$$

So, tendency to act as reducing agent decreases as:



Correct option: D

Question15

In the given electrochemical cell, $\text{Ag}(s)|\text{AgCl}(s)|\text{FeCl}_2(\text{aq}), \text{FeCl}_3(\text{aq}) | \text{Pt}(s)$ at 298 K, the cell potential (E_{cell}) will increase when :

A. Concentration of Fe^{2+} is increased.

B. Concentration of Fe^{3+} is decreased.

C. Concentration of Fe^{2+} is decreased.

D. Concentration of Fe^{3+} is increased.

E. Concentration of Cl^{-} is increased.

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

C, D and E Only

B.

A and B Only

C.

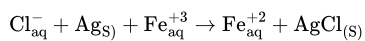
B Only

D.

A and E Only

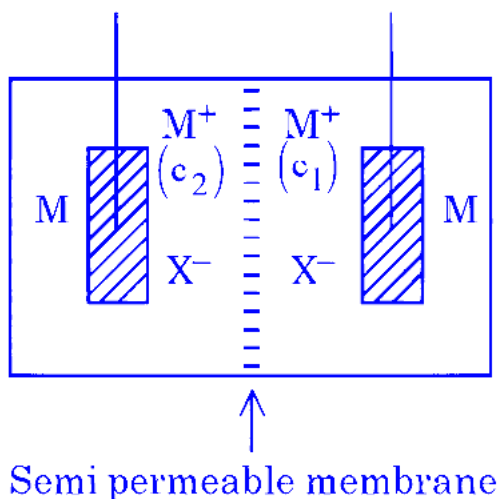
Answer: A

Solution:



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{1} \log \frac{[\text{Fe}^{2+}]}{[\text{Cl}^-] [\text{Fe}^{3+}]}$$

Question16



Consider the above electrochemical cell where a metal electrode (M) is undergoing redox reaction by forming M^+ ($\text{M} \rightarrow \text{M}^+ + \text{e}^-$). The cation M^+ is present in two different concentrations c_1 and c_2 as shown above. Which of the following statement is correct for generating a positive cell potential?

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Options:

A.

If c_1 is present at anode, then $c_1 > c_2$.

B.

If c_1 is present at cathode, then $c_1 > c_2$.

C.

If c_1 is present at cathode, then $c_1 < c_2$.

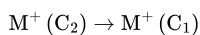
D.

If c_1 is present at anode, then $c_1 = c_2$.

Answer: B

Solution:

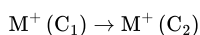
(1) If C_1 is at anode $\Rightarrow$ cell reaction



$$E_{\text{cell}} = -0.059 \log \frac{C_1}{C_2}$$

$$\therefore E_{\text{cell}} > 0 \Rightarrow C_1 < C_2$$

(2) If C_1 is at cathode

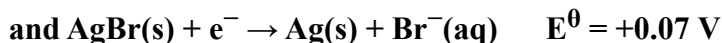
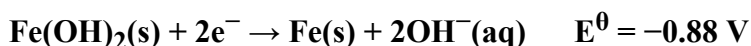


$$E_{\text{cell}} = -0.059 \log \frac{C_2}{C_1} > 0$$

$$C_2 < C_1$$

Question 17

An electrochemical cell is constructed using half cells in the direction of spontaneous change:



Which of the following option is correct?

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Options:



B.

$$E_{\text{cell}}^\theta = -0.95 \text{ V}$$

C.

Fe is reduced in the electrochemical cell

D.

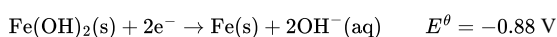
E_{cell}^θ is an extensive property

Answer: A

Solution:

For a spontaneous electrochemical cell, the half-reaction with higher standard reduction potential undergoes reduction, and the other one undergoes oxidation.

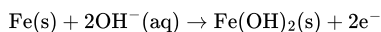
Given half-reactions:



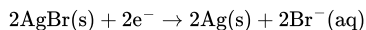


Since $+0.07 \text{ V} > -0.88 \text{ V}$, the AgBr/Ag half-cell will undergo reduction and the iron half-cell will undergo oxidation.

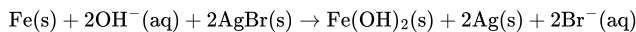
So reverse the iron half-reaction:



Multiply the silver half-reaction by 2:



Now add them:



This matches Option A.

Now calculate E_{cell}^θ :

$$E_{\text{cell}}^\theta = E_{\text{cathode}}^\theta - E_{\text{anode}}^\theta$$

Here,

- cathode: AgBr/Ag, so $E_{\text{cathode}}^\theta = +0.07 \text{ V}$
- anode: reduction potential of Fe(OH)₂/Fe is -0.88 V

Therefore,

$$E_{\text{cell}}^\theta = 0.07 - (-0.88) = 0.95 \text{ V}$$

So Option B is false.

Option C is false because Fe is oxidised, not reduced.

Option D is false because E_{cell}^θ is an intensive property, not an extensive property.

Hence, the correct option is:

Option A

Question18

One half cell in a voltaic cell is constructed by dipping silver rod in AgNO₃ solution of unknown concentration, other half cell is Zn rod dipped in 1 molar solution of ZnSO₄.

A voltage of 1.60 V is measured at 298 K for this cell. What is the concentration of Ag⁺ ions used in terms of log x ($x = [\text{Ag}^+]$)?

$$E_{\text{Zn}^{2+}/\text{Zn}}^\ominus = -0.76 \text{ V}, \quad E_{\text{Ag}^+/\text{Ag}}^\ominus = +0.80 \text{ V}, \quad \frac{2.303RT}{F} = 0.059 \text{ V}$$

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Options:

A.

$$\frac{2}{3.9}$$

B.

$$\frac{4}{5.9}$$

C.

$$\frac{2.9}{2}$$

D.

$$\frac{5.9}{4}$$

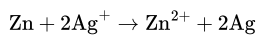
Answer: B

Solution:

For the cell,

- anode: $\text{Zn} \rightarrow \text{Zn}^{2+} + 2e^{-}$
- cathode: $\text{Ag}^{+} + e^{-} \rightarrow \text{Ag}$

So the overall cell reaction is



Now, standard cell potential:

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

$$E_{\text{cell}}^{\circ} = 0.80 - (-0.76) = 1.56 \text{ V}$$

Given:

$$E_{\text{cell}} = 1.60 \text{ V}$$

Using Nernst equation at 298 K:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

Here, $n = 2$.

Reaction quotient:

$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Ag}^{+}]^2}$$

Since ZnSO_4 is 1 M,

$$[\text{Zn}^{2+}] = 1$$

Let

$$[\text{Ag}^{+}] = x$$

Then

$$Q = \frac{1}{x^2}$$

Substitute in Nernst equation:

$$1.60 = 1.56 - \frac{0.059}{2} \log \left(\frac{1}{x^2} \right)$$

Now,

$$\log \left(\frac{1}{x^2} \right) = -2 \log x$$

So,

$$1.60 = 1.56 - \frac{0.059}{2} (-2 \log x)$$

$$1.60 = 1.56 + 0.059 \log x$$

$$1.60 - 1.56 = 0.059 \log x$$

$$0.04 = 0.059 \log x$$

$$\log x = \frac{0.04}{0.059} = \frac{4}{5.9}$$

Hence, the correct answer is

$$\boxed{\frac{4}{5.9}}$$

So, Option B is correct.

Question19

Consider the following data.

Electrolyte	$\Lambda_m^\circ (\text{S cm}^2 \text{ mol}^{-1})$
BaCl ₂	x_1
H ₂ SO ₄	x_2
HCl	x_3

BaSO₄ is sparingly soluble in water. If the conductivity of the saturated BaSO₄ solution is $x \text{ S cm}^{-1}$ then the solubility product of BaSO₄ can be given as (Here $\Lambda_m = \Lambda_m^\circ$)

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Options:

A.

$$\frac{10^6 x^2}{\alpha^2 (x_1 + x_2 - 2x_3)^2}$$

B.

$$\frac{x^2}{(x_1 + x_2 - 2x_3)^2}$$

C.

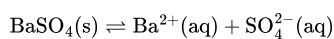
$$\frac{\alpha^2 (x_1 + x_2 - 2x_3)^2}{10^6 x^2}$$

D.

$$\frac{x^2}{(x_1 + x_2 + 2x_3)^2}$$

Answer: A

Solution:



If the solubility of BaSO₄ is $s \text{ mol L}^{-1}$, then in solution:

$$[\text{Ba}^{2+}] = s, \quad [\text{SO}_4^{2-}] = s$$

Thus, the solubility product is

$$K_{\text{sp}} = [\text{Ba}^{2+}][\text{SO}_4^{2-}] = s^2$$

Relation between molar conductivity and observed conductivity

By definition,

$$\kappa = \frac{\Lambda_m c}{1000}$$

where

- κ = conductivity in S cm^{-1} ,
- Λ_m = molar conductivity in $\text{S cm}^2 \text{ mol}^{-1}$,
- c = molar concentration in mol L^{-1} .

For the saturated solution of BaSO₄,

$$\kappa = x$$

and

$$c = s$$

Therefore,

$$\Lambda_m^{(\text{BaSO}_4)} = \frac{1000x}{s}$$

Determining Λ_m° for BaSO₄

We can find Λ_m° for BaSO₄ using **Kohlrausch's law of independent migration of ions**.

According to the law,

$\Lambda_m^\circ(\text{Electrolyte}) = \text{sum of molar ionic conductivities of its ions.}$

Now, from the data:

$$\Lambda_m^\circ(\text{BaCl}_2) = \lambda_{\text{Ba}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ = x_1$$

$$\Lambda_m^\circ(\text{H}_2\text{SO}_4) = 2\lambda_{\text{H}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = x_2$$

$$\Lambda_m^\circ(\text{HCl}) = \lambda_{\text{H}^+}^\circ + \lambda_{\text{Cl}^-}^\circ = x_3$$

To get $\Lambda_m^\circ(\text{BaSO}_4) = \lambda_{\text{Ba}^{2+}}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ$,

we combine the above equations:

$$(x_1 + x_2 - 2x_3) = (\lambda_{\text{Ba}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ) + (2\lambda_{\text{H}^+}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ) - 2(\lambda_{\text{H}^+}^\circ + \lambda_{\text{Cl}^-}^\circ)$$

Simplifying,

$$x_1 + x_2 - 2x_3 = \lambda_{\text{Ba}^{2+}}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = \Lambda_m^\circ(\text{BaSO}_4)$$

Thus,

$$\Lambda_m^\circ(\text{BaSO}_4) = x_1 + x_2 - 2x_3$$

Equating conductivities

Since it is given that ($\Lambda_m = \Lambda_m^\circ$) for the saturated solution:

$$\Lambda_m^\circ = \frac{1000x}{s}$$

$$s = \frac{1000x}{\Lambda_m^\circ} = \frac{1000x}{x_1 + x_2 - 2x_3}$$

Expression for solubility product

$$K_{\text{sp}} = s^2 = \left(\frac{1000x}{x_1 + x_2 - 2x_3} \right)^2 = \frac{10^6 x^2}{(x_1 + x_2 - 2x_3)^2}$$

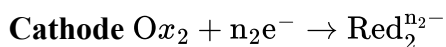
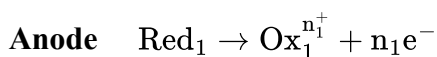
Final Answer

$$K_{\text{sp}} = \frac{10^6 x^2}{(x_1 + x_2 - 2x_3)^2}$$

Hence, **Option A** is correct.

Question 20

For a general redox reaction



Which of the following statement is incorrect?

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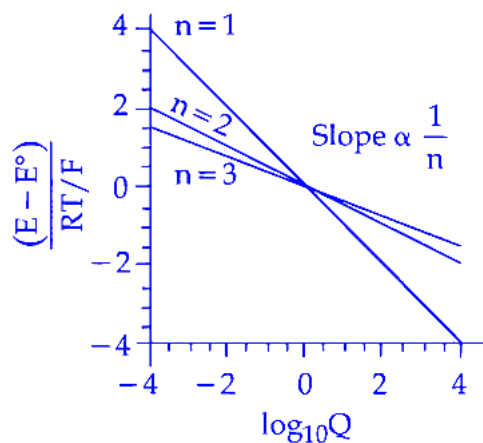
Options:

A.

The overall reaction can be written as $\text{n}_2 \text{Red}_1 + \text{n}_1 \text{Ox}_2 \rightleftharpoons \text{n}_2 \text{Ox}_1^{\text{n}_1^+} + \text{n}_1 \text{Red}_2^{\text{n}_2^-}$

B.

The electrons do not appear in the overall reaction because electrons produced at the anode are consumed at the cathode.



C.

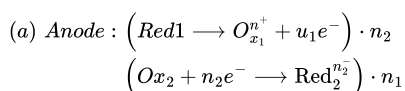
Here n is the number of electrons transferred in redox reaction.

D.

If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

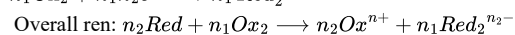
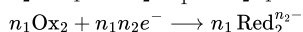
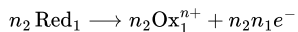
Answer: D

Solution:



To add the anode and cathode half-reactions, the number of electrons must be the same on both sides.

So, we multiply the anode reaction by n_2 and the cathode reaction by n_1 , so that both have $n_1 n_2$ electrons.



Now, when we add these two equations, the $n_1 n_2 e^-$ on the product side of the anode reaction cancels with the $n_1 n_2 e^-$ on the reactant side of the cathode reaction.

(b) The e^- produced at the anode are consumed at the cathode, that's why e^- do not appear in the overall reaction.

This statement is correct because electrons cancel during addition of the two half-reactions.

(c) We know,

$$E = E^\circ - \frac{RT}{nF} \log Q$$

$$E - E^\circ = -\frac{RT}{nF} \log Q$$

$$\frac{E - E^\circ}{RT/F} = -\frac{1}{n} \log Q$$

Here, E changes with $\log Q$ according to the Nernst equation, and n is the number of electrons transferred in the balanced redox reaction.

plotting a $y = mx + c$ graph,

$$y = \frac{E - E^\circ}{RT/F}$$

$$x = \log Q$$

$$\text{slope } (m) = -\frac{1}{n}$$

$$\text{slope} \propto \frac{1}{n}$$

So, the slope is $-\frac{1}{n}$. This statement is also correct.

Where, n = no. of e^- transferred in redox reaction.

(D) The electrical work done by a galvanic cell is given by the product of the total charge transferred and the potential difference, not their ratio.

This statement is correct because electrical work is given by $W = qV$ (charge multiplied by potential difference).

$$W_{\text{electrical}} = \text{charge} \times \text{Potential difference.}$$

$$= Q \times V$$

$$W_e = It \times V$$

Question21

Given at 298 K :

$$E_{\text{Fe}^{2+}/\text{Fe}}^{\ominus} = X \text{ Volt}$$

$$E_{\text{Fe}^{3+}/\text{Fe}}^{\ominus} = Y \text{ Volt}$$

The $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^{\ominus}$ in Volt at 298 K is given by :

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Options:

A.

$$2X - 3Y$$

B.

$$3Y - 2X$$

C.

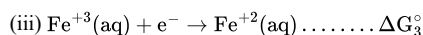
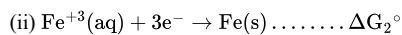
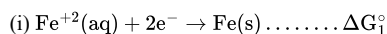
$$3Y + 2X$$

D.

$$Y + X$$

Answer: B

Solution:



$$iii = ii - i$$

$$-1 \times F \times E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^{\circ} = -3 \times F \times y - (-2 \times F \times x)$$

$$E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^{\circ} = 3y - 2x$$

Option (B) is correct

Question22

A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2 A . The amount of the aluminium deposited at the cathode is _____ .

[Given : molar mass of aluminium and chlorine are 27 g mol^{-1} and 35.5 g mol^{-1} respectively. Faraday constant = 96500 C mol^{-1}]

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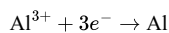
Options:

A. 1.660 g

- B. 1.007 g
- C. 0.336 g
- D. 0.441 g

Answer: C

Solution:



The mass of aluminium deposited can be calculated using Faraday's laws of electrolysis. The steps are as follows:

Compute the total charge passed:

$$Q = I \cdot t = 2 \text{ A} \cdot (30 \times 60 \text{ s}) = 3600 \text{ C}$$

Determine the moles of aluminium deposited. Since three electrons are required to deposit one mole of aluminium, the number of moles is given by:

$$n(\text{Al}) = \frac{Q}{3F} = \frac{3600}{3 \times 96500} \approx 0.01242 \text{ mol}$$

Finally, calculate the mass of aluminium using its molar mass ($M = 27 \text{ g/mol}$):

$$m(\text{Al}) = n(\text{Al}) \times M = 0.01242 \times 27 \approx 0.335 \text{ g}$$

Thus, the amount of aluminium deposited at the cathode is approximately

0.336 g

Question23

Which of the following electrolyte can be used to obtain $\text{H}_2 \text{ S}_2\text{O}_8$ by the process of electrolysis ?

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Options:

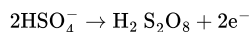
- A. Concentrated solution of sulphuric acid
- B. Dilute solution of sodium sulphate.
- C. Acidified dilute solution of sodium sulphate.
- D. Dilute solution of sulphuric acid

Answer: A

Solution:

$\text{H}_2 \text{ S}_2\text{O}_8$ is obtained by electrolysis of concentrated solution of sulphuric acid.

At anode :



Question24

Given below are two statements :

Statement (I) : Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

Statement (II) : The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

- A. Both Statement I and Statement II are true
- B. Both Statement I and Statement II are false
- C. Statement I is true but Statement II is false
- D. Statement I is false but Statement II is true

Answer: C

Solution:

Let's examine both statements one by one and see whether they hold true for typical corrosion processes (e.g., iron rusting):

Statement (I)

"Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode."

Corrosion is electrochemical

True. Corrosion, especially rusting of iron, involves oxidation at one region (the anode) and reduction at another region (the cathode) on the metal's surface.

Pure metal as anode, impurity as cathode

In many practical cases (e.g., iron containing small amounts of carbon or other impurities), the *relatively pure* region of the iron is more active (less noble) and tends to oxidize (lose electrons) — that is, it serves as the anode.

The impurity (e.g., carbon-rich region) often behaves more nobly and thus becomes the cathode region where reduction (e.g., oxygen reduction) occurs.

This difference in electrode potentials between the pure region and the impurity region drives the corrosion cell.

Hence, **Statement (I)** is generally considered **true** in the usual context of corrosion (like rusting of iron).

Statement (II)

"The rate of corrosion is more in alkaline medium than in acidic medium."

For most common metals (like iron), **acidic** media typically enhance corrosion because abundant H^+ ions (and possibly other acidic species) can accelerate the oxidation/dissolution of the metal.

In *mildly* alkaline or neutral environments, metals often form protective oxide or hydroxide layers that can slow down further corrosion.

While certain strong bases can attack specific metals (e.g., Al in strong NaOH), **in general** for iron and many other metals, corrosion is more severe in acidic environments than in alkaline ones.

Thus, **Statement (II)** is **false** under normal corrosion scenarios (e.g., rusting of iron).

Conclusion

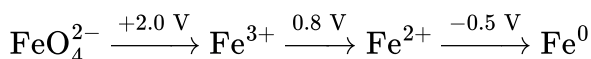
Statement (I): True

Statement (II): False

Therefore, the correct choice (matching these truth values) is:

(C) Statement I is true but Statement II is false.

Question 25



In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $E_{FeO_4^{2-}/Fe^{2+}}$ is :

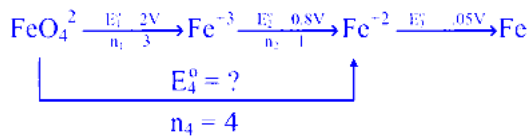
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Options:

- A. 1.2 V
B. 2.1 V
C. 1.4 V
D. 1.7 V

Answer: D

Solution:



$$\begin{aligned}\Delta G_4^0 &= \Delta G_1^0 + \Delta G_2^0 \\ \Rightarrow -n_4 \text{FE}_4^0 &= -n_1 \text{FE}_1^0 - n_2 \text{FE}_2^0 \\ \Rightarrow +4\text{E}_4^0 &= 3 \times 2 + (1 \times 0.8) \\ \Rightarrow \text{E}_4^0 &= \frac{6.8}{4} \text{ V} \\ \Rightarrow \text{E}_4^0 &= 1.7 \text{ V}\end{aligned}$$

Question26

Standard electrode potentials for a few half cells are mentioned below :

$$\begin{aligned} E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} &= 0.34 \text{ V}, E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} = -0.76 \text{ V} \\ E_{\text{Ag}^{+}/\text{Ag}}^{\circ} &= 0.80 \text{ V}, E_{\text{Mg}^{2+}/\text{Mg}}^{\circ} = -2.37 \text{ V} \end{aligned}$$

Which one of the following cells gives the most negative value of ΔG° ?

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Options:

- A. $\text{Zn} \mid \text{Zn}^{2+}(1\text{M}) \parallel \text{Ag}^{+}(1\text{M}) \mid \text{Ag}$
 B. $\text{Ag} \mid \text{Ag}^{+}(1\text{M}) \mid \text{Mg}^{2+}(1\text{M}) \mid \text{Mg}$
 C. $\text{Zn} \mid \text{Zn}^{2+}(1\text{M}) \mid \text{Mg}^{2+}(1\text{M}) \mid \text{Mg}$
 D. $\text{Cu} \mid \text{Cu}^{2+}(1\text{M}) \mid \text{Ag}^{+}(1\text{M}) \mid \text{Ag}$

Answer: A

Solution:

$$\therefore \Delta G^\circ = -nFE^\circ$$

$$\begin{aligned}\text{Option (1)} E^\circ &= 0.8 + 0.76 \\ &= 1.56 \text{ V}\end{aligned}$$

$$\begin{aligned}\therefore \Delta G^\circ &= -2 \times F \times 1.56 \\ &= -3.12 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{Option (2)} E^\circ &= -2.37 + 0.76 \\ &= -1.61 \text{ V}\end{aligned}$$

$$\begin{aligned}\therefore \Delta G^\circ &= -2 \times F \times (-1.61) \\ &= +3.22 \text{ V}\end{aligned}$$

$$\begin{aligned}\text{Option (3)} E^\circ &= -2.37 - 0.8 \\ &= -3.17 \text{ V}\end{aligned}$$

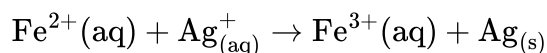
$$\begin{aligned}\therefore \Delta G^\circ &= -2 \\ &= F \times (-3.17) \\ &= +6.34\end{aligned}$$

$$\begin{aligned}\text{Option (4)} E^\circ &= 0.8 - 0.34 \\ &= 0.46 \text{ V}\end{aligned}$$

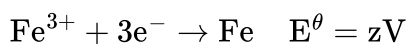
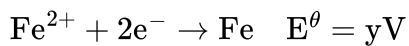
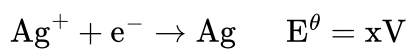
$$\begin{aligned}\Delta G^\circ &= -2 \times F \times 0.46 \\ &= -0.92 \text{ V}\end{aligned}$$

Question 27

For the given cell



The standard cell potential of the above reaction is Given:



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Options:

A. $x + 2y - 3z$

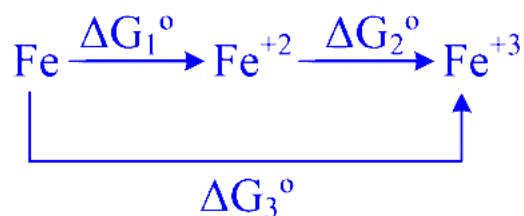
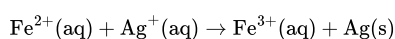
B. $x + 2y$

C. $y - 2x$

D. $x + y - z$

Answer: A

Solution:



$$\Delta G_3^0 = \Delta G_1^0 + \Delta G_2^0$$

$$-3F(-z) = -2F(-y) + \Delta G_2^0$$

$$\Delta G_2^0 = 3Fz - 2Fy$$

$$\text{Also } \Delta G_2^0 = -nFE_{\text{Fe}^{+2}/\text{Fe}^{+3}}^0$$

$$3Fz - 2Fy = -1F \left(E_{\text{Fe}^{+2}/\text{Fe}^{+3}}^0 \right)$$

$$E_{\text{Fe}^{+2}/\text{Fe}^{+3}}^0 = 2y - 3z$$

$$E_{\text{Cell}}^0 \text{ for reaction will be}$$

$$E_{\text{Ag}^+/\text{Ag}}^0 + E_{\text{Fe}^{+2}/\text{Fe}^{+3}}^0$$

$$= x + 2y - 3z$$

Option (2)

Question 28

Based on the data given below :

$$E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}}^{\circ} = 1.33 \text{ V} \quad E_{\text{Cl}_2/\text{Cl}^{(-)}}^{\circ} = 1.36 \text{ V}$$

$$E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 = 1.51 \text{ V} \quad E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = -0.74 \text{ V}$$

the strongest reducing agent is :

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Options:

A. Cl^-

B. MnO_4^-

C. Cr

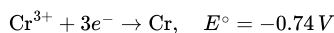
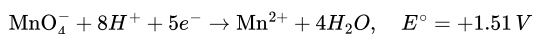
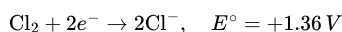
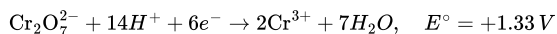
D. Mn^{2+}

Answer: C

Solution:

To determine the strongest reducing agent among the options, we need to consider which species most readily donates electrons (i.e., is most easily oxidized). One standard method is to compare the standard reduction potentials of the corresponding half-reactions. A lower (more negative) standard reduction potential indicates that the reduced species is less stable and more prone to oxidation.

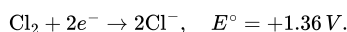
Here's the data provided:



Now, let's consider the species from the Options:

Option A: Cl^-

The relevant half-reaction is:



The reduced form here is Cl^- . Its corresponding oxidation (the reverse reaction) would have an electrode potential of

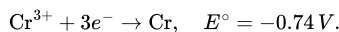
$$E_{\text{oxidation}}^{\circ} = -1.36 \text{ V}.$$

Option B: MnO_4^-

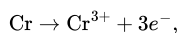
In the given half-reaction, MnO_4^- acts as the oxidizing agent (it is reduced to Mn^{2+}). Therefore, MnO_4^- is not a good reducing agent.

Option C: Cr (metallic chromium)

The relevant half-reaction is:



Here, Cr is the reduced species. Reversing this reaction gives:



with an effective oxidation potential of

$$E_{\text{oxidation}}^\circ = +0.74 \text{ V}.$$

Option D: Mn²⁺

In the MnO₄⁻/Mn²⁺ couple, Mn²⁺ is the reduced form. Its reverse (oxidation) potential would be:

$$E_{\text{oxidation}}^\circ = -1.51 \text{ V}.$$

When comparing the oxidation potentials (remember, a higher oxidation potential means the species is more willing to lose electrons), we have:

$$\text{Cl}^-: E_{\text{ox}}^\circ = -1.36 \text{ V}$$

$$\text{Cr}: E_{\text{ox}}^\circ = +0.74 \text{ V}$$

$$\text{Mn}^{2+}: E_{\text{ox}}^\circ = -1.51 \text{ V}$$

Among these, metallic Cr stands out because its corresponding reduction half-reaction has the most negative potential (-0.74 V). This negative value indicates that Cr (in its elemental form) is unstable relative to its oxidized form (Cr^{3+}) and, therefore, is the most eager to lose electrons. In other words, Cr is the strongest reducing agent.

Thus, the answer is:

Option C: Cr

Question29

The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

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A.

$$E_{\text{Sn}^{4+}/\text{Sn}^{2+}}^\circ = +1.15 \text{ V}$$

B.

$$E_{\text{Al}^{3+}/\text{Al}}^\circ = -1.66 \text{ V}$$

C.

$$E_{\text{Pb}^{4+}/\text{Pb}^{2+}}^\circ = +1.67 \text{ V}$$

D.

$$E_{\text{Tl}^{3+}/\text{Tl}}^\circ = +1.26 \text{ V}$$

Answer: C

Solution:

Oxidizing capacity is an element's tendency to donate electrons, or the atmosphere's ability to oxidize compounds.

A higher standard reduction potential indicates a stronger oxidizing capacity. A substance with a higher standard reduction potential is more likely to gain electrons and act as an oxidizing agent in a redox reaction. The more positive the reduction potential, the stronger the oxidizing power of a species.

From the given options, the higher standard reduction potential is for lead (Pb). ($+1.67$).

so, the p-block ion with strongest oxidizing capacity is Pb (lead).

Answer: Option 3) $E_{\text{Pb}^{4+}/\text{Pb}^{2+}}^0 = +1.67 \text{ V}$

Question 30

The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?

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Options:

A.

Molar conductivity increases sharply with increase in concentration.

B.

Molar conductivity decreases sharply with increase in concentration.

C.

A small increase in molar conductivity is observed at infinite dilution.

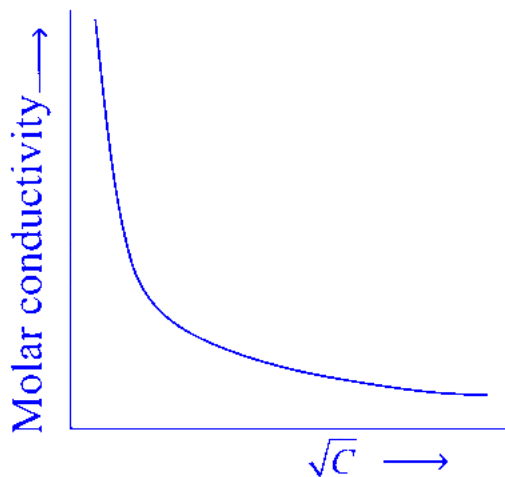
D.

A small decrease in molar conductivity is observed at infinite dilution.

Answer: B

Solution:

The molar conductivity of a weak electrolyte is plotted against the square root of its concentration:



c -concentration of weak electrolyte

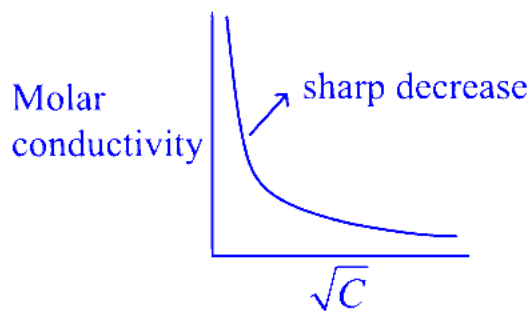
Weak electrolytes give curved decreasing graph.

1) Molar conductivity increases sharply with increase in concentration.

Incorrect.

With increase in concentration, molar conductivity decreases and it is lean from the graph of molar conductivity vs $\sqrt{c}$.

2) Molar conductivity decreases sharply with increase in concentration.



Correct

From the graph it is clear that, molar conductivity decreases sharply with increase in concentration.

3) A small increase in molar conductivity is observed at infinite dilution.

Incorrect

There is no small increase in molar conductivity with increase in concentration for a weak electrolyte. And there is no small increase in molar conductivity. Significant increase occurs.

4) A small decrease in molar conductivity is observed at infinite dilution.

Incorrect

For a weak electrolyte, the molar conductivity significantly increases with dilution, especially at infinite dilution.

Correct answer:

Option 2) Molar conductivity decreases sharply with increase in concentration.

Question 31

For a $\text{Mg} | \text{Mg}^{2+} (\text{aq}) || \text{Ag}^+ (\text{aq}) | \text{Ag}$ the correct Nernst Equation is :

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Options:

A. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Ag}^+]^2}{[\text{Mg}^{2+}]}$

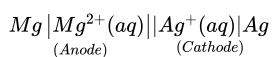
B. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]}$

C. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Ag}^+]}{[\text{Mg}^{2+}]}$

D. $E_{\text{cell}} = E_{\text{cell}}^{\circ} + \frac{RT}{2F} \ln \frac{[\text{Ag}^+]^2}{[\text{Mg}^{2+}]}$

Answer: D

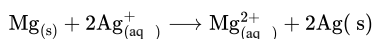
Solution:



Nernst equation (general)

$$E_{\text{Cell}} = E_{\text{Cell}}^{\circ} - \frac{RT}{nF} \ln Q$$

For the cell, the chemical equation can be written as



n (number of electrons transferred) = 2

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Mg}^{2+}][\text{Ag}]^2}{[\text{Mg}][\text{Ag}^+]^2}$$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{2F} \ln \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$$

$$E_{cell} = E_{cell}^o + \frac{RT}{2F} \ln \frac{[Ag^+]^2}{[Mg^{2+}]}$$

Concentration of solid species

$$[Mg] = [Ag] = 1$$

Correct answer :

$$\text{Option D) } E_{cell} = E_{cell}^o + \frac{RT}{2F} \ln \frac{[Ag^+]^2}{[Mg^{2+}]}$$

Question 32

O₂ gas will be evolved as a product of electrolysis of :

(A) an aqueous solution of AgNO₃ using silver electrodes.

(B) an aqueous solution of AgNO₃ using platinum electrodes.

(C) a dilute solution of H₂SO₄ using platinum electrodes.

(D) a high concentration solution of H₂SO₄ using platinum electrodes.

Choose the correct answer from the options given below :

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Options:

A. (B) and (C) only

B. (B) and (D) only

C. (A) and (D) only

D. (A) and (C) only

Answer: A

Solution:

(A) An aqueous solution of AgNO₃ using silver electrodes.

Cathode - Reduction - $Ag_{(aq)}^+ + e^- \rightarrow Ag(s)$

Anode - Oxidation - $Ag(s) \rightarrow Ag_{(aq)}^+ + e^-$

Solid silver will be deposited at the cathode. Solid anode (silver) will dissolve, releasing silver ions into the solution.

So, there is no formation of O₂ gas in this electrolysis.

(B) An aqueous solution of AgNO₃ using platinum electrodes.

Cathode - Reduction - $Ag_{(aq)}^+ + e^- \rightarrow Ag(s)$

Anode - Oxidation - $2H_2O \rightarrow 4H^+ + O_2 + 4e^-$

When platinum electrodes are used, Ag⁺ from solution is reduced and deposited at cathode whereas O₂ is produced at the anode.

(C) A dilute solution of H₂SO₄ using platinum electrodes.

Cathode - Reduction - $2H^+ + 2e^- \rightarrow H_2$

Anode - Oxidation - $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

H₂ gas is produced at the cathode and O₂ gas is produced at the anode.

(D) a high concentration solution of H₂SO₄ using platinum electrodes.

O₂ gas is not formed in this case.

Cathode - Reduction - The substance formed is H₂ gas.

Anode - Oxidation - The substance formed is not O₂ gas.

So, statements (B) and (C) are correct.

Question33

Match List - I with List - II :

List - I (Applications)	List - II (Batteries/Cell)
(A) Transistors	(I) Anode - Zn/Hg; Cathode - HgO + C
(B) Hearing aids	(II) Hydrogen fuel cell
(C) Inverters	(III) Anode - Zn; Cathode - Carbon
(D) Apollo space ship	(IV) Anode - Pb; Cathode - Pb PbO ₂

Choose the correct answer from the options given below :

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Options:

A.

(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

B.

(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C.

(A)-(IV), (B)-(III), (C)-(II), (D)-(I)

D.

(A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Answer: A

Solution:

For each application listed in List-I, match it with the appropriate battery or cell type from List-II based on their characteristics:

(A) Transistors: Generally use a dry cell or Leclanché cell, where the anode is zinc (Zn) and the cathode is a carbon rod surrounded by manganese dioxide (MnO₂).

Therefore, (A) Transistors match with (III) Anode - Zn; Cathode - Carbon.

(B) Hearing aids: Operate using a mercury cell. The anode in this cell is a zinc-mercury amalgam, and the cathode is composed of mercury oxide (HgO) with carbon.

Thus, (B) Hearing aids match with (I) Anode - Zn/Hg; Cathode - HgO + C.

(C) Inverters: Use a lead-acid storage cell. In this type of cell, the anode consists of lead (Pb), and the cathode is a lead dioxide (PbO₂) grid.

As a result, (C) Inverters match with (IV) Anode - Pb; Cathode - Pb | PbO₂.

(D) Apollo space ship: Utilizes a hydrogen-oxygen fuel cell. The hydrogen is oxidized at the anode, while oxygen is reduced at the cathode.

Therefore, (D) Apollo space ship matches with (II) Hydrogen fuel cell.

Summary of matches:

A - III

B - I

C - IV

D - II

This corresponds with Answer Option A.

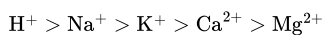
Question34

Correct order of limiting molar conductivity for cations in water at 298 K is :

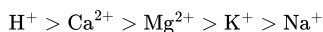
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Options:

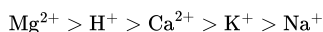
A.



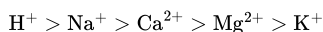
B.



C.



D.



Answer: B

Solution:

The limiting molar conductivities of ions are important in determining how well they conduct electricity in solution. Here are the values for some common cations at 298 K:

H^+ : 349.8 S cm² mol⁻¹

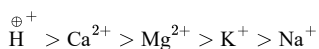
Na^+ : 50.11 S cm² mol⁻¹

K^+ : 73.52 S cm² mol⁻¹

Ca^{2+} : 119 S cm² mol⁻¹

Mg^{2+} : 106.12 S cm² mol⁻¹

Based on these values, the correct order of limiting molar conductivity for these cations is:



This order reflects the decreasing ability of each ion to conduct electricity in water under the given conditions.

Question35

The standard cell potential ($E_{\text{cell}}^{\ominus}$) of a fuel cell based on the oxidation of methanol in air that has been used to power television relay station is measured as 1.21 V . The standard half cell reduction potential for O_2 ($E_{\text{O}_2/\text{H}_2\text{O}}^{\circ}$) is 1.229 V .

Choose the correct statement :

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Options:

A. Reactants are fed at one go to each electrode.

B. Oxygen is formed at the anode.

C. The standard half cell reduction potential for the reduction of CO_2 ($E_{\text{CO}_2/\text{CH}_3\text{OH}}^{\circ}$) is 19 mV

D. Reduction of methanol takes place at the cathode.

Answer: C

Solution:

To determine the reaction potentials at the anode and cathode in the methanol fuel cell, we start with the given standard cell potential:

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Given:

$$E_{\text{cell}}^{\circ} = 1.21 \text{ V}$$

$$E_{\text{O}_2/\text{H}_2\text{O}}^{\circ} = 1.229 \text{ V}$$

Substituting the values,

$$1.21 = 1.229 - E_{\text{anode}}^{\circ}$$

Solving for E_{anode}° :

$$E_{\text{anode}}^{\circ} = 1.229 - 1.21 = 0.019 \text{ V}$$

Since this is a fuel cell based on the oxidation of methanol, methanol undergoes oxidation at the anode, while oxygen is reduced at the cathode. Therefore:

Anode Reaction: Oxidation of methanol occurs here, contributing to the fuel cell's electrical output.

Cathode Reaction: Reduction of oxygen occurs at the cathode, where the $E_{\text{O}_2/\text{H}_2\text{O}}^{\circ}$ potential is utilized.

Thus, we verify that the half-cell reduction potential for oxidation of methanol by CO_2 is indeed 19 mV, aligning with the calculated E_{anode}° . The processes correctly identify the roles of anode and cathode in this fuel cell.

Question36

On charging the lead storage battery, the oxidation state of lead changes from x_1 to y_1 at the anode and from x_2 to y_2 at the cathode. The values of x_1, y_1, x_2, y_2 are respectively :

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Options:

A. 0, +2, +4, +2

B. +2, 0, 0, +4

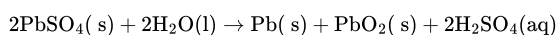
C. +4, +2, 0, +2

D. +2, 0, +2, +4

Answer: D

Solution:

For charging of lead storage battery cell reaction is



At anode PbSO_4 reduced back to Pb and at cathode PbSO_4 oxidised back to PbO_2 .

$$\therefore x_1 = +2, y_1 = 0$$

$$x_2 = +2, y_2 = 4$$

Question37

Given below are two statements :

1 M aqueous solutions of each of $\text{Cu}(\text{NO}_3)_2$, AgNO_3 , $\text{Hg}_2(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$ are electrolysed using inert electrodes. Given: $E^0_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$, $E^0_{\text{Hg}_2^{2+}/\text{Hg}} = 0.79 \text{ V}$, $E^0_{\text{Cu}^{2+}/\text{Cu}} = 0.24 \text{ V}$ and $E^0_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}$.

Statement (I) : With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu.

Statement (II) : Magnesium will not be deposited at the cathode instead oxygen gas will be evolved at the cathode.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are incorrect

B.

Statement I is incorrect but **Statement II** is correct

C.

Statement I is correct but **Statement II** is incorrect

D.

Both **Statement I** and **Statement II** are correct

Answer: C

Solution:

Let's analyze the situation step by step.

We have 1 M aqueous solutions of:

AgNO_3 with Ag^+/Ag $E^0 = +0.80 \text{ V}$

$\text{Hg}_2(\text{NO}_3)_2$ with $\text{Hg}_2^{2+}/\text{Hg}$ $E^0 = +0.79 \text{ V}$

$\text{Cu}(\text{NO}_3)_2$ with Cu^{2+}/Cu $E^0 = +0.24 \text{ V}$

$\text{Mg}(\text{NO}_3)_2$ with Mg^{2+}/Mg $E^0 = -2.37 \text{ V}$

In an electrolytic cell with inert electrodes, reduction occurs at the cathode. The ion with the highest (most positive) reduction potential will be reduced (or deposited) first. Therefore, as we increase the applied voltage (making the cathode sufficiently negative), the metals will begin to deposit in the order of their reduction potentials.

The order of reduction (deposition) will be:

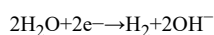
First, silver (+0.80V)

Next, mercury (+0.79V)

Then, copper (+0.24V)

So, Statement (I) which says "With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu" is correct.

For magnesium, the reduction potential is very negative (-2.37V). In aqueous solution, before reaching such a negative potential, water will be reduced. The cathodic reaction for water typically is:



which leads to the evolution of hydrogen gas, not oxygen. (Oxygen is produced at the anode during the oxidation of water.)

Hence, Statement (II) erroneously states that "oxygen gas will be evolved at the cathode" when in fact, if magnesium were to be reduced (which it won't be), water reduction would yield hydrogen gas. Thus, Statement (II) is incorrect.

Summary:

Statement (I) is correct.

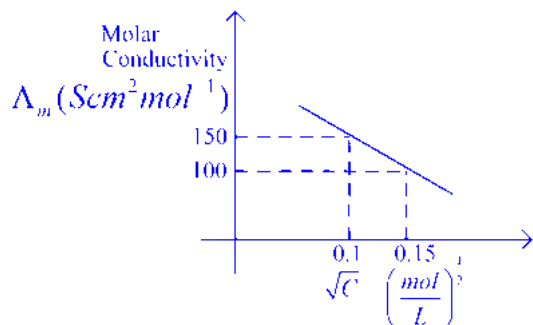
Statement (II) is incorrect.

Therefore, the most appropriate answer is:

Option C: Statement I is correct but Statement II is incorrect.

Question 38

Given below is the plot of the molar conductivity vs $\sqrt{\text{concentration}}$ for KCl in aqueous solution.



If, for the higher concentration of KCl solution, the resistance of the conductivity cell is , then the resistance of the same cell with the dilute solution is ' x '

The value of is _____ (Nearest integer)

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Answer: 150

Solution:**Resistance and Resistivity:**

$$R = \rho \frac{\ell}{A}$$

Where R is resistance, ρ is resistivity, ℓ is the length, and A is the cross-sectional area of the cell. The term $\frac{\ell}{A}$ is known as the cell constant, G^* .

Conductivity and Conductance:

$$\kappa = G \cdot G^*$$

$$G = \frac{1}{R} \quad ; \quad \kappa = \frac{1}{\rho}$$

Where G is conductance and κ is conductivity.

Molar Conductivity:

$$\lambda_m = \frac{\kappa \times 1000}{C}$$

Where λ_m is molar conductivity and C is concentration.

Given Information:

The resistance of the cell with the concentrated solution $R_c = 100 \, \Omega$.

We need to find the resistance R_d for the dilute solution.

Steps to Solution:

The conductivity ratio for concentrated (κ_c) and dilute (κ_d) solutions relates to resistance:

$$\frac{\kappa_c}{\kappa_d} = \frac{R_d}{R_c}$$

From the plot or given values (assuming values for molar conductivity):

$\lambda_{m,c}$ and $\lambda_{m,d}$ are the molar conductivities for the concentrated and dilute solutions.

If from the plot or data, $\lambda_{m,c} = 150 \text{ S cm}^2 \text{ mol}^{-1}$ and $\lambda_{m,d} = 100 \text{ S cm}^2 \text{ mol}^{-1}$.

Calculate the ratio of conductivities:

$$\frac{\kappa_c}{\kappa_d} = \frac{(\lambda_{m,c} C_c)}{(\lambda_{m,d} C_d)}$$

Substituting values into the equation:

$$\frac{100 \cdot (0.15)^2}{150 \cdot (0.1)^2} = \frac{R_d}{100}$$

Solving the equation gives:

$$R_d = 150 \Omega$$

Thus, the resistance of the conductivity cell with the dilute solution is $R_d = 150 \Omega$.

Question39

Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12 .

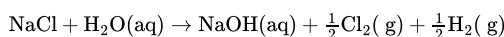
The current in Amperes used for the given electrolysis is _____ . (Nearest integer).

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Answer: 2

Solution:

Electrolysis of NaCl is



Since during electrolysis pH changes to 12

$$\text{So } [\text{OH}^\ominus] = 10^{-2} \text{ and } [\text{H}^+] = 10^{-12}$$

So by Faraday law

Gram amount of substance deposited = Amount of electricity passed

$$10^{-2} \times \frac{600}{1000} \times 96500 = I \times t$$

$$\frac{10^{-2} \times 600}{1000} \times 96500 = I \times 5 \times 60$$

$$I = \frac{10^{-2} \times 600 \times 96500}{1000 \times 5 \times 60}$$

$$I = 1.93 \text{ ampere}$$

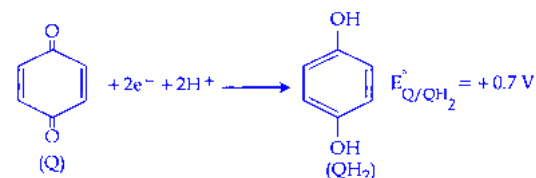
So, $I = 2$ ampere (nearest integer)

Question40

Consider the following electrochemical cell at standard condition.



The couple QH_2/Q represents quinhydrone electrode, the half cell reaction is given below:



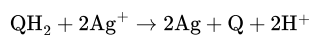
$$\left[\text{Given : } E_{\text{Ag}^+/\text{Ag}}^0 = +0.8 \text{ V and } \frac{2.303RT}{F} = 0.06 \text{ V} \right]$$

The pK_b value of the ammonium halide salt (NH_4X) used here is _____. (nearest integer)

Answer: 6

Solution:

The cell reaction is:



The Nernst equation for the reaction can be expressed as:

$$E = E^\circ - \frac{0.06}{2} \log [\text{H}^+]^2$$

Simplifying, we obtain:

$$E = E^\circ - 0.06 \log [\text{H}^+]$$

Given data includes:

$$E_{\text{cell}} = +0.4 \text{ V}$$

$$\text{Standard potential for } \text{Ag}^+/\text{Ag}, E_{\text{Ag}^+/\text{Ag}}^0 = +0.8 \text{ V}$$

Using the cell potential and the standard potential, we calculate:

$$\text{pH} = -\log (\text{H}^+) = \frac{E - E^\circ}{0.06} = \frac{0.4 - 0.8}{0.06}$$

$$\text{pH} = \frac{0.3}{0.06} = 5$$

This established pH must now relate to the buffer equation involving NH_4X , at

$$\text{pH} + \text{NH}_4\text{X} = 7 - \frac{1}{2} \text{pK}_b - \frac{1}{2} \log C$$

Given the concentration $C = 0.01 \text{ M} = 10^{-2}$, we substitute into the equation:

$$5 = 7 - \frac{1}{2} \times \text{pK}_b - \frac{1}{2} \log(10^{-2})$$

Simplifying further, the equation resolves to:

$$\text{pK}_b = 6$$

Thus, the calculated pK_b value of the ammonium halide salt NH_4X is 6.

Question41

0.2% (w/v) solution of NaOH is measured to have resistivity $870.0 \text{ m}\Omega \text{ m}$. The molar conductivity of the solution will be _____ $\times 10^2 \text{ mSdm}^2 \text{ mol}^{-1}$. (Nearest integer)

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Answer: 23

Solution:

Determine the conductivity from the resistivity.

The resistivity is given as

$$\rho = 870.0 \text{ m}\Omega \cdot \text{m}.$$

Convert this into ohm-meters:

$$870.0 \text{ m}\Omega \cdot \text{m} = 870.0 \times 10^{-3} \Omega \cdot \text{m} = 0.87 \Omega \cdot \text{m}.$$

Conductivity is the reciprocal of resistivity:

$$\kappa = \frac{1}{\rho} = \frac{1}{0.87} \approx 1.15 \text{ S/m}.$$

Find the concentration of NaOH in mol/m³.

A 0.2% (w/v) solution means there are 0.2 grams of NaOH per 100 mL of solution.

In 1 liter (1000 mL) there are:

$$\frac{0.2 \text{ g}}{100 \text{ mL}} \times 1000 \text{ mL} = 2 \text{ g/L}.$$

The molar mass of NaOH is approximately 40 g/mol. Thus, the molarity is:

$$\text{Molarity} = \frac{2 \text{ g/L}}{40 \text{ g/mol}} = 0.05 \text{ mol/L}.$$

Since 1 L = 0.001 m³, converting to SI units (mol/m³):

$$c = 0.05 \text{ mol/L} \times \frac{1}{0.001 \text{ m}^3/\text{L}} = 50 \text{ mol/m}^3.$$

Calculate the molar conductivity in SI units.

Molar conductivity Λ_m is given by:

$$\Lambda_m = \frac{\kappa}{c}.$$

Substituting the values:

$$\Lambda_m = \frac{1.15 \text{ S/m}}{50 \text{ mol/m}^3} = 0.023 \text{ S} \cdot \text{m}^2/\text{mol}.$$

Convert the molar conductivity to the desired units (mS dm² mol⁻¹).

First, convert the conductivity:

$$0.023 \text{ S} \cdot \text{m}^2/\text{mol} \times 1000 \frac{\text{mS}}{\text{S}} = 23 \text{ mS} \cdot \text{m}^2/\text{mol}.$$

Now, convert the area units. Recall that:

$$1 \text{ m}^2 = 100 \text{ dm}^2.$$

So,

$$23 \text{ mS} \cdot \text{m}^2/\text{mol} = 23 \times 100 \text{ mS} \cdot \text{dm}^2/\text{mol} = 2300 \text{ mS} \cdot \text{dm}^2/\text{mol}.$$

The problem asks for the answer in the form

$$[\text{blank}] \times 10^2 \text{ mS} \cdot \text{dm}^2/\text{mol}.$$

Expressing 2300 mS dm²/mol in this form:

$$2300 \text{ mS} \cdot \text{dm}^2/\text{mol} = 23 \times 10^2 \text{ mS} \cdot \text{dm}^2/\text{mol}.$$

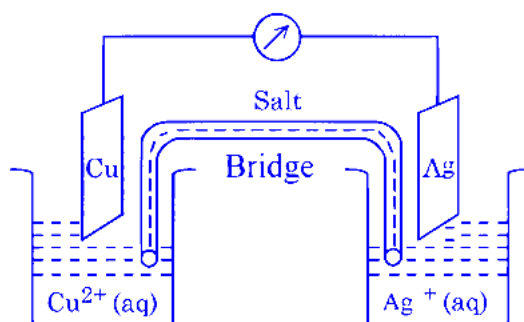
Thus, the molar conductivity is

$23 \times 10^2 \text{ mS} \cdot \text{dm}^2/\text{mol}$

.

Question42

1 Faraday electricity was passed through Cu^{2+} (1.5M, 1 L)/Cu and 0.1 Faraday was passed through Ag^+ (0.2M, 1 L)/Ag electrolytic cells. After this the two cells were connected as shown below to make an electrochemical cell. The emf of the cell thus formed at 298 K is _____ mV (nearest integer)



Given : $E^\circ \text{Cu}^{2+}/\text{Cu} = 0.34 \text{ V}$

$$E^\circ \text{Ag}^+/\text{Ag} = 0.8 \text{ V}$$

$$\frac{2 \cdot 303RT}{F} = 0.06 \text{ V}$$

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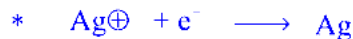
Answer: 400

Solution:

$$t = 0 \quad 1.5 \quad 1 \text{ mole}$$

$$t = t \quad 1 \quad - \quad 0.5 \text{ mole}$$

$$[\text{Cu}^{+2}] = 1 \text{ M after electrolysis}$$



$$t = 0 \quad 0.2 \quad 0.1 \text{ mole}$$

$$t = t \quad 0.1 \quad - \quad -$$

$$[\text{Ag}^{+}] = 0.1 \text{ M after electrolysis}$$



reaction

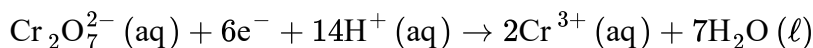
$$E = E^{\circ} - \frac{0.06}{n} \log \frac{[\text{Cu}^{+2}]}{[\text{Ag}^{+}]^2}$$

$$E = (0.8 - 0.34) - \frac{0.06}{2} \log \frac{1}{(0.1)^2} = 0.4 \text{ V}$$

Correct answer = 400mV

Question43

Consider the following half cell reaction



The reaction was conducted with the ratio of $\frac{[\text{Cr}^{3+}]^2}{[\text{Cr}_2\text{O}_7^{2-}][\text{H}^{+}]^{14}} = 10^{-6}$. The pH value at which the EMF of the half cell will become zero is _____.

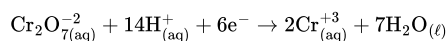
(nearest integer value)

[Given: standard half cell reduction potential $E_{\text{Cr}_2\text{O}_7^{2-}, \text{H}^{+}/\text{Cr}^{3+}}^{\circ} = 1.33 \text{ V}$, $\frac{2.303RT}{F} = 0.059 \text{ V}$.]

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Answer: 10

Solution:



$$E_{\text{R}} = E_{\text{R}}^{\circ} - \frac{0.059}{6} \log \frac{[\text{Cr}^{3+}]^2}{[\text{Cr}_2\text{O}_7^{2-}][\text{H}^{+}]^{14}}$$

$$0 = 1.33 - \frac{0.059}{6} \log \frac{10^{-6}}{[\text{H}^{+}]^{14}}$$

$$\frac{1.33 \times 6}{0.059} = \log \frac{10^{-6}}{[\text{H}^{+}]^{14}}$$

$$135.254 = -6 - 14 \log [\text{H}^{+}]$$

$$141.254 = 14 \text{pH}$$

$$\text{pH} = \frac{141.254}{14} = 10.08$$

Question44

The mass of silver (Molar mass of Ag : 108gmol^{-1}) displaced by a quantity of electricity which displaces 5600mL of O_2 at S.T.P. will be _____g.

[27-Jan-2024 Shift 1]

Answer: 108

Solution:

Eq. of Ag = Eq. of O_2

Let x gm silver displaced,

$$\frac{x \times 1}{108} = \frac{5.6}{22.7} \times 4$$

(Molar volume of gas at STP = 22.7 lit)

$$x = 106.57 \text{ gm}$$

Ans. 107

OR,

as per old STP data, molar volume = 22.4 lit

$$\frac{x \times 1}{108} = \frac{5.6}{22.4} \times 4, x = 108 \text{ gm}.$$

Ans. 108

Question45

Which of the following statements is not correct about rusting of iron?

[27-Jan-2024 Shift 2]

Options:

A.

Coating of iron surface by tin prevents rusting, even if the tin coating is peeling off.

B.

When pH lies above 9 or 10, rusting of iron does not take place.

C.

Dissolved acidic oxides SO_2 , NO_2 in water act as catalyst in the process of rusting.

D.

Rusting of iron is envisaged as setting up of electrochemical cell on the surface of iron object.

Answer: A

Solution:

As tin coating is peeled off, then iron is exposed to atmosphere.

Question46

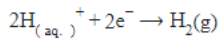
The hydrogen electrode is dipped in a solution of $\text{pH} = 3$ at 25°C . The potential of the electrode will be $\underline{\hspace{1cm}} \times 10^{-2}\text{V}$.

$$\left(\frac{2.303 RT}{F} = 0.059\text{V} \right)$$

[27-Jan-2024 Shift 2]

Answer: 18

Solution:



$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.059}{2} \log \frac{P_{\text{H}_2}}{[\text{H}^{+}]^2}$$

$$= 0 - 0.059 \times 3 = -0.177 \text{ volts.} = -17.7 \times 10^{-2}\text{V.}$$

Question47

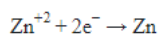
The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of 0.015 A for 15 minutes is $\underline{\hspace{1cm}} \times 10^{-4}\text{g}$.

(Atomic mass of zinc = 65.4amu)

[29-Jan-2024 Shift 1]

Answer: 45.75

Solution:



$$W = Z \times i \times t$$

$$= \frac{65.4}{2 \times 96500} \times 0.015 \times 15 \times 60$$

$$= 45.75 \times 10^{-4} \text{ gm}$$

Question48

A constant current was passed through a solution of AuCl_4^{-} ion between gold electrodes. After a period of 10.0 minutes , the increase in mass of cathode was 1.314g . The total charge passed through the solution is $\underline{\hspace{1cm}} \times 10^{-2}\text{F}$.

(Given atomic mass of $\text{Au} = 197$)

[29-Jan-2024 Shift 2]

Answer: 2

Solution:

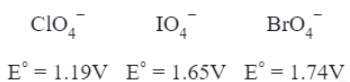
$$\frac{W}{E} = \frac{\text{charge}}{1F}$$

$$\frac{1.314}{\frac{197}{3}} = \frac{Q}{1F}$$

$$Q = 2 \times 10^{-2}F$$

Question49

Reduction potential of ions are given below:

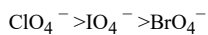


The correct order of their oxidising power is:

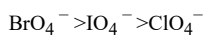
[30-Jan-2024 Shift 2]

Options:

A.



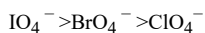
B.



C.



D.



Answer: B

Solution:

Higher the value of $\oplus$ ve SRP (Std. reduction potential) more is tendency to undergo reduction, so better is oxidising power of reactant.

Hence, ox. Power:- $\text{BrO}_4^- > \text{IO}_4^- > \text{ClO}_4^-$

Question50

Identify the factor from the following that does not affect electrolytic conductance of a solution.

[31-Jan-2024 Shift 1]

Options:

A.

The nature of the electrolyte added.

B.

The nature of the electrode used.

C.

Concentration of the electrolyte.

D.

The nature of solvent used.

Answer: B

Solution:

Conductivity of electrolytic cell is affected by concentration of electrolyte, nature of electrolyte and nature of solvent.

Question51

The metals that are employed in the battery industries are

A. Fe

B. Mn

C. Ni

D. Cr

E. Cd

Choose the correct answer from the options given below:

[31-Jan-2024 Shift 1]

Options:

A.

B, C and E only

B.

A, B, C, D and E

C.

A, B, C and D only

D.

B, D and E only

Answer: A

Solution:

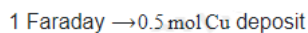
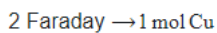
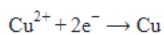
Mn, Ni and Cd metals used in battery industries.

Question52

One Faraday of electricity liberates $x \times 10^{-1}$ gram atom of copper from copper sulphate, x is _____

[31-Jan-2024 Shift 1]

Answer: 5

Solution:

$$0.5 \text{ mol} = 0.5 \text{ g atom} = 5 \times 10^{-1}$$

$$x = 5$$

Question53

The values of conductivity of some materials at $298.15\text{K}^{-1}\text{Sm}^{-1}$ are 2.1×10^3 , 1.0×10^{-16} , 1.2×10 , 3.91 , 1.5×10^{-2} , 1×10^{-7} , 1.0×10^3 . The number of conductors among the materials is _____

[31-Jan-2024 Shift 2]

Answer: 4

Solution:

Conductivity (Sm^{-1})

$$\left. \begin{array}{l} 2.1 \times 10^3 \\ 1.2 \times 10 \\ 3.91 \\ 1 \times 10^3 \end{array} \right\} \text{conductors at } 298.15\text{K}$$

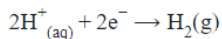
1×10^{-16} Insulator at 298.15K

$$\left. \begin{array}{l} 1.5 \times 10^{-2} \\ 1 \times 10^{-7} \end{array} \right\} \text{Semiconductor at } 298.15\text{K}$$

Therefore number of conductors is 4 .

Question54

The potential for the given half cell at 298K is $\times 10^{-2}\text{V}$.



$$[\text{H}^{+}] = 1\text{M}, P_{\text{H}_2} = 2 \text{ atm}$$

(Given: $2.303 \text{ RT}/\text{F} = 0.06\text{V}$, $\log 2 = 0.3$)

[1-Feb-2024 Shift 1]

Answer: 1

Solution:

$$E = E_{\text{H}^+/\text{H}_2}^\circ - \frac{0.06}{2} \log \frac{P_{\text{H}_2}}{[\text{H}^+]^2}$$

$$E = 0.00 - \frac{0.06}{2} \log \frac{2}{[1]^2}$$

$$E = -0.03 \times 0.3 = -0.9 \times 10^{-2} \text{V}$$

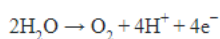
Question55

The amount of electricity in Coulomb required for the oxidation of 1mol of H_2O to O_2 is _____ $\times 10^5 \text{C}$.

[1-Feb-2024 Shift 2]

Answer: 2

Solution:



$$\frac{W}{E} = \frac{Q}{96500}$$

$$\text{mole} \times \text{n-factor} = \frac{Q}{96500}$$

$$1 \times 2 = \frac{Q}{96500}$$

$$Q = 2 \times 96500 \text{C}$$

$$= 1.93 \times 10^5 \text{C}$$

Question56

At 298K, a 1 litre solution containing 10 mmol of $\text{Cr}_2\text{O}_7^{2-}$ and 100 mmol of Cr^{3+} shows a pH of 3.0.

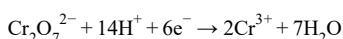
Given : $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}^{3+}$; $E^0 = 1.330 \text{V}$ and $\frac{2.303 RT}{F} = 0.059 \text{V}$

The potential for the half cell reaction is $x \times 10^{-3} \text{V}$. The value of x is _____

[24-Jan-2023 Shift 1]

Answer: 917

Solution:



$$E = 1.33 - \frac{0.059}{6} \log \frac{(0.1)^2}{(10^{-2})(10^{-3})^{14}}$$

$$E = 1.33 - \frac{0.059}{6} \times 42 = 0.917$$

$$E = 917 \times 10^{-3}$$

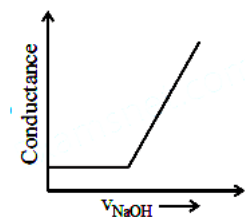
$$x = 917$$

Question57

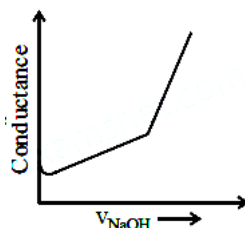
Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.
[24-Jan-2023 Shift 2]

Options:

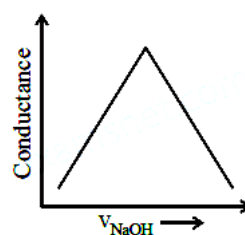
A.



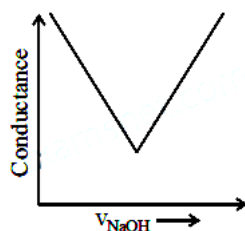
B.



C.

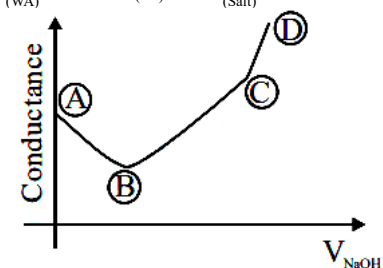
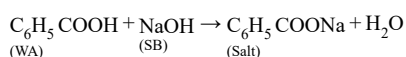


D.



Answer: B

Solution:



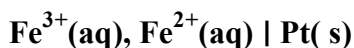
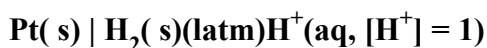
(A) → (B) Free H^+ ions are replaced by Na^+ which decreases conductance.

(B) → (C) Un-dissociated benzoic acid reacts with NaOH and forms salt which increases ions & conductance increases.

(C) → (D) After equivalence point at (3), NaOH added further increases Na^+ & OH^- ions which further increases the conductance.

Question58

Consider the cell

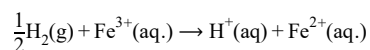


Given: $E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^\circ = 0.771\text{V}$ and $E_{\text{H}^+/\frac{1}{2}\text{H}_2}^\circ = 0\text{V}$, $T = 298\text{K}$

If the potential of the cell is 0.712V the ratio of concentration of Fe^{2+} to Fe^{3+} is _____ (Nearest integer)
[25-Jan-2023 Shift 1]

Answer: 10

Solution:



$$E = E^\circ - \frac{0.059}{1} \log \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]}$$

$$\Rightarrow 0.712 = (0.771 - 0) - \frac{0.059}{1} \log \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]}$$

$$\Rightarrow \log \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} = \frac{(0.771 - 0.712)}{0.059} = 1$$

$$\Rightarrow \frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} = 10$$

Question59



The E_{cell} for the given cell is 0.1115V at 298K when $\frac{[\text{M}^+(\text{aq})]}{[\text{M}^{3+}(\text{aq})]} = 10^a$

The value of a is _____

Given : $E^\circ_{\text{M}^{3+}/\text{M}^+} = 0.2\text{V}$

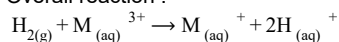
$$\frac{2.303RT}{F} = 0.059\text{V}$$

[25-Jan-2023 Shift 2]

Answer: 3

Solution:

Overall reaction :-



$$E_{\text{Cell}} = E_{\text{Cathode}}^\circ - E_{\text{anode}}^\circ - \frac{0.059}{2} \log \frac{[\text{M}^+] \times 1^2}{[\text{M}^{3+}]1}$$

$$0.1115 = 0.2 - \frac{0.059}{2} \log \frac{[\text{M}^+]}{[\text{M}^{3+}]}$$

$$3 = \log \frac{[\text{M}^+]}{[\text{M}^{3+}]}$$

$$\therefore a = 3$$

Question60

The standard electrode potential (M^{3+}/M^{2+}) for V, Cr, Mn & Co are $-0.26V$, $-0.41V$, $+1.57V$ and $+1.97V$, respectively. The metal ions which can liberate H_2 from a dilute acid are

[29-Jan-2023 Shift 1]

Options:

A. V^{2+} and Mn^{2+}

B. Cr^{2+} and Co^{2+}

C. V^{2+} and Cr^{2+}

D. Mn^{2+} and Co^{2+}

Official Ans. by NTA (3)

Answer: C

Solution:

Metal cation with $(-)$ value of reduction potential (M^{+3}/M^{+2}) or with $(+)$ value of oxidation potential (M^{+2}/M^{+3}) will liberate H_2

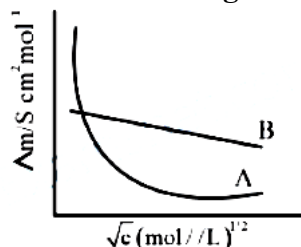
Therefore they will reduce H^+

i.e. V^{+2} and Cr^{+2}

Question61

Following figure shows dependence of molar conductance of two electrolytes on concentration.

Λ_m^0 is the limiting molar conductivity



The number of Incorrect statement(s) from the following is _____

(A) Λ_m for electrolyte A is obtained by extrapolation

(B) For electrolyte B, Λ_m vs $\sqrt{c}$ graph is a straight line with intercept equal to Λ_m^0

(C) At infinite dilution, the value of degree of dissociation approach zero for electrolyte B.

(D) Λ_m for any electrolyte A or B can be calculated using λ^0 for individual ions.

[29-Jan-2023 Shift 1]

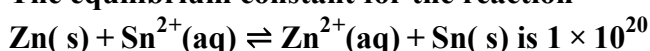
Answer: 2

Solution:

Statement (A) and Statement (C) are incorrect

Question62

The equilibrium constant for the reaction

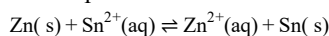


at 298K. The magnitude of standard electrode potential of Sn/Sn²⁺ if $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V}$ is _____
 $\times 10^{-2}\text{V}$. (Nearest integer)
 [29-Jan-2023 Shift 2]

Answer: 17

Solution:

$$\text{Given : } \frac{2.303 RT}{F} = 0.059\text{V}$$



$$\Delta G^\circ = -2.303 RT \log_{10} K_{\text{eq}}$$

$$-nF(E_{\text{cell}}^0) = -2.303 RT \log_{10} K_{\text{eq}}$$

$$0.76 + E_{\text{Sn}^{2+}/\text{Sn}}^0 = \frac{0.059}{2} \log_{10} 10^{20}$$

$$0.76 + E_{\text{Sn}^{2+}/\text{Sn}}^0 = \frac{0.059 \times 20}{2}$$

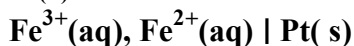
$$E_{\text{Sn}^{2+}/\text{Sn}}^0 = 0.59 - 0.76 = -0.17$$

$$E_{\text{Sn}/\text{Sn}^{2+}}^0 = 17 \times 10^{-2}\text{V}$$

$$\text{Ans.} = 17$$

Question 63

Consider the cell



When the potential of the cell is 0.712V at 298K, the ratio $[\text{Fe}^{2+}]/[\text{Fe}^{3+}]$ is _____.
 (Nearest integer)

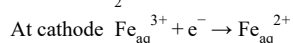
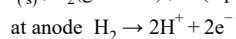
Given: $\text{Fe}^{3+} + \text{e}^- = \text{Fe}^{2+}$, $E^\circ \text{Fe}^{3+}, \text{Fe}^{2+} | \text{Pt} = 0.771$

$$\frac{2.303 RT}{F} = 0.06\text{V}$$

[30-Jan-2023 Shift 1]

Answer: 10

Solution:



$$E^\circ = E_{\text{H}_2 | \text{H}^+}^\circ + E_{\text{Fe}^{3+} | \text{Fe}^{2+}}^\circ = 0.771\text{V}$$

$$E = E^\circ - \frac{0.06}{1} \log \frac{\text{Fe}^{2+}}{\text{Fe}^{3+}}$$

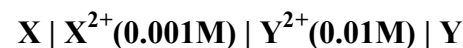
$$0.712 = (0 + 0.771) - \frac{0.06}{1} \log \frac{\text{Fe}^{2+}}{\text{Fe}^{3+}}$$

$$\log \frac{\text{Fe}^{2+}}{\text{Fe}^{3+}} = \frac{0.059}{0.06} \approx 1$$

$$\frac{\text{Fe}^{2+}}{\text{Fe}^{3+}} = 10$$

Question64

The electrode potential of the following half cell at 298K .



is _____ $\times 10^{-2}V$ (Nearest integer).

$$\text{Given: } E_{X^{2+} | X}^0 = -2.36V$$

$$E_{Y^{2+} | Y}^0 = +0.36V$$

$$\frac{2.303RT}{F} = 0.06V$$

[30-Jan-2023 Shift 2]

Answer: 275

Solution:

$$\begin{aligned} X + Y^{2+} &\rightarrow Y + X^{2+} \\ E_{\text{Cell}}^0 &= 0.36 - (-2.36) = 2.72V \\ E_{\text{Cell}} &= 2.72 - \frac{0.06}{2} \log \frac{0.001}{0.01} \\ &= 2.72 + 0.03 = 2.75V \\ &= 275 \times 10^{-2}V \end{aligned}$$

Question65

Which one of the following statements is correct for electrolysis of brine solution?

[31-Jan-2023 Shift 1]

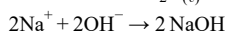
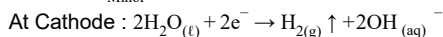
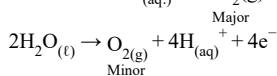
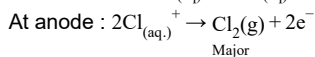
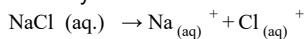
Options:

- A. Cl_2 is formed at cathode
- B. O_2 is formed at cathode
- C. H_2 is formed at anode
- D. OH^- is formed at cathode

Answer: D

Solution:

Electrolysis of brine solution



Question66

The resistivity of a 0.8M solution of an electrolyte is _____ $5 \times 10^{-3} \Omega \text{ cm}$. Its molar conductivity is $\times 10^4 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$. (Nearest integer)
[31-Jan-2023 Shift 2]

Answer: 25

Solution:

$$\Lambda_m = \frac{\kappa \times 1000}{M}$$

$$\Lambda_m = \frac{1}{\rho} \times \frac{1000}{M}$$

$$\frac{1}{5 \times 10^{-3}} \times \frac{1000}{0.8}$$

$$\text{Ans. } 25 \times 10^4 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

Question67

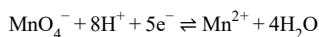
At what pH, given half cell $\text{MnO}_4^- (0.1\text{M}) \mid \text{Mn}^{2+} (0.001 \text{ M})$ will have electrode potential of 1.282 V ?
_____ (Nearest Integer)

Given $E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 = 1.54\text{V}$, $\frac{2.303RT}{F} = 0.059\text{V}$

[1-Feb-2023 Shift 1]

Answer: 3

Solution:



$$E = E^0 - \frac{0.059}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-][\text{H}^+]^8}$$

$$1.282 = 1.54 - \frac{0.059}{5} \log \frac{10^{-3}}{10^{-1} \times [\text{H}^+]^8}$$

$$\frac{0.258 \times 5}{0.059} = \log \frac{10^{-2}}{[\text{H}^+]^8}$$

$$\Rightarrow 21.86 = -2 + 8 \text{ pH}$$

$$\therefore \text{pH} = 2.98$$

$$\approx 3$$

Question68

$1 \times 10^{-5} \text{ M AgNO}_3$ is added to 1L of saturated solution of AgBr. The conductivity of this solution at 298K is _____ $\times 10^{-8} \text{ Sm}^{-1}$.

[Given : $K_{\text{sp}}(\text{AgBr}) = 4.9 \times 10^{-13}$ at 298K]

$$\lambda_{\text{Ag}^+}^0 = 6 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

$$\lambda_{\text{Br}^-}^0 = 8 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1}$$

$$\lambda_{\text{NO}_3^-}^0 = 7 \times 10^{-3} \text{ Sm}^2 \text{ mol}^{-1} \quad]$$

[1-Feb-2023 Shift 2]

Answer: 14

Solution:

$$[\text{Ag}^+] = 10^{-5}$$

$$[\text{NO}_3^-] = 10^{-5}$$

$$[\text{Br}^-] = \frac{K_{\text{sp}}}{[\text{Ag}^+]} = 4.9 \times 10^{-8}$$

$$\Lambda_m = \frac{k}{1000 \times M}$$

For Ag^+

$$6 \times 10^{-3} = \frac{K_{\text{Ag}^+}}{1000 \times 10^{-5}}$$

$$K_{\text{Ag}^+} = 6 \times 10^{-5}$$

$$\Rightarrow 6000 \times 10^{-8}$$

for Br^-

$$8 \times 10^{-3} = \frac{K_{\text{Br}^-}}{1000 \times 4.9 \times 10^{-8}}$$

$$K_{\text{Br}^-} = 39.2 \times 10^{-8}$$

for NO_3^-

$$7 \times 10^{-3} = \frac{K_{\text{NO}_3^-}}{1000 \times 10^{-5}}$$

$$K_{\text{NO}_3^-} = 7 \times 10^{-5}$$

$$= 7000 \times 10^{-8}$$

Conductivity of solution

$$\Rightarrow (6000 + 7000 + 39.2) \times 10^{-8}$$

$$\Rightarrow 13039.2 \times 10^{-8} \text{ Sm}^{-1}$$

Question 69

The standard electrode potential of M^+/M in aqueous solution does not depend on [6-Apr-2023 shift 1]

Options:

- A. Ionisation of a solid metal atom
- B. Sublimation of a solid metal
- C. Ionisation of a gaseous metal atom
- D. Hydration of a gaseous metal ion

Answer: A

Question 70

The product, which is not obtained during the electrolysis of brine solution is [6-Apr-2023 shift 2]

Options:

- A. NaOH

B. Cl_2

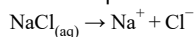
C. H_2

D. HCl

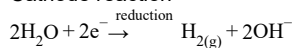
Answer: D

Solution:

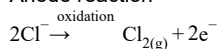
Brine is aq. Solution of NaCl



Cathode reaction



Anode reaction



So HCl will not form during electrolysis.

Question 71

The standard reduction potential at 298K for the following half cells are given below :-

$\text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO}(\text{g}) + 2\text{H}_2\text{O}$	$E^\circ = 0.97\text{V}$
$\text{V}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{V}(\text{s})$	$E^\circ = -1.19\text{V}$
$\text{Fe}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe}(\text{s})$	$E^\circ = -0.04\text{V}$
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$	$E^\circ = 0.80\text{V}$
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s})$	$E^\circ = 1.40\text{V}$

The number of metal(s) which will be oxidized by NO_3^- in aqueous solution is
[6-Apr-2023 shift 2]

Answer: 3

Solution:

Metal having lower SRP than 0.97V will be oxidised by NO_3^- .

Question 72

The reaction



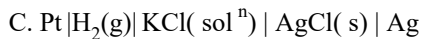
occurs in which of the given galvanic cell.

[8-Apr-2023 shift 1]

Options:

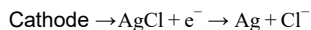
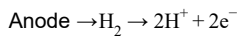
A. $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl}(\text{sol}^n) | \text{AgNO}_3(\text{sol}^n) | \text{Ag}$

B. $\text{Pt} | \text{H}_2(\text{g}) | \text{HCl}(\text{sol}^n) | \text{AgCl}(\text{s}) | \text{Ag}$



Answer: B

Solution:



Question73

The specific conductance of 0.0025M acetic acid is $5 \times 10^{-5} \text{Scm}^{-1}$ at a certain temperature. The dissociation constant of acetic acid is _____ $\times 10^{-7}$. (Nearest integer)

Consider limiting molar conductivity of CH_3COOH as $400 \text{Scm}^2 \text{mol}^{-1}$.

[10-Apr-2023 shift 2]

Answer: 66

Solution:

$$\Lambda_m = \frac{k}{C} \times 1000$$

Given $k = 5 \times 10^{-5} \text{Scm}^{-1}$

$C = 0.0025 \text{M}$

$$\Lambda_m = \frac{5 \times 10^{-5} \times 10^3}{0.0025} = \frac{5 \times 10^{-2}}{2.5 \times 10^{-3}}$$

$$= 20 \text{Scm}^2 \text{mol}^{-1}$$

$$\alpha = \frac{20}{400} = \frac{1}{20}$$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.0025 \times \frac{1}{20} \times \frac{1}{20}}{\frac{19}{20}}$$

$$= \frac{0.0025}{19 \times 20} = 6.6 \times 10^{-6}$$

$$= 66 \times 10^{-7}$$

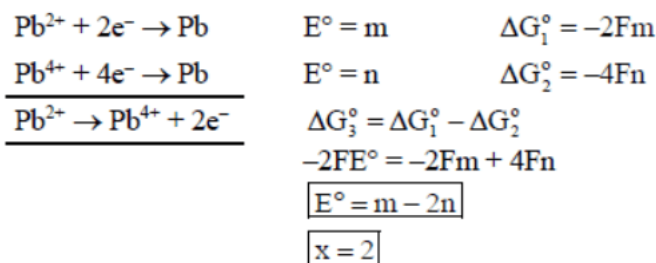
Question74

In an electrochemical reaction of lead, at standard temperature, if $E_{(\text{pb}^{2+}/\text{pb})}^\circ = m$ volt and $E_{(\text{pb}^+/\text{Pb}_b)}^\circ = n$ volt, then the value of $E_{(\text{Pb}^{2+}/\text{Pb}^+)}^\circ$ is given by $m - xn$. The value of x is _____ (Nearest integer)

[11-Apr-2023 shift 1]

Answer: 2

Solution:



Question 75

The number of correct statements from the following is _____

A. E_{cell} is an intensive parameter

B. A negative $E^\ominus$ means that the redox couple is a stronger reducing agent than the H^+/H_2 couple.

C. The amount of electricity required for oxidation or reduction depends on the stoichiometry of the electrode reaction.

D. The amount of chemical reaction which occurs at any electrode during electrolysis by a current is proportional to the quantity of electricity passed through the electrolyte.

[11-Apr-2023 shift 2]

Answer: 4

Solution:

Given statements A, B, C and D are correct.

Question 76

For lead storage battery pick the correct statements

A. During charging of battery, PbSO_4 on anode is converted into PbO_2

B. During charging of battery, PbSO_4 on cathode is converted into PbO_2

C. Lead storage battery consists of grid of lead packed with PbO_2 as anode

D. Lead storage battery has ~38% solution of sulphuric acid as an electrolyte Choose the correct answer from the options given below:

[12-Apr-2023 shift 1]

Options:

A. B, D only

B. B, C only

C. B, C, D only

D. A, B, D only

Answer: A

Solution:

Lead storage battery consists of lead anode and a grid of lead packed with lead oxide (PbO_2) as cathode, a 38% solution of H_2SO_4 is used as an electrolyte.

On charging the battery the reaction is reversed and $\text{PbSO}_4(\text{s})$ on anode and cathode is converted into Pb and PbO_2 respectively.

Question77

A metal surface of 100cm^2 area has to be coated with nickel layer of thickness 0.001 mm . A current of 2A was passed through a solution of $\text{Ni}(\text{NO}_3)_2$ for 'x' seconds to coat the desired layer. The value of x is (Nearest integer)

(ρ_{Ni} (.. density of Nickel) is 10gmL^{-1} , Molar mass of Nickel is _____ 60gmol^{-1} $F = 96500\text{Cmol}^{-1}$)
[13-Apr-2023 shift 1]

Answer: 161

Solution:

$$\begin{aligned}\text{Volume of nickel required} &= 100 \times 0.001 \times 10^{-3} \times 100 \\ &= 0.01\text{cm}^3 \\ \text{Mass of Nickel required} &= 0.01 \times 10 \\ &= 0.1\text{ gm} \\ \text{Moles} &= \frac{0.1}{60} = \frac{1}{600}\text{ mol} \\ \text{Ni}^{2+} + 2\text{e}^- &\rightarrow \text{Ni(s)} \\ \text{for coating of } 1\text{ mol Ni, charge required} &= 2 \times 96500\text{C} \\ \text{for coating of } \frac{1}{600}\text{ mol, charge required} &= 2 \times 96500 \times \frac{1}{600}\text{C} \\ &= \frac{965}{3}\text{C} \\ I &= \frac{q}{t} \\ t &= \frac{965/3}{2} = 160.83\text{ sec} \approx 161\end{aligned}$$

Question78

At 298K , the standard reduction potential for Cu^{2+}/Cu electrode is 0.034V .

Given : $K_{\text{sp}} \text{Cu}(\text{OH})_2 = 1 \times 10^{-20}$

Take $\frac{2.303RT}{F} = 0.059\text{V}$

The reduction potential at $\text{pH} = 14$ for the above couple is $(-)\text{x} \times 10^{-2}\text{V}$. The value of x is _____.
[13-Apr-2023 shift 2]

Answer: 25

Solution:

$$\begin{aligned}\text{Cu}(\text{OH})_2(\text{s}) &\rightleftharpoons \text{Cu}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \\ K_{\text{sp}} &= [\text{Cu}^{2+}][\text{OH}^-]^2 \\ \text{pH} = 14; \text{pOH} = 0; [\text{OH}^-] &= 1\text{M} \\ \therefore [\text{Cu}^{2+}] &= \frac{K_{\text{sp}}}{[\text{OH}^-]^2} = 10^{-20}\text{M} \\ \text{Cu}^{2+}(\text{aq}) + 2\text{e}^- &\rightarrow \text{Cu(s)} \\ E &= E^\circ - \frac{0.059}{2} \log_{10} \frac{1}{[\text{Cu}^{2+}]} \\ &= 0.34 - \frac{0.059}{2} \log_{10} \frac{1}{10^{-20}} \\ &= -0.25 = -25 \times 10^{-2}\end{aligned}$$

Question79

The number of correct statements from the following is _____.

- (A) Conductivity always decreases with decrease in concentration for both strong and weak electrolysis.
- (B) The number of ions per unit volume that carry current in a solution increases on dilution.
- (C) Molar conductivity increases with decrease in concentration.
- (D) The variation in molar conductivity is different for strong and weak electrolysis.
- (E) For weak electrolysis, the change in molar conductivity with dilution is due to decrease in degree of dissociation.

[15-Apr-2023 shift 1]

Answer: 3

Solution:

- (A) Conductivity decreases with dilution for strong electrolyte as well as weak electrolyte.
 - (B) On dilution, The number of ions per unit volume that carry current in a solution decreases.
 - (C) Molar conductivity increases with dilution.
 - (D) Molar conductivity of strong electrolyte follows DHO equation but it is not applicable for weak electrolyte.
 - (E) On dilution degree of dissociation of weak electrolyte increases.
- So answer is (A), (C) & (D)

Question80

The cell potential for the following cell



is 0.576V at 298K. The pH of the solution is _____ (Nearest integer)

(Given : $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34\text{V}$ and $\frac{2.303RT}{F} = 0.06\text{V}$)

[24-Jun-2022-Shift-1]

Answer: 5

Solution:

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{2} \log \frac{[\text{H}^\oplus]^2}{[\text{Cu}^{+2}]}$$

$$0.576 = 0.34 - 0.03 \log \frac{[\text{H}^\oplus]^2}{[0.01]}$$

$$0.576 - 0.34 = -0.03 \log [\text{H}^\oplus]^2 + 0.03 \log(0.01)$$

$$= 0.06 \text{pH} - 0.06$$

$$\text{pH} \approx 4.93 \approx 5$$

Question81

The resistance of a conductivity cell containing 0.01M KCl solution at 298K is 1750Ω. If the conductivity of 0.01M KCl solution at 298K is $0.152 \times 10^{-3} \text{Scm}^{-1}$, then the cell constant of the conductivity cell is _____ $\times 10^{-3} \text{cm}^{-1}$

[24-Jun-2022-Shift-2]

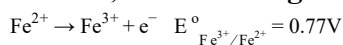
Answer: 266

Solution:

$$\begin{aligned}\text{Molarity of KCl solution} &= 0.1\text{M} \\ \text{Resistance} &= 1750\text{ ohm} \\ \text{Conductivity} &= 0.152 \times 10^{-3} \text{Scm}^{-1} \\ \text{Conductivity} &= \frac{\text{Cell constant}}{\text{Resistance}} \\ \therefore \text{Cell constant} &= 0.152 \times 10^{-3} \times 1750 \\ &= 266 \times 10^{-3} \text{cm}^{-1}\end{aligned}$$

Question82

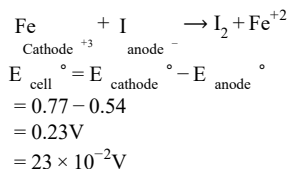
In a cell, the following reactions take place



The standard electrode potential for the spontaneous reaction in the cell is $x \times 10^{-2}\text{V}$ at 298K. The value of x is ____ (Nearest Integer)
[25-Jun-2022-Shift-1]

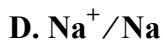
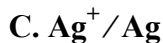
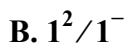
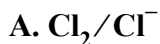
Answer: 23

Solution:



Question83

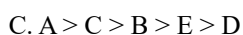
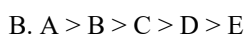
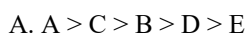
The correct order of reduction potentials of the following pairs is



Choose the correct answer from the options given below.

[25-Jun-2022-Shift-2]

Options:



D. $A > B > C > E > D$

Answer: A

Solution:

$$E_{\text{C}_2/\text{Cl}}^\circ = +1.36\text{V}$$

$$E_{\text{I}_2/\text{I}^-}^\circ = +0.54\text{V}$$

$$E_{\text{Ag}^+/\text{Ag}}^\circ = +0.80\text{V}$$

$$E_{\text{Na}^+/\text{Na}}^\circ = -2.71\text{V}$$

$$E_{\text{Li}^+/\text{Li}}^\circ = -3.05\text{V}$$

Question84

A solution of $\text{Fe}_2(\text{SO}_4)_3$ is electrolyzed for 'x' min with a current of 1.5A to deposit 0.3482g of Fe. The value of x is [nearest integer]

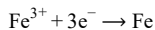
Given : $1\text{F} = 96500\text{Cmol}^{-1}$

Atomic mass of Fe = 56gmol^{-1}

[25-Jun-2022-Shift-2]

Answer: 20

Solution:



$3\text{F} \rightarrow 1 \text{ mole Fe is deposited}$

For 56g $\rightarrow 3 \times 96500$ (required charge)

For 1g $\rightarrow \frac{3 \times 96500}{56}$ (required charge)

For 0.3482g $\rightarrow \frac{3 \times 96500}{56} \times 0.3482$

$$= 1800.06$$

$Q = it$

$$1800.06 = 1.5t$$

$$t = 20 \text{ min}$$

Question85

The $\left(\frac{\partial E}{\partial T}\right)_P$ of different types of half cells are as follows:

A	B	C	D
1×10^{-4}	2×10^{-4}	0.1×10^{-4}	0.2×10^{-4}

(Where E is the electromotive force)

Which of the above half cells would be preferred to be used as reference electrode?

[26-Jun-2022-Shift-1]

Options:

A. A

B. B

C. C

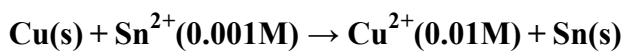
D. D

Answer: C

Solution:

A cell with less variation in EMF with temperature is preferred as a reference electrode because it can be used for a wider range of temperatures without much derivation from standard value so a cell with less $\left(\frac{\partial E}{\partial T}\right)_p$ is preferred.

Question86



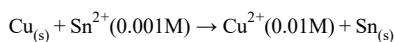
The Gibbs free energy change for the above reaction at 298K is $x \times 10^{-1} \text{ kJ mol}^{-1}$. The value of x is _____ [nearest integer]

[Given : $E_{\text{Cu}^{2+}/\text{Cu}}^{\ominus} = 0.34\text{V}$; $E_{\text{Sn}^{2+}/\text{Sn}}^{\ominus} = -0.14\text{V}$; $F = 96500 \text{ C mol}^{-1}$]

[26-Jun-2022-Shift-2]

Answer: 983

Solution:



$$\begin{aligned} E_{\text{cell}}^{\circ} &= E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} \\ &= -0.14 - (0.34) \\ &= -0.48\text{V} \end{aligned}$$

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Sn}^{2+}]} \\ &= -0.48 - \frac{0.059}{2} \log \frac{0.01}{0.001} \\ &= -0.509 \end{aligned}$$

$$\begin{aligned} \Delta G &= -nF E_{\text{cell}} \\ &= -2 \times 96500 \times (-0.5095) \\ &= 98333.5 \text{ J/mol} \\ &= 98.335 \text{ kJ/mol} \\ &= 983.35 \times 10^{-1} \text{ kJ/mol} \end{aligned}$$

Question87

In 3d series, the metal having the highest M^{2+}/M standard electrode potential is [27-Jun-2022-Shift-2]

Options:

A. Cr

B. Fe

C. Cu

D. Zn

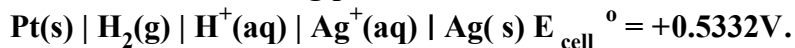
Answer: C

Solution:

$\text{Cr}^{+2}/\text{Cr} \rightarrow -0.90\text{V}$
 $\text{Fe}^{+2}/\text{Fe} \rightarrow -0.44\text{V}$
 $\text{Cu}^{+2}/\text{Cu} \rightarrow +0.34\text{V}$
 $\text{Zn}^{+2}/\text{Zn} \rightarrow -0.76\text{V}$
 So Ans. is Cu^{+2}/Cu

Question88

For the reaction taking place in the cell :

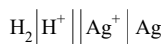


The value of $\Delta_r G^{\circ}$ is _____ kJ mol^{-1} . (in nearest integer)

[27-Jun-2022-Shift-2]

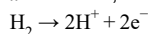
Answer: 103

Solution:

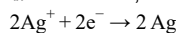


Anode Cathode

At anode, oxidation occur



\# At cathode, reduction occur



Adding equation (1) and (2), we get $n = 2$, where $n =$ cancelled out electron

Now,

$$\Delta G^{\circ} = -nF E_{\text{cell}}^{\circ}$$

$$= -2 \times 96500 \times 0.5332$$

$$= -102907.6$$

$$= -102.9 \text{ kJ/mol}$$

$$= -103 \text{ kJ/mol}$$

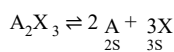
Question89

The solubility product of a sparingly soluble salt A_2X_3 is 1.1×10^{-23} . If specific conductance of the solution is $3 \times 10^{-5} \text{Sm}^{-1}$, the limiting molar conductivity of the solution is $x \times 10^{-3} \text{Sm}^2 \text{mol}^{-1}$. The value of x is _____

[28-Jun-2022-Shift-1]

Answer: 3

Solution:



$$K_{\text{sp}} = (2\text{s})^2 (3\text{s})^3 = 1.1 \times 10^{-23}$$

$$\text{S} \approx 10^{-5}$$

For sparingly soluble salts

$$\Lambda_{\text{m}} = \Lambda_{\text{m}}^{\circ}$$

$$\Lambda_{\text{m}} = \frac{k}{\text{S} \times 10^3}$$

$$= \frac{3 \times 10^{-5}}{10^{-5}} \times 10^{-3}$$

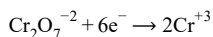
$$= 3 \times 10^{-3} \text{Sm}^2 \text{mol}^{-1}$$

Question90

The quantity of electricity in Faraday needed to reduce 1 mol of $\text{Cr}_2\text{O}_7^{2-}$ to Cr^{3+} is _____
[28-Jun-2022-Shift-1]

Answer: 6

Solution:

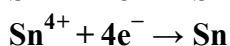
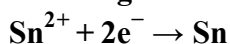


1 mol 6 mol

$\Rightarrow$ number of faradays = moles of electrons = 6

Question91

For the given reactions

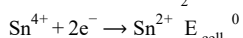
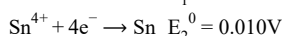
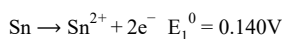


the electrode potentials are ; $E_{\text{Sn}^{2+}/\text{Sn}}^0 = -0.140\text{V}$ and $E_{\text{Sn}^{4+}/\text{Sn}}^0 = -0.010\text{V}$. The magnitude of standard electrode potential for $\text{Sn}^{4+}/\text{Sn}^{2+}$ i.e. $E_{\text{Sn}^{4+}/\text{Sn}^{2+}}^0$ is _____ $\times 10^{-2}\text{V}$. (Nearest integer)

[28-Jun-2022-Shift-2]

Answer: 16

Solution:



$$E_{\text{cell}}^0 = \frac{n_2 E_2^0 + n_1 E_1^0}{2} = \frac{4(0.010) + 2(0.140)}{2}$$

$$E_{\text{cell}}^0 = 0.16\text{V} = 16 \times 10^{-2}\text{V}$$

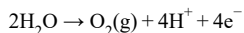
Question92

A dilute solution of sulphuric acid is electrolysed using a current of 0.10A for 2 hours to produce hydrogen and oxygen gas. The total volume of gases produced at STP is _____ cm^3 . (Nearest integer)
[Given : Faraday constant $F = 96500\text{Cmol}^{-1}$ at STP, molar volume of an ideal gas is 22.7Lmol^{-1}]
[29-Jun-2022-Shift-1]

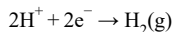
Answer: 127

Solution:

At anode



At cathode



$$\text{Now number of gm eq.} = \frac{i \times t}{96500}$$

$$= \frac{0.1 \times 2 \times 60 \times 60}{96500}$$

$$= 0.00746$$

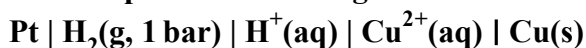
$$V_{\text{O}_2} = \frac{0.00746}{4} \times 22.7 = 0.0423$$

$$V_{\text{H}_2} = \frac{0.00746}{2} \times 22.7 = 0.0846$$

$$V_{\text{Total}} \approx 127 \text{ ml or cc}$$

Question93

The cell potential for the given cell at 298K



is 0.31V. The pH of the acidic solution is found to be 3, whereas the concentration of Cu^{2+} is 10^{-x}M .

The value of x is _____

(Given : $E_{\text{Cu}^{2+}/\text{Cu}}^\ominus = 0.34\text{V}$ and $\frac{2.303RT}{F} = 0.06\text{V}$)

[29-Jun-2022-Shift-2]

Answer: 7

Solution:

$$Q = \frac{[\text{H}^+]^2}{[\text{Cu}^{2+}]\text{pH}_2} = \frac{10^{-6}}{C} \quad \text{pH}_2 = 1$$

$$E = E_{\text{cell}}^\ominus - \frac{0.06}{n} \log Q$$

$$0.31 = 0.34 - \frac{0.06}{2} \log \frac{10^{-6}}{C}$$

$$\log \frac{10^{-6}}{C} = 1$$

$$C = 10^{-7}\text{M}$$

$$x = 7$$

Question94

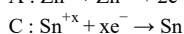
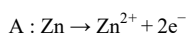
The cell potential for $\text{Zn} | \text{Zn}^{2+}(\text{aq}) || \text{Sn}^{x+} | \text{Sn}$ is 0.801V at 298K. The reaction quotient for the above reaction is 10^{-2} . The number of electrons involved in the given electrochemical cell reaction is

(Given : $E_{\text{Zn}^{2+} | \text{Zn}}^\ominus = -0.763\text{V}$, $E_{\text{Sn}^{x+} | \text{Sn}}^\ominus = +0.008\text{V}$ and $\frac{2.303RT}{F} = 0.06\text{V}$)

[25-Jul-2022-Shift-1]

Answer: 4

Solution:



$$E_{\text{Cell}}^{\circ} = E_{\text{Zn} | \text{Zn}^{2+}}^{\circ} + E_{\text{Sn}^{+2} | \text{Sn}}^{\circ}$$

$$\Rightarrow 0.763 + 0.008 = 0.771 \text{ V}$$

From the Nernst equation,

$$E_{\text{Cell}} = E_{\text{Cell}}^{\circ} - \frac{2.303 RT}{nF} \log Q$$

$$0.801 = 0.771 - \frac{0.06}{n} \log 10^{-2}$$

$$0.03 = \frac{0.06}{n} \times 2$$

$$n = 4$$

Question95

The molar conductivity of a conductivity cell filled with 10 moles of 20mLN aCl solution is Λ_{m1} and that of 20 moles another identical cell heaving 80mLN aCl solution is Λ_{m2} . The conductivities exhibited by these two cells are same. The relationship between Λ_{m2} and Λ_{m1} is
[25-Jul-2022-Shift-2]

Options:

A. $\Lambda_{m2} = 2\Lambda_{m1}$

B. $\Lambda_{m2} = \Lambda_{m1}/2$

C. $\Lambda_{m.2} = \Lambda_{m1}$

D. $\Lambda_{m2} = 4\Lambda_{m1}$

Answer: A

Solution:

$$\Lambda_{m1} = \frac{k_1 \times 1000}{M_1} = \frac{k \times 1000}{\frac{10}{0.02}}$$

$$\Lambda_{m2} = \frac{k_2 \times 1000}{\frac{20}{0.08}}$$

It is given that $k_1 = k_2$

$$k_1 = \frac{\Lambda_{m1}}{2} \quad k_2 = \frac{\Lambda_{m2}}{4}$$

Applying the given condition on conductivity.

$$\frac{\Lambda_{m1}}{2} = \frac{\Lambda_{m2}}{4}$$

$$\Lambda_{m2} = 2\Lambda_{m1}$$

Question96

The spin-only magnetic moment value of M^{3+} ion (in gaseous state) from the pairs $\text{Cr}^{3+}/\text{Cr}^{2+}$, $\text{Mn}^{3+}/\text{Mn}^{2+}$, $\text{Fe}^{3+}/\text{Fe}^{2+}$ and $\text{Co}^{3+}/\text{Co}^{2+}$ that has negative standard electrode potential, is _____ B.M.
[Nearest integer]
[25-Jul-2022-Shift-2]

Answer: 4

Solution:

Among the pairs given, $\text{Cr}^{3+}/\text{Cr}^{2+}$ has negative reduction potential which is -0.41V .

$\text{Cr(III)} \Rightarrow d^3$

Number of unpaired electrons = 3

$\mu = \sqrt{3(3+2)} = \sqrt{15} \approx 4 \text{ B.M.}$

Question97

The amount of charge in F (Faraday) required to obtain one mole of iron from Fe_3O_4 is _____.

(Nearest Integer)

[26-Jul-2022-Shift-1]

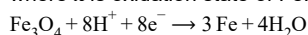
Answer: 3

Solution:

For Fe_3O_4 ,

$$x = \frac{+8}{3}$$

where x is oxidation state of Fe.



$$\text{Charge required} = \frac{8}{3} \times F = \frac{8F}{3} \approx 3F$$

Question98

Given below are two statements :

Statement I : For KI, molar conductivity increases steeply with dilution

Statement II : For carbonic acid, molar conductivity increases slowly with dilution

In the light of the above statements, choose the correct answer from the options given below:

[27-Jul-2022-Shift-2]

Options:

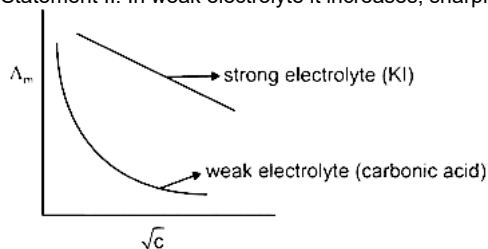
- A. Both Statement I and Statement II are true
- B. Both Statement I and Statement II are false
- C. Statement I is true but Statement II is false
- D. Statement I is false but Statement II is true

Answer: B

Solution:

Statement I: KI is strong electrolyte thus almost constant on dilution.

Statement II: In weak electrolyte it increases, sharply.



Question99

Match List - I with List - II.

List - I	List - II
(A) $\text{Cd}(s) + 2 \text{Ni}(\text{OH})_3(s) \rightarrow \text{CdO}(s) + 2 \text{Ni}(\text{OH})_2(s) + \text{H}_2\text{O}(l)$	(I) Primary battery
(B) $\text{Zn}(Hg) + \text{HgO}(s) \rightarrow \text{ZnO}(s) + \text{Hg}(l)$	(II) Discharging of secondary battery
(C) $2\text{PbSO}_4(s) + 2\text{H}_2\text{O}(l) \rightarrow \text{Pb}(s) + \text{PbO}_2(s) + 2\text{H}_2\text{SO}_4(aq)$	(III) Fuel cell
(D) $2\text{H}_2(g) + \text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(l)$	(IV) Charging of secondary battery

Choose the correct answer from the options given below:
[28-Jul-2022-Shift-1]

Options:

- A. (A) – (I), (B) – (II), (C) – (II), (D) – (IV)
 B. (A) – (IV), (B) – (I), (C) – (I), (D) – (II)
 C. (A) – (II), (B) – (I), (C) – (IV), (D) – (III)
 D. (A) – (II), (B) – (I), (C) – (II), (D) – (IV)

Answer: C

Solution:

- (a) $\text{Cd}(s) + 2 \text{Ni}(\text{OH})_3(s) \rightarrow \text{CdO}(s) + 2 \text{Ni}(\text{OH})_2(s) + \text{H}_2\text{O}(l)$ Discharge of secondary Battery
 (b) $\text{Zn}(Hg) + \text{HgO}(s) \rightarrow \text{ZnO}(s) + \text{Hg}(l)$ (Primary Battery Mercury cell)
 (c) $2\text{PbSO}_4(s) + 2\text{H}_2\text{O}(l) \rightarrow \text{Pb}(s) + \text{PbO}_2(s) + 2\text{H}_2\text{SO}_4(aq)$ (Charging of secondary Battery)
 (d) $2\text{H}_2(g) + \text{O}_2(g) \rightarrow 2\text{H}_2\text{O}(l)$ (Fuel cell)

Question100

Resistance of a conductivity cell (cell constant 129m^{-1}) filled with 74.5 ppm solution of KCl is 100Ω (labelled as solution 1). When the same cell is filled with KCl solution of 149 ppm, the resistance is 50Ω (labelled as solution 2). The ratio of molar conductivity of solution 1 and solution 2 is i.e. $\frac{\Lambda_1}{\Lambda_2} = x \times 10^{-3}$.

The value of x is _____.(Nearest integer)

Given, molar mass of KCl is 74.5gmol^{-1} .

[29-Jul-2022-Shift-1]

Answer: 1000

Solution:

$$\frac{l}{A} = 129\text{m}^{-1}$$

KCl solution 1 $\Rightarrow$ 74.5 ppm, $R_1 = 100\Omega$

KCl solution 2 $\Rightarrow$ 149 ppm, $R_2 = 50\Omega$

$$\text{Here, } \frac{\text{ppm}_1}{\text{ppm}_2} = \frac{M_1}{M_2} = \left(\frac{w_1/M_0 \times V}{V \times w_2/M_0} \right)$$

$$\frac{\Lambda_1}{\Lambda_2} = \frac{k_1 \times \frac{1000}{M_1}}{k_2 \times \frac{1000}{M_2}}$$

$$= \frac{k_1 \times M_2}{k_2 \times M_1}$$

$$= \frac{50}{100} \times 2$$

$$= \frac{\Lambda_1}{\Lambda_2} = 1000 \times 10^{-3}$$

$$= 1000$$

Question101

For a cell, $\text{Cu(s)} | \text{Cu}^{2+}(0.001\text{M}) | \text{Ag}^+(0.01\text{M}) | \text{Ag(s)}$ the cell potential is found to be 0.43V at 298K. The magnitude of standard electrode potential for Cu^{2+}/Cu is $\text{_____} \times 10^{-2}\text{V}$

[Given : $E_{\text{Ag}^+/\text{Ag}}^\ominus = 0.80\text{V}$ and $\frac{2.303RT}{F} = 0.06\text{V}$]
 [29-Jul-2022-Shift-2]

Answer: 34

Solution:

Anode : $\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$

Cathode: $[\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag(s)}] \times 2$

$\text{Cu(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag(s)}$

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2}$$

$$0.43 = E_{\text{cell}}^0 - \frac{0.06}{2} \log \left(\frac{10^{-3}}{(10^{-2})^2} \right)$$

$$0.43 = E_{\text{cell}}^0 - 0.03 \log 10$$

$$E_{\text{cell}}^0 = 0.46\text{V}$$

$$E_{\text{cell}}^0 = E_{\text{Ag}^+/\text{Ag}}^0 - E_{\text{Cu}^{2+}/\text{Cu}}^0$$

$$E_{\text{Cu}^{2+}/\text{Cu}}^0 = (0.80 - 0.46) = 0.34\text{V} = 34 \times 10^{-2}$$

Question102

Emf of the following cell at 298K in V is $x \times 10^{-2}$, $\text{Zn} | \text{Zn}^{2+}(0.1\text{M}) | \text{Ag}^+(0.01\text{M}) | \text{Ag}$ The value of x is

(Rounded off to the nearest integer).

[Given, $E_{\text{Zn}^{2+}/\text{Zn}}^\ominus = -0.76\text{V}$, $E_{\text{Ag}^+/\text{Ag}}^\ominus = +0.80\text{V}$, $\frac{2.303RT}{F} = 0.059$]

[26 Feb 2021 Shift 2]

Answer: 147

Solution:

$\text{Zn} | \text{Zn}^{2+}(0.1\text{M}) | \text{Ag}^+(0.01\text{M}) | \text{Ag}$

Cell reaction : $\text{Zn(s)} + 2\text{Ag}^+ \rightleftharpoons \text{Zn}^{2+} + 2\text{Ag(s)}$

$$\Rightarrow Q = \frac{[\text{Zn}^{2+}][\text{Ag}]^2}{[\text{Zn}][\text{Ag}^+]^2} = \frac{0.1\text{M}}{1 \times (0.01)^2} = 10^3$$

$$\begin{aligned}
 n = 2 \text{ and } E_{\text{cell}}^{\circ} &= E_{\text{Ag}^+/\text{Ag}}^{\circ} - E_{\text{Zn}^{2+}/\text{Zn}}^{\circ} \\
 &= 0.8 - (-0.76) = 1.56\text{V} \\
 \Rightarrow E_{\text{cell}} &= E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q = 1.56 - \frac{0.059}{2} \log 10^3 \\
 &= 1.4715\text{V} \\
 &= 147.15 \times 10^{-2}\text{V} = 147 \times 10^{-2}\text{V} \\
 &= x \times 10^{-2}\text{V} \\
 x &= 147
 \end{aligned}$$

Question 103

Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution. (Assuming fixed $[\text{Cu}^{2+}]$ and $p_{\text{NO}} = p_{\text{NO}_2}$), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is 10^xM . The value of $2x$ is
 .(Rounded off to the nearest integer)

[Given, $E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = 0.34\text{V}$, $E_{\text{NO}_3^-/\text{NO}}^{\circ} = 0.96\text{V}$

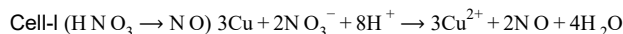
$E_{\text{NO}_3^-/\text{NO}_2}^{\circ} = 0.79\text{V}$ and at 298K , $\frac{RT}{F}$

(2.303) = 0.059]

[25 Feb 2021 Shift 2]

Answer: 4

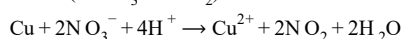
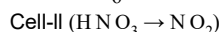
Solution:



$$Q_1 = \frac{[\text{Cu}^{2+}]^3 \times (p_{\text{NO}})^2}{[\text{NO}_3^-]^2 \times [\text{H}^+]^8}$$

$$\therefore E_1^{\circ} = 0.96 - (-0.34) = 1.3\text{V}$$

$$E_1 = 1.3 - \frac{0.059}{6} \log Q_1$$



$$Q_2 = \frac{[\text{Cu}^{2+}] \times (p_{\text{NO}_2})^2}{[\text{NO}_3^-]^2 \times [\text{H}^+]^4}$$

$$E_2^{\circ} = 0.79 - (-0.34)\text{V} = 1.13\text{V}$$

$$E_2 = 1.13 - \frac{0.059}{2} \log Q_2$$

$$\therefore E_2$$

Now, $E_1 = E_2$

$$1.3 - \frac{0.059}{6} \log Q_1 = 1.13 - \frac{0.059}{2} \log Q_2$$

$$0.17 = \frac{0.059}{6} [\log Q_1 - 3 \log Q_2] = \frac{0.059}{6} \log \frac{Q_1}{Q_2}$$

$$= \frac{0.059}{6} \log \frac{[\text{Cu}^{2+}]^3 \times (p_{\text{NO}})^2}{[\text{NO}_3^-]^2 \times [\text{H}^+]^8} \times \frac{[\text{NO}_3^-]^6 \times [\text{H}^+]^{12}}{[\text{Cu}^{2+}]^3 \times (p_{\text{NO}_2})^6} = \frac{0.059}{6} \log \frac{[\text{H}^+]^4 \times [\text{NO}_3^-]^4}{(p_{\text{NO}_2})^4} \quad [\because p_{\text{NO}} = p_{\text{NO}_2}]$$

$$= \frac{0.059}{6} \log \frac{[\text{HNO}_3]^4}{(p_{\text{NO}_2})^4}$$

Now, $p_{\text{NO}_2} = [\text{HNO}_3]$

$$\text{So, } 0.17 = \frac{0.059}{6} \log [\text{HNO}_3]^8$$

$$= \frac{0.059}{6} \times 8 \log [\text{HNO}_3]$$

$$\log [\text{HNO}_3] = 2.16$$

$$[\text{HNO}_3] = 10^{2.16} \text{M} = 10^x \text{M}$$

$$\therefore x = 2.16$$

$$\Rightarrow 2x = 2 \times 2.16 = 4.32 \sim 4$$

Question 104

The magnitude of the change in oxidising power of the $\text{MnO}_4^-/\text{Mn}^{2+}$ couple is $x \times 10^{-4} \text{ V}$, if the H^+ concentration is decreased from 1M to 10^{-4}M at 25°C . (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H^+ concentration). The value of x is (Rounded off to the nearest integer). [

Given, $\frac{2.303RT}{F} = 0.059$]

[24 Feb 2021 Shift 2]

Answer: 3776

Solution:

Reaction, $\text{MnO}_4^- + \text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$

Applying Nernst equation, $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log \frac{[\text{P}]}{[\text{R}]}$

or $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-]} \left[\frac{1}{[\text{H}^+]} \right]^8$

(I) Given, $[\text{H}^+] = 1\text{M}$

$E_1 = E^\circ - \frac{0.0591}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-]}$

(II) Now, $[\text{H}^+] = 10^{-4}\text{M}$

$E_2 = E^\circ - \frac{0.0591}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-]} \times \frac{1}{(10^{-4})^8}$

$\therefore |E_1 - E_2|$

$|E_1 - E_2| = \frac{0.0591}{5} \times 32 = 0.3776\text{V} = 3776 \times 10^{-4}$

$x = 3776$

Question 105

Compound A used as a strong oxidising agent is amphoteric in nature. It is the part of lead storage batteries. Compound A is
[26 Feb 2021 Shift 1]

Options:

A. PbO_2

B. PbO

C. PbSO_4

D. Pb_3O_4

Answer: A

Solution:

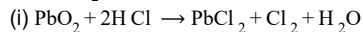
In the set of four lead compounds Pb (II) compounds are PbO and PbSO_4 . PbO_2 is a Pb (IV) compound whereas Pb_3O_4 is a mixed oxide of Pb (II) and Pb (IV) i.e. $2\text{PbO} \cdot \text{PbO}_2$.

Pb is a member of group 14 and it shows +2 and +4 oxidation states. But due to inert pair effect, Pb^{2+} is more stable than Pb^{4+} . So, Pb (IV) compounds are strong oxidising agents as Pb^{4+} gets easily reduced to more stable Pb^{2+} .

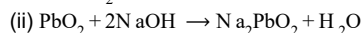
$\text{Pb}^{4+} + 2\text{e}^- \rightarrow \text{Pb}^{2+}$, $\Delta C^\circ < 0$ (spontaneous)

So, PbO_2 or Pb_3O_4 can be the compound A. But out of these two compounds only PbO_2 is used in lead storage batteries where a grid of lead packed

with PbO_2 acts as cathode and also it is amphoteric in nature. It reacts with both acids and alkali.



Here, PbO_2 acts as a basic oxide as well as an oxidising agent.



Here, PbO_2 acts as an acidic oxide.

So, the compound A is PbO_2 (option-a).

Question106

Consider the following reaction,



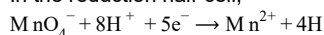
The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is

[26 Feb 2021 Shift 1]

Answer: 25

Solution:

In the reduction half-cell,



1mol 5F

5mol 25F

As one mole of MnO_4^- required 5F of charge,

5 moles of MnO_4^- will require charge,

$$Q = 5 \times 5 \text{ Faraday} = 25 \text{ Faraday}$$

Question107

A 5.0mmol dm^{-3} aqueous solution of KCl has a conductance of 0.55mS when measured in a cell constant 1.3cm^{-1} .

The molar conductivity of this solution is $\text{mSm}^2\text{mol}^{-1}$.

(Round off to the nearest integer)

[16 Mar 2021 Shift 2]

Answer: 14

Solution:

$$\Lambda_m = \frac{K}{1000 \times C}$$

where, Λ_m = molar conductivity κ = conductivity, C = molarity

$$\text{Conductivity } (\kappa) = G \frac{l}{A} = 0.55 \times (1.3 \times 100)$$

$$= 55 \times 1.3$$

$$\Lambda_m = \frac{55 \times 1.3}{1000 \times 5 \times 10^{-3}}$$

$$\Lambda_m = 14.3$$

Closest integer, $\Lambda_m = 14$

Question108

The molar conductivities at infinite dilution of barium chloride, sulphuric acid and hydrochloric acid are $280, 860$ and $426 \text{ Scm}^2 \text{ mol}^{-1}$ respectively. The molar conductivity at infinite dilution of barium sulphate is $\text{Scm}^2 \text{ mol}^{-1}$ (Round off to the nearest Integer).
[18 Mar 2021 Shift 2]

Answer: 288

Solution:

Molar conductivity of $\text{BaCl}_2 = 280 \text{ Scm}^2 \text{ mol}^{-1}$

Molar conductivity of $\text{H}_2\text{SO}_4 = 860 \text{ Scm}^2 \text{ mol}^{-1}$

Molar conductivity of $\text{HCl} = 426 \text{ Scm}^2 \text{ mol}^{-1}$

Molar conductivity of $\text{BaSO}_4 = ?$

From Kohlrausch's law,

$$\Lambda_m^\infty(\text{BaSO}_4) = \lambda_m^\infty(\text{Ba}^{2+}) + \lambda_m^\infty(\text{SO}_4^{2-})$$

$$\Lambda_m^\infty(\text{BaSO}_4) = \Lambda_m^\infty(\text{BaCl}_2) + \Lambda_m^\infty(\text{H}_2\text{SO}_4) - 2\Lambda_m^\infty(\text{HCl})$$

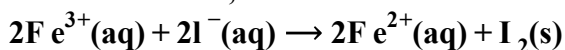
On putting given values we get,

$$= 280 + 860 - 2(426)$$

$$= 288 \text{ Scm}^2 \text{ mol}^{-1}$$

Question 109

For the reaction,



The magnitude of the standard molar Gibbs free energy change,

$$\Delta_r G_m^\circ = -\text{..... kJ}$$

(Round off to the nearest integer).

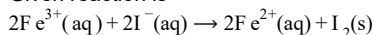
$$\left[\begin{array}{l} E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^\circ = -0.440\text{V}; \quad E_{\text{Fe}^{3+}/\text{Fe}}^\circ = -0.036\text{V} \\ E_{\text{I}_2/2\text{I}^{-}}^\circ = 0.539\text{V}; \quad F = 96500\text{C} \end{array} \right]$$

[18 Mar 2021 Shift 1]

Answer: 45

Solution:

Given reaction is



$$\text{Now, } \text{Fe}^{3+} \xrightarrow{E_1^\circ} \text{Fe}^{2+} \xrightarrow{E_2^\circ} \text{Fe}$$

$$\overbrace{nE_1^\circ + nE_2^\circ = nE_3^\circ}^{E_3^\circ} \quad [n = \text{number of electron transferred}]$$

$$E_1^\circ + 2E_2^\circ = 3E_3^\circ$$

$$E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^\circ + 2E_{\text{Fe}^{2+}/\text{Fe}}^\circ = 3E_{\text{Fe}^{3+}/\text{Fe}}^\circ$$

$$E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^\circ + (-0.440) \times 2 = (-0.036) \times 3$$

$$E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^\circ = 0.772\text{V}$$

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

$$= 0.772 - 0.539 = 0.233\text{V}$$

Using standard Gibbs free energy,

$$\Delta G^\circ = nF E_{\text{cell}}^\circ = +2 \times 96500 \times 0.233$$

$$\Delta G^\circ = 44969\text{J} = 44.9\text{kJ} = 45\text{kJ}$$

Question110

Potassium chlorate is prepared by electrolysis of KCl in basic solution as shown by following equation.



A current of x A has to be passed for 10h to produce 10.0g of potassium chlorate. the value of x is _____. (Nearest integer)

(Molar mass of KClO₃ = 122.6g mol⁻¹, F = 96500C)

[20 Jul 2021 Shift 2]

Answer: 1

Solution:

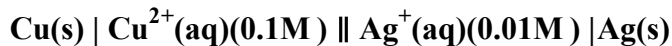
$$W = \frac{E}{F} \times I \times t$$

$$10 = \frac{122.6}{96500 \times 6} \times x \times 10 \times 3600$$

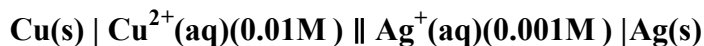
$$x = 1.311$$

Question111

For the cell



the cell potential E₁ = 0.3095V For the cell



the cell potential = ____10⁻²V . (Round off the Nearest Integer).

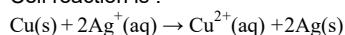
$$\left[\text{Use : } \frac{2.303RT}{F} = 0.059 \right]$$

[27 Jul 2021 Shift 2]

Answer: 28

Solution:

Cell reaction is :



$$\text{Now, } E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2} \dots\dots(1)$$

$$\therefore E_1 = 0.3095 = E_{\text{cell}}^\circ - \frac{0.059}{2} \log \frac{0.01}{(0.001)^2}$$

$$\text{From (1) and (2), } E_2 = 0.28\text{V} = 28 \times 10^{-2}\text{V}$$

Question112

Consider the cell at 25°C



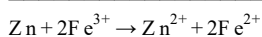
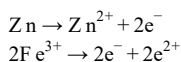
The fraction of total iron present as Fe^{3+} ion at the cell potential of 1.500V is $x \times 10^{-2}$. The value of x is _____ . (Nearest integer)

(Given : $E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.77\text{V}$, $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V}$)

[25 Jul 2021 Shift 1]

Answer: 24

Solution:



$$\begin{aligned}E^\circ_{\text{cell}} &= 0.77 - (-0.76) \\ &= 1.53\text{V}\end{aligned}$$

$$1.50 = 1.53 - \frac{0.06}{2} \log \left(\frac{[\text{Fe}^{2+}]^2}{[\text{Fe}^{3+}]^2} \right)$$

$$\log \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right) = \frac{0.03}{0.06} = \frac{1}{2}$$

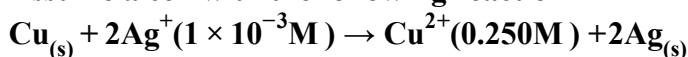
$$\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} = 10^{1/2} = \sqrt{10}$$

$$\frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]} = \frac{1}{\sqrt{10}}$$

$$\begin{aligned}\frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}] + [\text{Fe}^{3+}]} &= \frac{1}{1 + \sqrt{10}} = \frac{1}{4.16} \\ &= 0.2402 \\ &= 24 \times 10^{-2}\end{aligned}$$

Question 113

Assume a cell with the following reaction



$$E^\circ_{\text{cell}} = 2.97\text{V}$$

E_{cell} for the above reaction is _____ V.

(Nearest integer)

[Given : $\log 2.5 = 0.3979$, $T = 298\text{K}$]

[22 Jul 2021 Shift 2]

Answer: 3

Solution:

$$\begin{aligned}E &= E^\circ - \frac{0.059}{2} \log \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2} \\ &= 2.97 - \frac{0.059}{2} \log \frac{0.25}{(10^{-3})^2} = 2.81\text{V}\end{aligned}$$

Question114

The conductivity of a weak acid HA of concentration 0.001 mol L^{-1} is $2.0 \times 10^{-5} \text{ Scm}^{-1}$.

If $\Lambda_m^\circ(\text{HA}) = 190 \text{ Scm}^2 \text{ mol}^{-1}$, the ionization constant (K_a) of HA is equal to _____ $\times 10^{-6}$.

(Round off to the Nearest Integer)

[27 Jul 2021 Shift 1]

Answer: 12

Solution:

$$\begin{aligned}\Lambda_m &= 1000 \times \frac{K}{M} \\ &= 1000 \times \frac{2 \times 10^{-5}}{0.001} = 20 \text{ Scm}^2 \text{ mol}^{-1} \\ \Rightarrow \alpha &= \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{20}{190} = \left(\frac{2}{19}\right) \\ \text{HA} &\rightleftharpoons \text{H}^+ + \text{A}^- \\ 0.001(1-\alpha) & \quad 0.001\alpha \quad 0.001\alpha \\ \Rightarrow K_a &= 0.001 \left(\frac{\alpha^2}{1-\alpha} \right) = \frac{0.001 \times \left(\frac{2}{19}\right)^2}{1 - \left(\frac{2}{19}\right)} \\ &= 12.3 \times 10^{-6}\end{aligned}$$

Question115

Given below are two statements.

Statement I The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3COOH (weak electrolyte).

Statement II Molar conductivity decreases with decrease in concentration of electrolyte. In the light of the above statements, choose the most appropriate answer from the options given below

[26 Aug 2021 Shift 1]

Options:

- A. Statement I is true but statement II is false.
- B. Statement I is false but statement II is true.
- C. Both statement I and statement II are true.
- D. Both statement I and statement II are false.

Answer: D

Solution:

Limiting molar conductivity of KCl

$$\begin{aligned}\Lambda_m^\circ \text{KCl} &= \Lambda_m^\circ(\text{K}^+) + \Lambda_m^\circ(\text{Cl}^-) \\ &= 73.55 \text{ S cm}^2/\text{mol} + 76.3 \text{ S cm}^2/\text{mol} \\ &= 149.3 \text{ S cm}^2/\text{mol}\end{aligned}$$

Limiting molar conductivity of CH_3COOH

$$\begin{aligned}\Lambda_m^\circ(\text{CH}_3\text{COOH}) &= \Lambda_m^\circ(\text{CH}_3\text{COO}^-) + \Lambda_m^\circ(\text{H}^+) \\ &= 349.8 \text{ S cm}^2/\text{mol} + 40.9 \text{ S cm}^2/\text{mol} = 390.7 \text{ S cm}^2/\text{mol} \\ \therefore \text{Limiting molar conductivity of } \text{CH}_3\text{COOH} &\text{ is more than KCl.}\end{aligned}$$

Hence, statement I is false.

Also, as the concentration decreases, dilution of the electrolyte increases which will increase the degree of dissociation of weak electrolyte. Thus, the number of ions in the solution increases and hence, the molar conductance of electrolyte increases. Therefore, we can see that with decrease in

concentration of electrolyte, molar conductance increases.
Hence, statement II is also false.

Question116

These are physical properties of an element.

A. Sublimation enthalpy

B. Ionisation enthalpy

C. Hydration enthalpy

D. Electron gain enthalpy

The total number of above properties that affect the reduction potential is..... (Integer answer)

[26 Aug 2021 Shift 1]

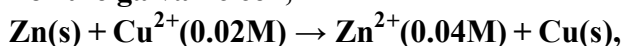
Answer: 3

Solution:

The physical properties of an element is affected by sublimation enthalpy, ionisation enthalpy and hydration enthalpy as these 3 enthalpies will affect the reduction potential.

Question117

For the galvanic cell,



$$E_{\text{cell}} = \dots \times 10^{-2} \text{V. (Nearest integer)}$$

[Use $E^\circ_{\text{Cu}/\text{Cu}^{2+}} = -0.34\text{V}$, $E^\circ_{\text{Zn}/\text{Zn}^{2+}} = +0.76\text{V}$,

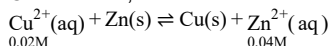
$$\frac{2.303RT}{F} = 0.059\text{V}]$$

[26 Aug 2021 Shift 2]

Answer: 109

Solution:

Cell reaction,



$$E^\circ_{\text{cell}} = E^\circ_{\text{Cu}^{2+}/\text{Cu}} - E^\circ_{\text{Zn}^{2+}/\text{Zn}}$$

$$= 0.34 - (-0.75) = 1.1\text{V}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

$$= 1.1 - \frac{0.059}{2} \log \frac{[0.04]}{[0.02]}$$

$$= 1.1 - 0.03 \log 2 = 1.1 - 0.03 \times 0.30$$

$$= 1.09\text{V} = 109 \times 10^{-2}\text{V}$$

$\therefore$ Answer is 109.

Question118

The resistance of a conductivity cell with cell constant 1.14cm^{-1} , containing 0.001 M KCl at 298K is 1500Ω . The molar conductivity of 0.001 M KCl solution at 298K in $\text{Scm}^2\text{mol}^{-1}$ is (Integer answer)

[27 Aug 2021 Shift 2]

Answer: 760

Solution:

Given, $C^* = 1.14 \text{ cm}^{-1} = 1.14 \times 10^3 \text{ m}^{-1}$

$R = 1500 \, \Omega$

$C = 0.001 \text{ M}$

$G^* = R \times \kappa$

[where, R = resistance

G^* = cell constant

κ = conductivity

C = concentration]

$$\therefore \kappa = \frac{G^*}{R} = \frac{1.14 \times 10^3 \text{ m}^{-1}}{1500}$$

$$\Lambda_m = \frac{\kappa}{C}$$

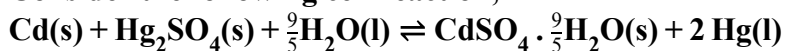
[where, Λ_m = molar conductivity]

$$= \frac{1.14 \times 10^3}{1500 \times 0.001} = 760 \text{ S cm}^2 \text{ mol}^{-1}.$$

Question 119

Section B : Numerical Type Questions

Consider the following cell reaction,



The value of E_{cell}° is 4.315 V at 25°C . If $\Delta H^\circ = -825.2 \text{ kJ mol}^{-1}$, the standard entropy change ΔS° in JK^{-1} is.... (Nearest integer)

[Given, Faraday constant = 96487 C mol^{-1}]

[31 Aug 2021 Shift 1]

Answer: 25

Solution:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \dots (i)$$

where, ΔG° = standard Gibb's free energy change

ΔH° = standard change in enthalpy

ΔS° = standard entropy change

Also, $\Delta G^\circ = -nFE^\circ$

where, n = number of electrons

F = Faraday constant

E° = standard cell potential

$\therefore$ Equation (i) is,

$$-nFE^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$-nFE^\circ - \Delta H^\circ = T\Delta S^\circ$$

$$\Rightarrow \Delta S^\circ = \frac{nFE^\circ + \Delta H^\circ}{T} \quad (2 \times 96487 \text{ C mol}^{-1} \times 4.315 \text{ V})$$

$$= \frac{+(-825.2 \text{ kJ/mol})}{298 \text{ K}}$$

$$= \frac{832.682 \times 10^3 \text{ J/mol} - 825.2 \times 10^3 \text{ J/mol}}{298 \text{ K}}$$

$$= 25.11 \text{ J/Kmol}$$

Question 120

Match List-I with List-II.

List-I (Parameter)	List-II (Unit)
A. Cell constant	1. $\text{Scm}^2\text{mol}^{-1}$
B. Molar conductivity	2. Dimensionless
C. Conductivity	3. m^{-1}
D. Degree of dissociation of electrolyte	4. $\Omega^{-1}\text{m}^{-1}$

Choose the most appropriate answer from the options given below
[31 Aug 2021 Shift 2]

Options:

A. A-3 B-1 C-4 D-2

B. A-3 B-1 C-2 D-4

C. A-1 B-4 C-3 D-2

D. A-2 B-1 C-3 D-4

Answer: A

Solution:

(A) Cell constant $= \frac{l}{A} \text{m}^{-1}$

(B) Molar conductivity $(\lambda_m) = \frac{\kappa \times 1000}{\text{molarity}} \text{Scm}^2\text{mol}^{-1}$

(C) Conductivity $(\kappa) = \frac{1}{\rho} = \frac{1}{RA} \Omega^{-1}\text{m}^{-1}$

(D) Degree of dissociation of electrolyte = Number of moles dissociated out of one mole.

It is a ratio. Hence, it is dimensionless.

Thus, the correct match is

A $\rightarrow$ 3, B $\rightarrow$ 1, C $\rightarrow$ 4, D $\rightarrow$ 2.

Question 121

If the conductivity of mercury at 0°C is $1.07 \times 10^6 \text{Sm}^{-1}$ and the resistance of a cell containing mercury is 0.243Ω , then the cell constant of the cell is $x \times 10^4 \text{m}^{-1}$. The value of x is..... .(Nearest integer)
[1 Sep 2021 Shift 2]

Answer: 26

Solution:

Conductance (G) is reciprocal of resistance (R).

$$R = \frac{1}{G} \text{ or } G = \frac{1}{R} = \frac{1}{0.243\Omega} = 4.115\Omega^{-1}$$

Relation between conductance (G),

conductivity (κ) and cell constant $\left(\frac{l}{A}\right)$ is given as

$$\kappa = \frac{Gl}{A}$$

$$\Rightarrow \frac{l}{A} = \frac{\kappa}{G} = \frac{1.07 \times 10^6 \text{Sm}^{-1}}{4.115\Omega^{-1}} = 26 \times 10^4 \text{m}^{-1}$$

$$\Rightarrow x = 26$$

$\therefore$ Answer is 26.

Question122

The equation that is incorrect is:

[Jan. 07, 2020 (II)]

Options:

A. $(\Lambda_m^0)_{\text{NaBr}} - (\Lambda_m^0)_{\text{NaCl}} = (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{KCl}}$

B. $(\Lambda_m^0)_{\text{KCl}} - (\Lambda_m^0)_{\text{NaCl}} = (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{NaBr}}$

C. $(\Lambda_m^0)_{\text{H}_2\text{O}} = (\Lambda_m^0)_{\text{HCl}} + (\Lambda_m^0)_{\text{NaOH}} - (\Lambda_m^0)_{\text{NaCl}}$

D. $(\Lambda_m^0)_{\text{NaBr}} - (\Lambda_m^0)_{\text{NaI}} = (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{NaBr}}$

Answer: D

Solution:

$$\begin{aligned} & (\Lambda_m^0)_{\text{NaBr}} - (\Lambda_m^0)_{\text{NaI}} \\ &= \Lambda_m^0 \text{Na}^+ + \Lambda_m^0 \text{Br}^- - (\Lambda_m^0 \text{Na}^+ + \Lambda_m^0 \text{I}^-) \\ &= \Lambda_m^0 \text{Na}^+ + \Lambda_m^0 \text{Br}^- - \Lambda_m^0 \text{Na}^+ - \Lambda_m^0 \text{I}^- \\ &= \Lambda_m^0 \text{Br}^- + \Lambda_m^0 \text{I}^- \\ & (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{NaBr}} \\ &= \Lambda_m^0 \text{K}^+ + \Lambda_m^0 \text{Br}^- - (\Lambda_m^0 \text{Na}^+ + \Lambda_m^0 \text{Br}^-) \\ &= \Lambda_m^0 \text{K}^+ + \Lambda_m^0 \text{Br}^- - \Lambda_m^0 \text{Na}^+ - \Lambda_m^0 \text{Br}^- \\ &= \Lambda_m^0 \text{K}^+ + \Lambda_m^0 \text{Na}^+ \\ & \therefore (\Lambda_m^0)_{\text{NaBr}} - (\Lambda_m^0)_{\text{NaI}} \neq (\Lambda_m^0)_{\text{KBr}} - (\Lambda_m^0)_{\text{NaBr}} \end{aligned}$$

Question123

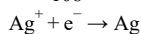
108g of silver (molar mass 108gmol^{-1}) is deposited at cathode from $\text{AgNO}_3(\text{aq})$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273K and 1 bar pressure from water by the same quantity of electricity is
[NV, Jan. 09, 2020(I)]

Answer: 5.68

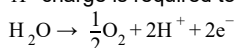
Solution:

No. of moles of silver deposited.

$$= \frac{108}{108} = 1\text{mol}$$



1F charge is required to deposit 1mol e of Ag



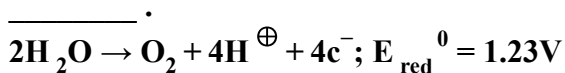
2F charge deposit $\rightarrow \frac{1}{2}$ moles of oxygen

1F charge will deposit $\rightarrow \frac{1}{4}$ moles of oxygen

$$\begin{aligned} V_{\text{O}_2} &= \frac{nRT}{P} \\ &= \frac{1}{4} \times \frac{0.08314\text{LbarK}^{-1}\text{mol}^{-1} \times 273\text{K}}{1\text{bar}} \\ &= \frac{1}{4} \times 22.7 \\ V_{\text{O}_2} &= 5.675\text{L} \end{aligned}$$

Question 124

What would be the electrode potential for the given half cell reaction at pH = 5 ?



(R = 8.314 J mol⁻¹ K⁻¹; T emp = 298 K ; oxygen under std. atm. pressure of 1 bar)

[NV, Jan. 08, 2020 (I)]

Answer: 1.52

Solution:

$$\begin{aligned} E &= 1.23 - \frac{0.0591}{4} \log[\text{H}^+]^4 \\ &= 1.23 + 0.0591 \times \text{pH} \\ &= 1.23 + 0.0591 \times 5 \\ &= 1.23 + 0.2955 \\ &= 1.52\text{V} \end{aligned}$$

Question 125

For an electrochemical cell

$\text{Sn(s)} | \text{Sn}^{2+}(\text{aq}, 1\text{M}) | \text{Pb}^{2+}(\text{aq}, 1\text{M}) | \text{Pb(s)}$ the ratio $\frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$

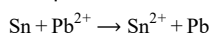
when this cell attains equilibrium is

[NV, Jan. 08, 2020 (II)]

Answer: 2.15

Solution:

At equilibrium state $E_{\text{cell}} = 0$; $E_{\text{cell}}^0 = 0.01\text{V}$



$$E = E_{\text{cell}}^0 - \frac{0.06}{n} \log \frac{[\text{P}]}{[\text{R}]}$$

$$0 = 0.01 - \frac{0.06}{2} \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$$

$$-0.01 = -\frac{0.06}{2} \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$$

$$\frac{1}{3} = \log \frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]}$$

$$\frac{[\text{Sn}^{2+}]}{[\text{Pb}^{2+}]} = 10^{1/3} = 2.15$$

Question 126

Given that the standard potentials (E) of Cu^{2+}/Cu and Cu^+/Cu are 0.34V and 0.522V respectively, the E^0 of Cu^{2+}/Cu is:

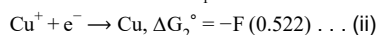
[Jan. 07, 2020 (I)]

Options:

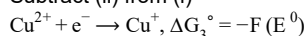
- A. +0.182V
- B. +0.158V
- C. -0.182V
- D. -0.158V

Answer: B

Solution:



Subtract (ii) from (i)



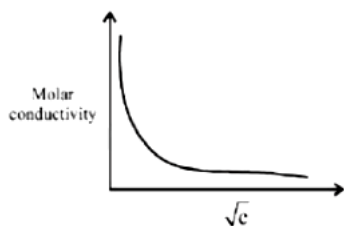
$$\therefore \Delta G_1^\circ - \Delta G_2^\circ = \Delta G_3^\circ$$

$$\Rightarrow -F E^\circ = -2F(0.34) + F(0.522)$$

$$\Rightarrow E^\circ = 0.68 - 0.522 = 0.158\text{V}$$

Question127

The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.



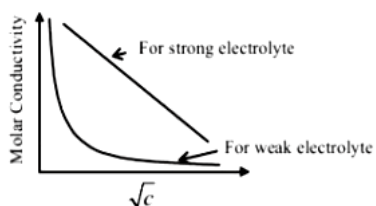
The electrolyte X is :
[Sep. 05, 2020 (II)]

Options:

- A. HCl
- B. NaCl
- C. KNO₃
- D. CH₃COOH

Answer: D

Solution:



Among given electrolytes, CH₃COOH is weak electrolyte.

Question128

Let C_{NaCl} and C_{BaSO_4} be the conductances (in S) measured for saturated aqueous solutions of NaCl and BaSO_4 , respectively, at a temperature T. Which of the following is false?

[Sep. 03, 2020 (I)]

Options:

A. Ionic mobilities of ions from both salts increase with T

B. $C_{\text{BaSO}_4}(T_2) > C_{\text{BaSO}_4}(T_1)$ for $T_2 > T_1$

C. $C_{\text{NaCl}}(T_2) > C_{\text{NaCl}}(T_1)$ for $T_2 > T_1$

D. $C_{\text{NaCl}} \gg C_{\text{BaSO}_4}$ at a given T

E. None of above

Answer: E

Solution:

(N) Ionic mobility increases with increase in temperature, which increases the conductance of the solution, while conductance of NaCl solution is independent of temperature above 400°C . Because the temperature is not given, so none of the option is correct.

Question 129

Potassium chlorate is prepared by the electrolysis of KCl in basic solution

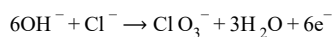
$6\text{OH}^- + \text{Cl}^- \rightarrow \text{ClO}_3^- + 3\text{H}_2\text{O} + 6\text{e}^-$ If only 60% of the current is utilized in the reaction, the time (rounded to the nearest hour) required to produce 10 g of KClO_3 using a current of 2A is _____.

(Given : $F = 96,500 \text{ C mol}^{-1}$; molar mass of $\text{KClO}_3 = 122 \text{ g mol}^{-1}$)

[NV, Sep. 06, 2020 (I)]

Answer: 11

Solution:



For synthesis of 1 mole of ClO_3^- , 6F of charge is required.

$\therefore$ Current efficiency = 60%

$\therefore$ To synthesis 1 mole of ClO_3^- , 10F of charge is required.

To synthesis $\frac{10}{122}$ moles of KClO_3 , charge = $\frac{10 \times 10}{122} F$ $Q = I \cdot t$

$$t = \frac{100 \times 96500}{122 \times 2} = 39549.18 \text{ s}$$

$$= \frac{79098.365}{3600 \text{ s}} = 10.99 \text{ h}$$

$\therefore t = 11 \text{ h}$.

Question 130

250mL of a waste solution obtained from the workshop of a goldsmith contains 0.1M AgNO_3 and 0.1M AuCl . The solution was electrolyzed at 2V by passing a current of 1 A for 15 minutes. The metal/metals electrodeposited will be:

[Sep. 04, 2020(II)]

$$(E_{\text{Ag}^+/\text{Ag}}^0 = 0.80\text{V}, E_{\text{Au}^+/\text{Au}}^0 = 1.69\text{V})$$

Options:

- A. only gold
- B. silver and gold in proportion to their atomic weights
- C. only silver
- D. silver and gold in equal mass proportion

Answer: A

Solution:

Millimoles of $\text{Au}^+ = 0.1 \times 250 = 25$

Mole of $\text{Au}^+ = \frac{25}{1000} = \frac{1}{40} = 0.025$

Similarly, moles of $\text{Ag}^+ = 0.025$

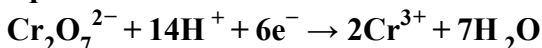
Charge passed $= I \times t = 1 \times 15 \times 60 = 900\text{C}$

Moles of e^- passed $= \frac{900}{96500} = 0.0093\text{mol}$.

Species with higher value of SRP will get deposited first at cathode.

Question 131

An acidic solution of dichromate is electrolyzed for 8 minutes using 2 A current. As per the following equation



The amount of Cr^{3+} obtained was 0.104g. The efficiency of the process (in %) is (Take : $F = 96000\text{C}$, At. mass of chromium = 52) _____.

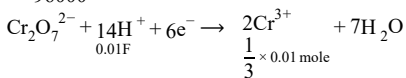
[NV, Sep. 03, 2020 (II)]

Answer: 60

Solution:

Charge (Q) $= I t = 2 \times 8 \times 60 = 960\text{C}$

$\Rightarrow \frac{960}{96000} = 0.01F$

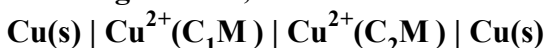


Theoretical mass of $\text{Cr}^{3+} = \frac{1}{3} \times \frac{960}{96000} \times 52 = 0.173\text{g}$

So, efficiency $= \frac{W_{\text{actual}}}{W_{\text{Theoretical}}} \times 100 = \frac{0.104}{0.173} \times 100 = 60\%$

Question 132

For the given cell;



change in Gibbs energy (ΔG) is negative, if:

[Sep. 06, 2020 (II)]

Options:

A. $C_1 = C_2$

B. $C_2 = \frac{C_1}{\sqrt{2}}$

C. $C_1 = 2C_2$

D. $C_2 = \sqrt{2}C_1$

Answer: D

Solution:

For the concentration cell, $E_{\text{cell}}^0 = 0$

Anode : $\text{Cu(s)} \rightarrow \text{Cu}^{2+}(\text{aq})_{\text{A}}$

Cathode : $\text{Cu}^{2+}(\text{aq})_{\text{e}} \rightarrow \text{Cu(s)}$

Overall $\text{Cu}^{2+}(\text{aq})_{\text{C}} \rightarrow \text{Cu}^{2+}(\text{aq})_{\text{A}}$
(C_2M) (C_1M)

As $\Delta G = -nF E$

If $\Delta G = -ve$, then E_{cell} is $+ve$.

$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{RT}{2F} \ln \frac{C_1}{C_2}$$

$$E_{\text{cell}} = 0 - \frac{RT}{2F} \ln \frac{C_1}{C_2}$$

$$E_{\text{cell}} = \frac{RT}{2F} \ln \frac{C_2}{C_1}$$

So, $C_2 > C_1$.

Thus, $C_2 = \sqrt{2}C_1$ relation is correct.

Question133

An oxidation-reduction reaction in which 3 electrons are transferred has a ΔG^0 of 17.37kJ mol^{-1} at 25°C . The value of E_{cell}^0 (inV) is $\text{_____} \times 10^{-2}$

($1F = 96,500\text{Cmol}^{-1}$)

[NV, Sep. 05, 2020(I)]

Answer: -6

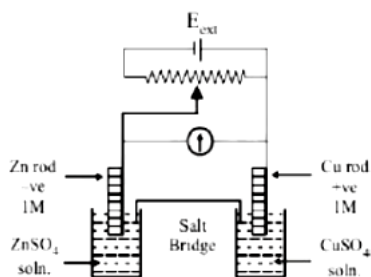
Solution:

$$\Delta G^0 = -nF E_{\text{cell}}^0$$

$$17.37 \times 10^3 = -3 \times 96500 \times E_{\text{cell}}^0$$

$$E_{\text{cell}}^0 = -0.06V \sim \text{eq} - 6.0 \times 10^{-2}V$$

Question134



$$E_{\text{Cu}^{2+} | \text{Cu}}^{\circ} = +0.34\text{V} ; E_{\text{Zn}^{2+} | \text{Zn}}^{\circ} = -0.76\text{V}$$

Identify the incorrect statement from the options below for the above cell :
[Sep. 04, 2020(I)]

Options:

- A. If $E_{\text{ext}} > 1.1\text{V}$, e^{-} flows from Cu to Zn
- B. If $E_{\text{ext}} > 1.1\text{V}$, Zn dissolves at Zn electrode and Cu deposits at Cu electrode
- C. If $E_{\text{ext}} < 1.1\text{V}$, Zn dissolves at anode and Cu deposits at cathode
- D. If $E_{\text{ext}} = 1.1\text{V}$, no flow of e^{-} or current occurs

Answer: B

Solution:

$$E_{\text{cell}}^{\circ} = E_{\text{Cu}^{2+} | \text{Cu}}^{\circ} - E_{\text{Zn}^{2+} | \text{Zn}}^{\circ} = 1.1\text{V}$$

So, if $E_{\text{ext}} = 1.1\text{V}$ no electron will flow

At, $E_{\text{ext}} > 1.1\text{V}$ cell act as electrolytic cell and electron will flow from Cu to Zn.

At, $E_{\text{ext}} < 1.1\text{V}$ cell act as electrochemical cells so Zn dissolve and Cu deposit.

Question 135

The photoelectric current from Na (work function, $w_0 = 2.3\text{eV}$) is stopped by the output voltage of the cell $\text{Pt(s)} | \text{H}_2(\text{g}, 1\text{bar}) | \text{HCl}(\text{aq}, \text{pH} = 1) | \text{AgCl(s)} | \text{Ag(s)}$. The pH of aq. HCl required to stop the photoelectric current from K ($w_0 = 2.25\text{eV}$), all other conditions remaining the same, is $\text{_____} \times 10^{-2}$ (to the nearest integer).

Given,

$$2.303 \frac{RT}{F} = 0.06\text{V} ; E_{\text{AgCl} | \text{Ag} | \text{Cl}^{-}}^{\circ} = 0.22\text{V}$$

[NV, Sep. 03, 2020(I)]

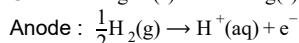
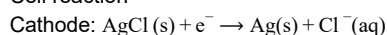
Answer: 142

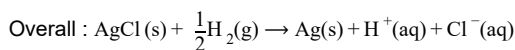
Solution:

Sodium metal :

$$E = E_0 + (K E)_{\text{max}} ; E_{\text{cell}}^{\circ} = 0.22\text{V}$$

Cell reaction





$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log[\text{H}^+][\text{Cl}^-]$$

$$E_{\text{cell}} = 0.22 - \frac{0.06}{1} \log[10^{-1}][10^{-1}]$$

$$= 0.22 + 0.12 = 0.34\text{V}$$

$$(\text{K E})_{\text{max}} = E_{\text{cell}} = 0.34\text{eV}$$

So, $E = 2.3 + 0.34 = 2.64\text{eV}$ = Energy of photon incident

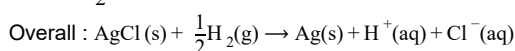
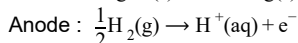
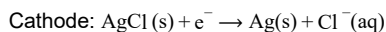
For potassium metal :

$$E = E_0 + (\text{K E})_{\text{max}}$$

$$2.64 = 2.25 + (\text{K E})_{\text{max}}$$

$$(\text{K E})_{\text{max}} = 0.39 = E_{\text{cell}}$$

Cell reaction



$$E_{\text{cell}} = E_{\text{cell}}^0 - \frac{0.06}{1} \log[\text{H}^+][\text{Cl}^-]$$

$$0.39 = 0.22 - 0.06 \log[\text{H}^+]^2$$

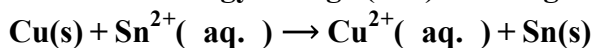
$$0.39 = 0.22 - 0.12 \log[\text{H}^+]$$

$$0.17 = 0.12 \times \text{pH}$$

$$\text{pH} = 17/12 = 1.4166 \sim \text{eq}1.42 = 142 \times 10^{-2}$$

Question136

The Gibbs energy change (in J) for the given reaction at $[\text{Cu}^{2+}] = [\text{Sn}^{2+}] = 1\text{M}$ and 298K is :



$$(E_{\text{Sn}^{2+}|\text{Sn}}^0 = -0.16\text{V}, E_{\text{Cu}^{2+}|\text{Cu}}^0 = 0.34\text{V}, \text{ Take } F = 96500\text{cmol}^{-1})$$

[NV, Sep. 02, 2020(I)]

Answer: 96500

Solution:

$$\text{Sol. } E_{\text{cell}}^0 = E_{\text{Sn}^{2+}/\text{Sn}}^0 - E_{\text{Cu}^{2+}/\text{Cu}}^0$$

$$= -0.16 - 0.34$$

$$= -0.50\text{V}$$

$$\Delta G^0 = -nF E_{\text{cell}}^0$$

$$= -2 \times 96500 \times (-0.5)$$

$$= 96500\text{J}$$

$$= 96.5\text{KJ} = 96500\text{J}$$

Question137

For the disproportionation reaction $2\text{Cu}^+(\text{aq}) \rightleftharpoons \text{Cu(s)} + \text{Cu}^{2+}(\text{aq})$ at 298K , $\ln K$ (where K is the equilibrium constant) is $\times 10^{-1}$.

Given

[NV, Sep. 02, 2020(II)]

$$(E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = 0.16\text{V}; E_{\text{Cu}^+/\text{Cu}}^0 = 0.52\text{V}; \frac{RT}{F} = 0.025)$$

Answer: 144

Solution:

$$\begin{aligned}2\text{Cu}^+(\text{aq}) &\rightleftharpoons \text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \\E_{\text{cell}}^0 &= E_{\text{Cu}^+/\text{Cu}}^0 - E_{\text{Cu}^{2+}/\text{Cu}^+}^0 = 0.52 - 0.16 = 0.36\text{V} \\E_{\text{cell}}^0 &= \frac{RT}{nF} \ln K_{\text{eq}} \\\Rightarrow 0.36 &= \frac{0.025}{1} \ln K \\\Rightarrow \ln K &= 14.4 = 144 \times 10^{-1}\end{aligned}$$

Question138

$\Lambda_{\text{m}}^{\circ}$ for NaCl, HCl and NaA are 126.4, 425.9 and 100.5 $\text{Scm}^2\text{mol}^{-1}$, respectively. If the conductivity of 0.001 M HA is $5 \times 10^{-5} \text{Scm}^{-1}$, degree of dissociation of HA is :
[Jan. 12, 2019 (II)]

Options:

- A. 0.50
- B. 0.25
- C. 0.125
- D. 0.75

Answer: C

Solution:

$$\begin{aligned}\text{Given: } \Lambda_{\text{m}}^{\circ}(\text{NaCl}) &= 126.4 \text{Scm}^2\text{mol}^{-1}, \Lambda_{\text{m}}^{\circ}(\text{HCl}) = 425.9 \text{Scm}^2\text{mol}^{-1} \text{ and } \Lambda_{\text{m}}^{\circ}(\text{NaA}) = 100.5 \text{Scm}^2\text{mol}^{-1} \\\Lambda_{\text{m}}^{\circ}(\text{HA}) &= \lambda_{\text{H}^+}^{\circ} + \lambda_{\text{A}^-}^{\circ} \\&= \lambda_{\text{H}^+}^{\circ} + \lambda_{\text{Cl}^-}^{\circ} + \lambda_{\text{A}^+}^{\circ} + \lambda_{\text{Na}^+}^{\circ} - \lambda_{\text{Cl}^-}^{\circ} - \lambda_{\text{Na}^+}^{\circ} \\&= \Lambda_{\text{m}}^{\circ}(\text{HCl}) + \Lambda_{\text{m}}^{\circ}(\text{NaA}) - \Lambda_{\text{m}}^{\circ}(\text{NaCl}) \\\Lambda_{\text{m}}^{\circ}(\text{HA}) &= 429.5 - 126.4 + 100.5 = 400 \text{Scm}^2\text{mol}^{-1} \\\kappa(\text{HA}) &= 5 \times 10^{-5} \text{Scm}^{-1} \\(\text{HA}) &= \frac{\kappa(\text{HA}) \times 1000}{\text{Molality of HA}} = \frac{5 \times 10^{-5} \times 1000}{0.001} = 50 \text{Scm}^2\text{mol}^{-1} \\\alpha &= \frac{\Lambda_{\text{m}}(\text{HA})}{\Lambda_{\text{m}}^{\circ}(\text{HA})} = \frac{50}{400} = 0.125\end{aligned}$$

Question139

The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of PbSO_4 electrolyzed in g during the process is :

(Molar mass of $\text{PbSO}_4 = 303 \text{gmol}^{-1}$)

[Jan. 9, 2019 (I)]

Options:

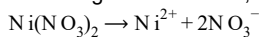
- A. 22.8
- B. 15.2
- C. 7.6
- D. 11.4

Answer: A

Solution:

According to the Faraday's law of electrolysis, nF of current is required for the deposition of 1mol

According to the reaction,



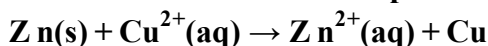
$2F$ of current deposits = 1mol

$$\therefore 0.1F \text{ of current deposits} = \frac{0.1}{2} = 0.05 \text{ mol}$$

Question 140

The standard electrode potential E° and its temperature coefficient $\left(\frac{dE^\circ}{dT}\right)$ for a cell are 2V and

$-5 \times 10^{-4} \text{ V K}^{-1}$ at 300K respectively. The cell reaction is:



The standard reaction enthalpy ($\Delta_r H^\circ$) at 300K in kJ mol^{-1} is,

[Use $R = 8 \text{ J K}^{-1} \text{ mol}^{-1}$ and $F = 96,000 \text{ C mol}^{-1}$]

[Jan. 12, 2019 (I)]

Options:

A. -412.8

B. -384.0

C. 192.0

D. 206.4

Answer: A

Solution:

$$\Delta G^\circ = \Delta_r H^\circ - T \Delta S^\circ$$

$$\Delta_r H^\circ = \Delta G^\circ + T \Delta S^\circ$$

$$\Delta_r H^\circ = -nF E^\circ + T nF \frac{dE^\circ}{dT}$$

$$\Delta_r H^\circ = -nF E^\circ + nF T \frac{dE^\circ}{dT}$$

Cell reaction: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

$$\Delta_r H^\circ = -nF \left(E^\circ - T \frac{dE^\circ}{dT} \right)$$

$$\Delta_r H^\circ = -2 \times 96000 (2 - 300 \times -5 \times 10^{-4})$$

$$\Delta_r H^\circ = -2 \times 96000 (2 + 300 \times 5 \times 10^{-4})$$

$$= -2 \times 96000 (2 + 0.15)$$

$$= -412.8 \times 10^3 \text{ J/mol}; = -412.8 \text{ kJ/mol}$$

Question 141

For the cell $Zn(s) | Zn^{2+}(aq) || M^{x+}(aq) | M(s)$, different half cells and their standard electrode potentials are given below:

$M^{x+}(aq)/$	$Au^{3+}(aq)$	$Ag^+(aq)/$	$Fe^{3+}(aq)$	$Fe^{2+}(aq)$
$M(s)$	$Au(s)$	$Ag(s)$	$Fe^{2+}(aq)$	$Fe(s)$
$E^\circ M^{x+/M} / (V)$	1.40	0.80	0.77	0.44

If $E_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V}$, which cathode will give a maximum value of E_{cell}° per electron transferred?
[Jan. 11, 2019 (I)]

Options:

- A. Ag^{+}/Ag
- B. $\text{Fe}^{3+}/\text{Fe}^{2+}$
- C. Au^{3+}/Au
- D. Fe^{2+}/Fe

Answer: A

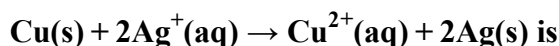
Solution:

$$\begin{aligned}\text{Zn} + 2\text{Ag}^{+} &\rightarrow 2\text{Ag} + \text{Zn}^{2+} (n = 2e^{-}) \\ E_{\text{cell}}^{\circ} &= (E_{\text{R.P.}}^{\circ})_{\text{cathode}} - (E_{\text{R.P.}}^{\circ})_{\text{anode}} \\ &= 0.80 - (-0.76) = 1.56\text{V for } 2e^{-} \\ \therefore E_{\text{cell}}^{\circ} \text{ for } 1e^{-} &= \frac{1.56}{2} = 0.78\text{V}\end{aligned}$$

Question 142

Given the equilibrium constant:

K_{C} of the reaction:



10×10^{15} calculate the E_{cell}° of this reaction at 298K

$$\left[2.303 \frac{RT}{F} \text{ at } 298\text{K} = 0.059\text{V} \right]$$

[Jan. 11, 2019 (II)]

Options:

- A. 0.04736mV
- B. 0.4736mV
- C. 0.4736V
- D. 0.04736V

Answer: C

Solution:

$$\begin{aligned}E_{\text{cell}}^{\circ} &= \frac{2.303RT}{nF} \log K_{\text{C}} \text{ or } E_{\text{cell}}^{\circ} = \frac{0.059\text{V}}{n} \log K_{\text{C}} \\ &= \frac{0.059\text{V}}{2} \log 10^{16} = 0.4736\text{V}\end{aligned}$$

Question 143

In the cell

$\text{Pt(s)} \mid \text{H}_2(\text{g}, 1 \text{ bar}) / \text{HCl(aq)} \mid \text{AgCl(s)} / \text{Ag(s)} \mid \text{Pt(s)}$, the cell potential is 0.92V when a 10^{-6} molal

HCl solution is used. The standard electrode potential of $(\text{AgCl} / \text{Ag}, \text{Cl}^{-})$ electrode is:

{ Given : $\frac{2.303RT}{F} = 0.06V$ at 298K }

[Jan. 10, 2019 (II)]

Options:

A. 0.94V

B. 0.76V

C. 0.40V

D. 0.20V

Answer: D

Solution:

Given that:

$\text{Pt(s)} \mid \text{H}_2(\text{g}, 1\text{bar}) / \text{HCl(aq)} \mid \text{AgCl(s)} / \text{Ag(s)} \mid \text{Pt(s)}$

$E_{\text{cell}} = 0.92V$

Now, $E_{\text{cell}} = E_{\text{H}_2(\text{g})/\text{H}^+(\text{aq})}^\circ + E_{\text{AgCl(s)}/\text{Ag(s)Cl}^-}^\circ - \frac{0.06}{n} \log Q$

Cell reaction: $\frac{1}{2}\text{H}_2(\text{g}) \rightarrow \text{H}^+(\text{aq}) + \text{e}^-$

$\text{AgCl(s)} + \text{e}^- \rightarrow \text{Ag(s)} + \text{Cl}^-(\text{aq})$

Net cell reaction:

$\frac{1}{2}\text{H}_2(\text{g}) + \text{AgCl(s)} \rightarrow \text{Ag(s)} + \text{H}^+ + \text{Cl}^-(\text{aq})$

$\therefore Q = \frac{[\text{H}^+][\text{Cl}^-]}{(P_{\text{H}_2})^{1/2}}$

Here, 10^{-6} molal HCl solution is used

So $Q = \frac{10^{-6} \times 10^{-6}}{1} = 10^{-12}$

(assuming molality = molarity)

Now, $0.92 = E_{\text{AgCl(s)}/\text{Ag(s)Cl}^-}^\circ - \frac{0.06}{1} \log 10^{-12}$

$E_{\text{AgCl(s)}/\text{Ag(s)Cl}^-}^\circ = 0.92 + [0.06 \times (-12)]$

$= 0.92 - 0.72 = 0.20V$

Question144

If the standard electrode potential for a cell is 2V at 300K, the equilibrium constant (K) for the reaction

$\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$

at 300K is approximately ($R = 8\text{J K}^{-1}\text{mol}^{-1}$, $F = 96000\text{Cmol}^{-1}$)

[Jan. 9, 2019 (II)]

Options:

A. e^{-80}

B. e^{-160}

C. e^{320}

D. e^{160}

Answer: D

Solution:

We know that,

$$\ln K = \frac{nF E_{\text{cell}}^{\circ}}{RT}$$

$$\Delta G = -nF E_{\text{cell}}^{\circ}$$

$$\Delta G = -RT \ln k$$

Now,

$$\text{After putting the given values, we get } \ln K = \frac{2 \times 96000 \times 2}{8 \times 300} = 160$$

$$\therefore K = e^{160}$$

Question145

The decreasing order of electrical conductivity of the following aqueous solutions is:

0.1 M Formic acid (A),

0.1 M Acetic acid (B),

0.1 M Benzoic acid (C).

[April 12, 2019 (II)]

Options:

A. $A > C > B$

B. $C > B > A$

C. $A > B > C$

D. $C > A > B$

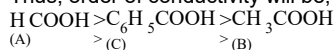
Answer: A

Solution:

Order of acidic strength is $\text{HCOOH} > \text{C}_6\text{H}_5\text{COOH} > \text{CH}_3\text{COOH}$

More the acidic strength more will be the dissociation of acid into ions and more will be the conductivity.

Thus, order of conductivity will be,



Question146

Consider the statements S1 and S2 :

S1: Conductivity always increases with decrease in the concentration of electrolyte. S2: Molar conductivity always increases with decrease in the concentration of electrolyte. The correct option among the following is:

[April 10, 2019 (I)]

Options:

A. Both S1 and S2 are wrong

B. S1 is wrong and S2 is correct

C. Both S1 and S2 are correct

D. S1 is correct and S2 is wrong

Answer: B

Solution:

Conductivity of an electrolyte is the conductance of 1cm^3 of the given electrolyte. It increases with the increase in concentration of electrolyte due to increase in the number of ions per unit volume. Molar conductivity (λ_m) is the conductance of a solution containing 1 mole of the electrolyte. It

increases with the decrease of concentration due to increase in the total volume having one mole of electrolyte. Thus, interionic attraction increases and degree of ionisation decreases. Therefore, (S_1) is wrong and (S_2) is correct.

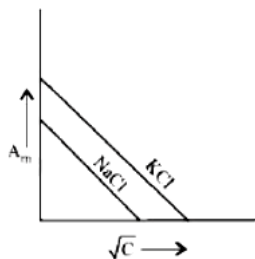
Question147

Which one of the following graphs between molar conductivity (Λ_m) versus $\sqrt{C}$ is correct?

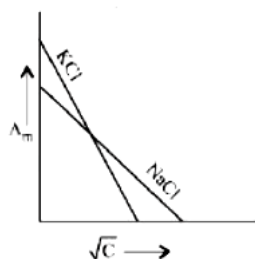
[April 10, 2019 (II)]

Options:

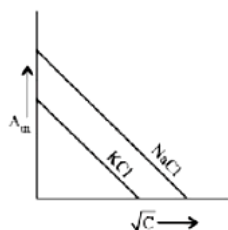
A.



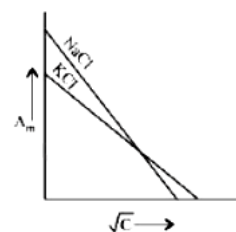
B.



C.



D.



Answer: A

Solution:

Since, KCl is more conducting than NaCl, therefore, graph (a) is correct.

Question148

A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many mole of Ni will be deposited at the cathode?

[April 9, 2019 (II)]

Options:

A. 0.05

B. 0.20

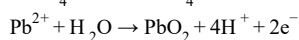
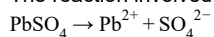
C. 0.15

D. 0.10

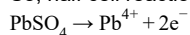
Answer: C

Solution:

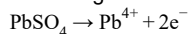
The reaction involved in the process is given below



So, half cell reaction is



According to the reaction:



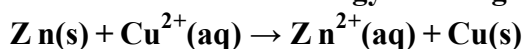
We require 2F for the electrolysis of 1mol or 303g of PbSO_4

$\therefore$ Amount of PbSO_4 electrolysed by 0.05F

$$= \frac{303}{2} \times 0.05 = 7.575\text{g} \approx 7.6\text{g}$$

Question 149

The standard Gibbs energy for the given cell reaction in kJ mol^{-1} at 298K is:



$E^\circ = 2\text{V}$ at 298K

(Faraday's constant, $F = 96000\text{Cmol}^{-1}$)

[April 9, 2019(I)]

Options:

A. -384

B. 384

C. 192

D. -192

Answer: A

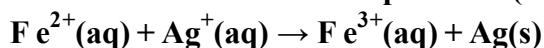
Solution:

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ$$

$$= -2 \times (96000) \times 2\text{V} = -384000\text{J/mol} = -384\text{kJ/mol}$$

Question 150

Calculate the standard cell potential (in V) of the cell in which following reaction takes place:



Given that

$$E_{\text{Ag}^+/\text{Ag}}^0 = xV; E_{\text{Fe}^{2+}/\text{Fe}}^0 = yV; E_{\text{Fe}^{3+}/\text{Fe}}^0 = zV$$

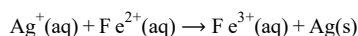
[April 8, 2019 (II)]

Options:

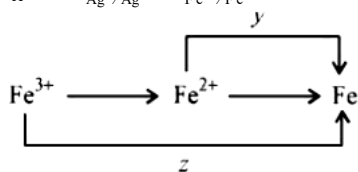
- A. $x - z$
- B. $x - y$
- C. $x + 2y - 3z$
- D. $x + y - z$

Answer: C

Solution:



$$E_{\text{cell}}^0 = E_{\text{Ag}^+/\text{Ag}}^0 - E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0$$



$$E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 = 3z - 2y$$

$$E_{\text{Ag}^+/\text{Ag}}^0 = x \quad (\text{given})$$

$$\therefore E_{\text{cell}}^0 = x - 3z + 2y$$

Question151

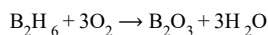
How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66g of diborane? (Atomic weight of B = 10.8u)
[2018]

Options:

- A. 6.4 hours
- B. 0.8 hours
- C. 3.2 hours
- D. 1.6 hours

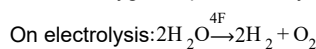
Answer: C

olution:



27.66g of B_2H_6 (1mol e) requires 3 moles of oxygen (O_2) for complete burning.

Now the oxygen is produced by the electrolysis of H_2O .



1 mole O_2 is produced by 4F charge

$\therefore$ 3mol O_2 will be produced by 12F charge.

$$\therefore Q = It$$

$$12 \times 96500\text{C} = I \times t$$

$$12 \times 96500\text{C} = 100 \times t$$

$$t = \frac{12 \times 96500}{100} \text{second}$$

$$= \frac{12 \times 96500}{100 \times 3600} \text{ hour}$$

$$= 3.2 \text{ hours}$$

Question 152

When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is
[Online April 16, 2018]

Options:

- A. 109.0g
- B. 98.1g
- C. 9.81g
- D. 10.9g

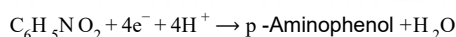
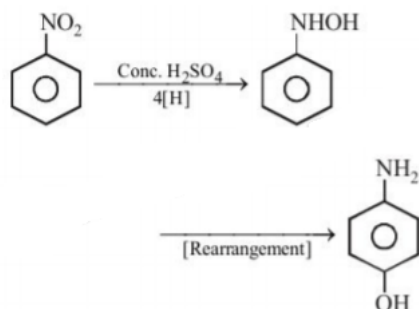
Answer: C

Solution:

9.65 ampere current was passed for 1.0 hour (3600 seconds)

Number of moles of electrons passed

$$= \frac{I(A) \times t(s)}{96500} = \frac{9.65A \times 3600s}{96500} = 0.36 \text{ mol es}$$



∴ 4 moles of electrons reduces 1 mole of nitrobenzene to p-aminophenol. ∴ 0.36 moles of electrons will reduce $\frac{0.36}{4} = 0.09$ moles of nitrobenzene to p-aminophenol.

p-aminophenol molar mass = 109.14g/mol

Mass of p-aminophenol obtained = 109.14g/mol $\times$ 0.09mol = 9.81g

Question 153

When an electric current is passes through acidified water, 112mL of hydrogen gas at N.T.P. was collected at the cathode in 965 seconds. The current passed, in ampere, is :
[Online April 15,2018 (I)]

Options:

- A. 2.0
- B. 0.1
- C. 0.5
- D. 1.0

Answer: D

Solution:

Reduction at cathode:



At NTP 22400mL of $H_2 = 1 \text{ mol e of } H_2$

$$112 \text{ mL of } H_2 = \frac{1}{22400} \times 112 = 0.005 \text{ mol e of } H_2$$

$$\text{Moles of } H_2 \text{ produced} = \frac{I \times t}{96500} \times \text{molar ratio}$$

$$0.005 = \frac{I \times 965}{96500} \times \frac{1 \text{ mol e of } H_2}{2 \text{ mol e of } e^-} \quad I = 1.0 \text{ A}$$

Question 154

Given

$$E_{Cl_2/Cl^-}^0 = 1.36 \text{ V}, E_{Cr^{3+}/Cr}^0 = -0.74 \text{ V}$$

$$E_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33 \text{ V}, E_{MnO_4^-/Mn^{2+}}^0 = 1.51 \text{ V}$$

Among the following, the strongest reducing agent is
[2017]

Options:

A. Cr

B. Mn^{2+}

C. Cr^{3+}

D. Cl

Answer: A

Solution:

$$E_{MnO_4^-/Mn^{2+}}^0 = 1.51 \text{ V}$$

$$E_{Cl_2/Cl^-}^0 = 1.36 \text{ V}$$

$$E_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33 \text{ V}$$

$$E_{Cr^{3+}/Cr}^0 = -0.74$$

Since Cr^{3+} is having least reduction potential, so Cr is the best reducing agent.

Question 155

Consider the following standard electrode potentials (E^0 in volts) in aqueous solution:

Element	M^{3+}/M	M^+/M
Al	-1.66	+0.55
Tl	+1.26	-0.34

Based on these data, which of the following statements is correct?
[Online April 8, 2017]

Options:

A. Tl^+ is more stable than Al^{3+}

B. Al^+ is more stable than Al^{3+}

C. Tl^+ is more stable than Al^+

Answer: C

(i) Al^{3+} more stable $\xrightarrow{E^\circ = -1.66}$ Al $\xleftarrow{E^\circ = +0.55}$ Al^+ Less stable

(ii) Tl^{3+} Less stable $\xrightarrow{E^\circ = +1.26}$ Tl $\xleftarrow{E^\circ = -0.34}$ Tl^+ more stable

Tl^+ has negative electrode potential ($E^\circ = -0.34$) means, it does not prefer to convert into Tl but reverse must be preferred that's why it is more stable than Tl^{3+} ($E^\circ = +1.26$). In Al , Al^{3+} is more stable ($E^\circ = -1.66$) than Al^+ ($E^\circ = +0.55$) and also from Tl^+ due to more negative value of E° . Therefore, by comparison it confirms that Tl^+ is more stable than Al^+ .

Question156

What is the standard reduction potential (E°) for $\text{Fe}^{3+} \rightarrow \text{Fe}$?

$$\text{Fe}^{2+} + 2\text{e}^{-} \rightarrow \text{Fe}; E^{\circ} \text{Fe}^{2+}/\text{Fe} = -0.47\text{V}$$
$$\text{Fe}^{3+} + \text{e}^{-} \rightarrow \text{Fe}^{2+}; E^{\circ} \text{Fe}^{3+}/\text{Fe}^{2+} = +0.77\text{V}$$

Options:

- A. -0.057V
B. $+0.057\text{V}$
C. $+0.30\text{V}$
D. -0.30V

Answer: A

Solution:

$$\Delta G^\circ = -nF E^\circ$$

(i) $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}; E^0 = -0.47\text{V};$

(ii) $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}; E^0 = +0.77\text{V} :$

$$\text{(iii) } \text{Fe}^{3+} + 3\text{e}^{-} \rightarrow \text{Fe}$$
$$(i) \Delta G^0 = -nF E^0 = -2(-0.47)F = 0.94F$$
$$(ii) \Delta G^0 = -nF E^0 = -1(+0.77)F = -0.77F$$

(iii) On adding : $\Delta G^0 = +0.17F$

$$\Delta G^0 = -nF E^0$$

$$E^0 \text{ for } (F e^{3+} \rightarrow F e) = \frac{\Delta G^0}{-nF} = \frac{0.17F}{-3F} = -0.057V$$

Question157

To find the standard potential of M^{3+}/M electrode, the following cell is constituted :

$$\text{Pt/M} / \text{M}^{3+} (0.001 \text{ mol L}^{-1}) \vee \text{Ag}^{+} (0.01 \text{ mol L}^{-1}) / \text{Ag}$$

The emf of the cell is found to be 0.421 volt at 298K . The standard potential of half reaction

$M^{3+} + 3e \rightarrow M$ at 298K will be:

(Given $E_{\text{Ag}^+/\text{Ag}}^\circ$ at 298K = 0.80V ol t)

[Online April 9, 2017]

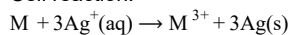
Options:

- A. 0.38 Volt
- B. 0.32 Volt
- C. 1.28 Volt
- D. 0.66 Volt

Answer: B

Solution:

Cell reaction:



$$E = E^0 - \frac{0.0591}{n} \log \left[\frac{\text{Reduced state}}{\text{Oxidised state}} \right]$$

$$0.421 = E^0 - \frac{0.0591}{3} \log \frac{0.001}{(0.01)^3}$$

$$E^0 = 0.48$$

$$E^0 = E_{Ag^+/Ag}^0 - E_{M^{3+}/M}^0$$

$$E_{M^{3+}/M}^0 = 0.8V - 0.48V = 0.32 \text{ volt}$$

Question 158

What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M ZnSO₄?

[Online April 9, 2016]

Options:

- A. The copper metal will dissolve with evolution of oxygen gas
- B. The copper metal will dissolve with evolution of hydrogen gas
- C. No reaction will occur
- D. The copper metal will dissolve and zinc metal will be deposited.

Answer: C

Solution:

No reaction will occur as the Zn is placed above Cu in electrochemical series. Hence there will be no displacement reaction.

Question 159

Galvanization is applying a coating of:
[2016]

Options:

- A. Cu
- B. Zn
- C. Pb
- D. Cr

Answer: B

Solution:

Galvanization is the process by which zinc is coated over corrosive (easily rusted) metals to prevent them from corrosion.

Question160

Identify the correct statement:
[Online April 10,2016]

Options:

- A. Corrosion of iron can be minimized by forming a contact with another metal with a higher reduction potential
- B. Iron corrodes in oxygen free water
- C. Corrosion of iron can be minimized by forming an impermeable barrier at its surface
- D. Iron corrodes more rapidly in salt water because its electrochemical potential is higher

Answer: C

Solution:

Corrosion of iron can be minimized by forming an impermeable barrier at its surface.

Question161

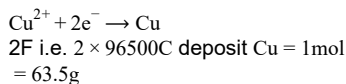
Two Faraday of electricity is passed through a solution of CuSO_4 . The mass of copper deposited at the cathode is
(at. mass of Cu = 63.5amu)
[2015]

Options:

- A. 2g
- B. 127g
- C. 0g
- D. 63.5g

Answer: D

Solution:



Question162

A variable, opposite external potential (E_{ext}) is applied to the cell $\text{Zn} | \text{Zn}^{2+}(1\text{M}) | \text{Cu}^{2+}(1\text{M}) | \text{Cu}$, of potential 1.1V When $E_{\text{ext}} < 1.1\text{V}$ and $E_{\text{ext}} > 1.1\text{V}$, respectively electrons flow from:
[Online April 10, 2015]

Options:

- A. anode to cathode in both cases
- B. cathode to anode and anode to cathode

C. anode to cathode and cathode to anode

D. cathode to anode in both cases

Answer: C

Solution:

EMF of galvanic cell = 1.1 volt

If $E_{\text{ext}} < E_{\text{M F}}$ then electrons flow steadily from anode to cathode, while if $E_{\text{ext}} > E_{\text{M F}}$ then electrons flow from cathode to anode as polarity is changed.

Question163

At 298K, the standard reduction potentials are 1.51V for $\text{MnO}_4^- | \text{Mn}^{2+}$, 1.36V for $\text{Cl}_2 | \text{Cl}^-$, 1.07V for $\text{Br}_2 | \text{Br}^-$ and 0.54V for $\text{I}_2 | \text{I}$. At pH = 3, permanganate is expected to oxidize : $\left(\frac{RT}{F} = 0.059\text{V} \right)$
[Online April 11, 2015]

Options:

A. Cl , Br and I

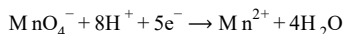
B. Br and I^-

C. Cl^- and Br^-

D. I^- only

Answer: B

Solution:



$$E = 1.51 - \frac{0.059}{5} \log \frac{[\text{Mn}^{2+}]}{[\text{MnO}_4^-][\text{H}^+]^8}$$

Taking Mn^{2+} and MnO_4^- in standard state i.e. 1M,

$$\begin{aligned} E &= 1.51 - \frac{0.059}{5} \times 8 \log \frac{1}{[\text{H}^+]^8} \\ &= 1.51 - \frac{0.059}{5} \times 8 \times 3 = 1.2268\text{V} \end{aligned}$$

Hence at this pH, MnO_4^- will oxidise only Br^- and I^- as SRP of Cl_2/Cl^- is 1.36V which is greater than that for $\text{MnO}_4^-/\text{Mn}^{2+}$

Question164

Resistance of 0.2M solution of an electrolyte is 50Ω. The specific conductance of the solution is 1.4Sm^{-1} . The resistance of 0.5M solution of the same electrolyte is 280 Ω. The molar conductivity of 0.5M solution of the electrolyte in $\text{Sm}^2\text{mol}^{-1}$ is:
[2014]

Options:

A. 5×10^{-4}

B. 5×10^{-3}

C. 5×10^3

D. 5×10^2

Answer: A

Solution:

Given for 0.2M solution

$$R = 50\Omega$$

$$\kappa = 1.4\text{Sm}^{-1} = 1.4 \times 10^{-2}\text{Scm}^{-1}$$

$$\text{Now, } R = \rho \frac{l}{a} = \frac{1}{\kappa} \times \frac{l}{a}$$

$$\Rightarrow \frac{1}{a} = R \times \kappa = 50 \times 1.4 \times 10^{-2}\text{cm}^{-1}$$

For 0.5 M solution

$$R = 280\Omega$$

$$\kappa = ?$$

$$\frac{1}{a} = 50 \times 1.4 \times 10^{-2}\text{cm}^{-1}$$

$$\Rightarrow R = \rho \frac{l}{a} = \frac{1}{\kappa} \times \frac{l}{a}$$

$$\Rightarrow \kappa = \frac{1}{280} \times 50 \times 1.4 \times 10^{-2}$$

$$= \frac{1}{280} \times 70 \times 10^{-2}$$

$$= 2.5 \times 10^{-3}\text{Scm}^{-1}$$

$$\text{Now, } \Lambda_m = \frac{\kappa \times 1000}{M}$$

$$= \frac{2.5 \times 10^{-3}\text{Scm}^{-1} \times 1000\text{cm}^3/\text{L}}{0.5\text{mol/L}}$$

$$= 5\text{Scm}^2\text{mol}^{-1} = 5 \times 10^{-4}\text{Sm}^2\text{mol}^{-1}$$

Question165

The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ_∞ , respectively. The correct relationship between λ_C and λ_∞ is given as:

(Where the constant B is positive)

[2014]

Options:

A. $\lambda_C = \lambda_\infty + (B)C$

B. $\lambda_C = \lambda_\infty - (B)C$

C. $\lambda_C = \lambda_\infty - (B)\sqrt{C}$

D. $\lambda_C = \lambda_\infty + (B)\sqrt{C}$

Answer: C

Solution:

According to Debye Huckle onsager equation,

$$\lambda_C = \lambda_\infty - B\sqrt{C}$$

Question166

The standard electrode potentials ($E_{M^+/M}^\circ$) of four metals A, B, C and D are -1.2V , 0.6V , 0.85V and -0.76V , respectively. The sequence of deposition of metals on applying potential is:

[Online April 9, 2014]

Options:

A. A, C, B, D

B. B, D, C, A

C. C, B, D, A

D. D, A, B, C

Answer: C

Solution:

As the value of reduction potential decreases the reducing power increases i.e.

$C < B < D < A$

$(0.85) \cdot (0.6) \cdot (-0.76)(-1.2)$

Question167

A current of 10.0A flows for 2.00h through an electrolytic cell containing a molten salt of metal X . This results in the decomposition of 0.250 mol of metal X at the cathode. The oxidation state of X in the molten salt is: (F = 96, 500C)

[Online April 9, 2014]

Options:

A. 1+

B. 2+

C. 3+

D. 4

Answer: C

Solution:

According to Faraday's first law of electrolysis

$$W = \frac{E \times i \times t}{96500}$$

Where E = equivalent weight

$$= \frac{\text{mol. mass of metal (M)}}{\text{oxidation state of metal (x)}}$$

Substituting the value in the formula

$$W = \frac{M}{x} \times \frac{i \times t}{96500}$$

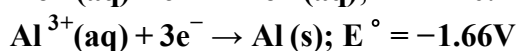
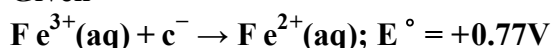
$$\text{or } x = \frac{M}{W} \times \frac{i \times t}{96500} = \frac{10 \times 2 \times 60 \times 60}{96500 \times 0.250} = 3$$

$$[\text{ Given : no. of moles } = \frac{W}{M} = 0.250]$$

Hence oxidation state of metal is (+3)

Question168

Given

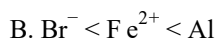


Considering the electrode potentials, which of the following represents the correct order of reducing power?

[Online April 11, 2014]

Options:

A. $\text{Fe}^{2+} < \text{Al} < \text{Br}^{-}$



Answer: D

Solution:

Reducing character decreases down the series. Hence the correct order is
 $\text{Al} < \text{F e}^{2+} < \text{Br}^-$

Question169

How many electrons would be required to deposit 6.35g of copper at the cathode during the electrolysis of an aqueous solution of copper sulphate? (Atomic mass of copper = 63.5u, N_A = Avogadro's constant)

[Online April 12, 2014]

Options:

A. $\frac{N_A}{20}$

B. $\frac{N_A}{10}$

C. $\frac{N_A}{5}$

D. $\frac{N_A}{2}$

Answer: C

Solution:



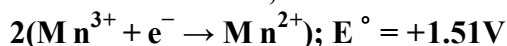
i.e, to deposit 1 mole of Cu at cathode from $\text{Cu}^{2+}\text{SO}_4^{2-}$ solution = 2 moles of electrons are required

$$\text{i.e, To deposit } 6.35\text{g} = \frac{6.35}{63.5} \times 2 = \frac{2}{10} = \frac{1}{5}\text{mol}$$

$$\text{Thus total no. of electrons required} = \frac{N_A}{5}$$

Question170

Given below are the half-cell reactions:



The E for $3\text{M n}^{2+} \rightarrow \text{M n} + 2\text{M n}^{3+}$ will be:
[2014]

Options:

A. -2.69V ; the reaction will not occur

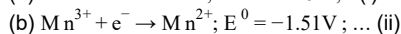
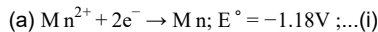
B. -2.69V ; the reaction will occur

C. -0.33V ; the reaction will not occur

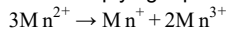
D. -0.33V ; the reaction will occur

Answer: A

Solution:



Now multiplying equation (ii) by two and subtracting from equation (i)



$$E^\circ = E_{\text{Ox.}} + E_{\text{Red}}$$

$$= -1.18 + (-1.51) = -2.69\text{V}$$

(-ve value of EMF (i.e. $\Delta G = +ve$) shows that the reaction is non-spontaneous)

Question 171

Electrode potentials (E°) are given below:

$$\text{Cu}^+/\text{Cu} = +0.52\text{V}$$

$$\text{Fe}^{3+}/\text{Fe}^{2+} = +0.77\text{V}$$

$$\frac{1}{2}\text{I}_2(\text{s})/\text{I}^- = +0.54\text{V}$$

$$\text{Ag}^+/\text{Ag} = +0.88\text{V}$$

Based on the above potentials, strongest oxidizing agent will be:

[Online April 9, 2013]

Options:



Answer: C

Solution:

Higher the value of reduction potential stronger will be the oxidising agent, hence based on the given values Ag^+ will be strongest oxidizing agent.

Question 172

Given:

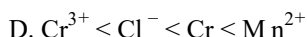
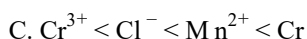
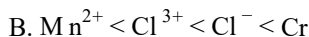
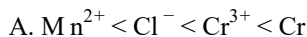
$$E_{\frac{1}{2}\text{Cl}_2/\text{Cl}^-} = 1.36\text{V}, E_{\text{Cr}^{3+}/\text{Cr}}^\circ = -0.74\text{V}$$

$$E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}}^\circ = 1.33\text{V}, E_{\text{MnO}_4^-/\text{Mn}^{2+}}^\circ = 1.51\text{V}$$

The correct order of reducing power of the species (Cr , Cr^{3+} , Mn^{2+} and Cl^-) will be:

[Online April 23, 2013]

Options:



Answer: A

Solution:

Lower the value of reduction potential higher will be reducing power hence the correct order will be $\text{Mn}^{2+} < \text{Cl}^- < \text{Cr}^{3+} < \text{Cr}$

Question 173

A solution of copper sulphate (CuSO_4) is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at the cathode (at. mass of Cu = 63u) is:
[Online April 25, 2013]

Options:

- A. 0.3892g
- B. 0.2938g
- C. 0.2398g
- D. 0.3928g

Answer: B

Solution:

$$W = Z it$$

where Z = Electrochemical equivalent

$$\text{Eq. wt. of copper} = \frac{63}{2} = 31.5$$

$$Z = \frac{31.5}{96500}$$

$$W = Z it = \frac{31.5}{96500} \times 1.5 \times 10 \times 60 = 0.2938\text{g}$$

Question 174

Given:

$$E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = -0.74\text{V}; E_{\text{MnO}_4^-/\text{Mn}^{2+}}^{\circ} = 1.51\text{V}$$

$$E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}} = 1.33\text{V}; E_{\text{Cl}/\text{Cl}^-}^{\circ} = 1.36\text{V}$$

Based on the data given above, strongest oxidising agent will be:
[2013]

Options:

- A. Cl
- B. Cr^{3+}
- C. Mn^{2+}
- D. MnO_4^-

Answer: D

Solution:

Higher the value of standard reduction potential, stronger is the oxidising agent, hence MnO_4^- is the strongest oxidising agent.

Question 175

The standard reduction potentials for Zn^{2+}/Zn , Ni^{2+}/Ni and Fe^{2+}/Fe are -0.76 , -0.23 and -0.44V respectively.

The reaction $\text{X} + \text{Y}^{2+} \rightarrow \text{X}^{2+} + \text{Y}$ will be spontaneous when :
[2012]

Options:

A. $\text{X} = \text{Ni}$, $\text{Y} = \text{Fe}$

B. $\text{X} = \text{Ni}$, $\text{Y} = \text{Zn}$

C. $\text{X} = \text{Fe}$, $\text{Y} = \text{Zn}$

D. $\text{X} = \text{Zn}$, $\text{Y} = \text{Ni}$

Answer: D

Solution:

For a spontaneous reaction ΔG° must be -ve

Since $\Delta G^\circ = -nFE^\circ$

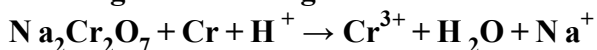
Hence for ΔG° to be -ve, ΔE° has to be positive. Which is possible when $\text{X} = \text{Zn}$, $\text{Y} = \text{Ni}$

$\text{Zn} + \text{Ni}^{2+} \rightarrow \text{Zn}^{2+} + \text{Ni}$

$$E_{\text{cell}}^\circ = E_{\text{Ni}^{2+}/\text{Ni}}^\circ - E_{\text{Zn}^{2+}/\text{Zn}}^\circ$$
$$= -0.23 - (-0.76) = +0.53 \text{ (positive)}$$

Question 176

A battery is constructed of Cr and $\text{Na}_2\text{Cr}_2\text{O}_7$. The unbalanced chemical equation when such a battery discharges is following:



If one Faraday of electricity is passed through the battery during the charging, the number of moles of Cr^{3+} removed from the solution is
[Online May 7, 2012]

Options:

A. $\frac{4}{3}$

B. $\frac{1}{3}$

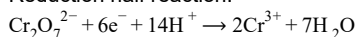
C. $\frac{3}{3}$

D. $\frac{2}{3}$

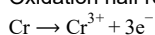
Answer: C

Solution:

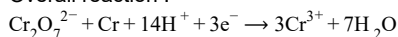
Reduction half reaction:



Oxidation half reaction:



Overall reaction :



$$3F \text{ of electricity} = 3 \text{ mol of } \text{Cr}^{3+}$$

$$1F \text{ of electricity} = \frac{3}{3} \text{ moles of } \text{Cr}^{3+}$$

Question 177

The standard potentials of Ag^+/Ag , $\text{Hg}_2^{2+}/2\text{Hg}$, Cu^{2+}/Cu and Mg^{2+}/Mg electrodes are 0.80, 0.79, 0.34 and -2.37V , respectively. An aqueous solution which contains one mole per litre of the salts of each of the four metals is electrolyzed. With increasing voltage, the correct sequence of deposition of the metals at the cathode is
[Online May 19, 2012]

Options:

- A. Ag, Hg, Cu, Mg
- B. Cu, Hg, Ag only
- C. Ag, Hg, Cu only
- D. Mg, Cu, Hg, Ag

Answer: C

Solution:

More the reduction potential, more is the deposition of metals at cathode. Cation having E° value less than -0.83V (reduction potential of H_2O) will not deposit from aqueous solution. Hence correct order of deposition of the metal at the cathode is
 $\text{Ag} > \text{Hg} > \text{Cu}$

Question 178

Given

$$E_{\text{Cu}^{2+}/\text{Cu}}^\circ = 0.34\text{V}, E_{\text{Cu}^{2+}/\text{Cu}^+}^\circ = 0.15\text{V}$$

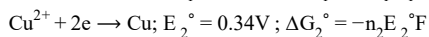
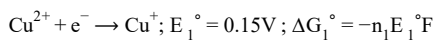
Standard electrode potential for the half cell Cu^+/Cu is
[Online May 26, 2012]

Options:

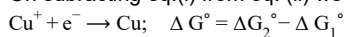
- A. 0.38V
- B. 0.53V
- C. 0.19V
- D. 0.49V

Answer: B

Solution:



On subtracting eq. (i) from eq. (ii) we get



$$-nE^\circ F = -(n_2 E_2^\circ F - n_1 E_1^\circ F)$$

$$E^\circ = \frac{n_2 E_2^\circ F - n_1 E_1^\circ F}{nF}$$

$$= \frac{2 \times 0.34 - 0.15}{1} = 0.53\text{V}$$

Question179

Resistance of 0.2M solution of an electrolyte is 50Ω. The specific conductance of the solution is 1.3Sm⁻¹. If resistance of the 0.4M solution of the same electrolyte is 260 Ω, its molar conductivity is :
[2011RS]

Options:

- A. $6.25 \times 10^{-4} \text{Sm}^2 \text{mol}^{-1}$
- B. $625 \times 10^{-4} \text{Sm}^2 \text{mol}^{-1}$
- C. $62.5 \text{Sm}^2 \text{mol}^{-1}$
- D. $6250 \text{Sm}^2 \text{mol}^{-1}$

Answer: A

Solution:

$$k = \frac{1}{R} \times \frac{1}{a}$$

$$1.3 = \frac{1}{50} \times \frac{1}{a}$$

$$\frac{1}{a} = 65 \text{m}^{-1}$$

$$\Lambda_m = \frac{k \times 1000}{\text{molarity}}$$

[molarity is in moles/litre but 1000 is used to convert litre into cm³]

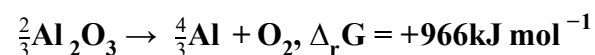
$$= \frac{\left(\frac{1}{260} \times 65 \text{m}^{-1}\right) \times 1000 \text{cm}^3}{0.4 \text{mol}}$$

$$= \frac{65 \text{m}^{-1}}{260 \times 0.4 \text{mol}} \times \frac{1}{1000} \text{m}^3$$

$$= 6.25 \times 10^{-4} \text{Sm}^2 \text{mol}^{-1}$$

Question180

The Gibbs energy for the decomposition of Al₂O₃ at 500 °C is as follows:



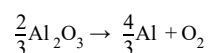
The potential difference needed for electrolytic reduction of Al₂O₃ at 500°C is at least
[2010]

Options:

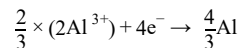
- A. 4.5V
- B. 3.0V
- C. 2.5V
- D. 5.0V

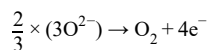
Answer: C

Solution:



The ionic reactions are





Thus, no. of electron transferred $\Rightarrow n = 4$

$$\Delta G = -nFE$$

$$\text{or } E = \frac{\Delta G}{-nF} = \frac{966 \times 10^3}{4 \times 96500} = -2.5\text{V}$$

$\therefore$ The potential difference needed for the reduction = 2.5V .

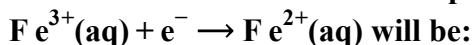
Question181

Given:

$$E_{\text{Fe}^{3+}/\text{Fe}}^\circ = -0.036\text{V}$$

$$E_{\text{Fe}^{2+}/\text{Fe}}^\circ = -0.439\text{V}$$

The value of standard electrode potential for the change,



[2009]

Options:

A. 0.385V

B. 0.770V

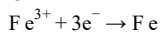
C. -0.270V

D. -0.072V

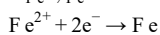
Answer: B

Solution:

Given

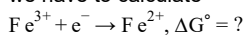


$$E_{\text{Fe}^{3+}/\text{Fe}}^\circ = -0.036\text{V} \dots(\text{i})$$

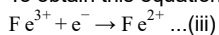


$$E_{\text{Fe}^{2+}/\text{Fe}}^\circ = -0.439\text{V} \dots(\text{ii})$$

we have to calculate



To obtain this equation subtract equ. (ii) from (i) we get



As we know that $\Delta G^\circ = -nFE^\circ$

Thus for reaction (iii)

$$\Delta G_3 = \Delta G_1 - \Delta G_2$$

$$-nFE_3^\circ = -nFE_1^\circ - (-nFE_2^\circ)$$

$$-nFE_3^\circ = nFE_2^\circ - nFE_1^\circ$$

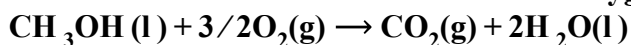
$$-1FE_3^\circ = 2 \times 0.439F - 3 \times 0.036F$$

$$-1FE_3^\circ = 0.770F$$

$$\therefore E_3^\circ = -0.770\text{V}$$

Question182

In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is



At 298K standard Gibb's energies of formation for $\text{CH}_3\text{OH}(\text{l})$, $\text{H}_2\text{O}(\text{l})$ and $\text{CO}_2(\text{g})$ are

-166.2 - 237.2 and -394.4 kJ mol⁻¹ respectively. If standard enthalpy of combustion of methanol is -726 kJ mol⁻¹, efficiency of the fuel cell will be:

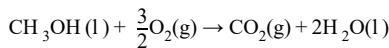
[2009]

Options:

- A. 87%
- B. 90%
- C. 97%
- D. 80%

Answer: C

Solution:



$$\Delta G_{\text{r}} = \Delta G_{\text{f}}(\text{CO}_2, \text{g}) + 2 \Delta G_{\text{f}}(\text{H}_2\text{O}, \text{l}) -$$

$$\Delta G_{\text{f}}(\text{CH}_3\text{OH}, \text{l}) - \frac{3}{2} \Delta G_{\text{f}}(\text{O}_2, \text{g})$$

$$= -394.4 + 2(-237.2) - (-166.2) - 0$$

$$= -394.4 - 474.4 + 166.2 = -702.6 \text{ kJ}$$

$$\% \text{ efficiency} = \frac{702.6}{726} \times 100 = 97\%$$

Question 183

Given $E^\circ \text{Cr}^{3+}/\text{Cr} = -0.72 \text{ V}$

$E^\circ \text{Fe}^{2+}/\text{Fe} = -0.42 \text{ V}$.

The potential for the cell

$\text{Cr} | \text{Cr}^{3+} (0.1 \text{ M}) | \text{Fe}^{2+} (0.01 \text{ M}) | \text{Fe}$ is
[2008]

Options:

- A. 0.26V
- B. 0.336V
- C. -0.339
- D. 0.26V

Answer: D

Solution:

From the given representation of the cell, E_{cell} can be found as follows.

$$E_{\text{cell}} = E_{\text{Fe}^{2+}/\text{Fe}}^\circ - E_{\text{Cr}^{3+}/\text{Cr}}^\circ - \frac{0.059}{6} \log \frac{[\text{Cr}^{3+}]^2}{[\text{Fe}^{2+}]^3}$$

[Nernst -Equ.]

$$= -0.42 - (-0.72) - \frac{0.059}{6} \log \frac{(0.1)^2}{(0.01)^3}$$

$$= -0.42 + 0.72 - \frac{0.059}{6} \log \frac{0.1 \times 0.1}{0.01 \times 0.01 \times 0.01}$$

$$= 0.3 - \frac{0.059}{6} \log \frac{10^{-2}}{10^{-6}} = 0.3 - \frac{0.059}{6} \times 4$$

$$= 0.30 - 0.0393 = 0.26 \text{ V}$$

Question 184

The equivalent conductances of two strong electrolytes at infinite dilution in H_2O (where ions move freely through a solution) at 25°C are given below:

$$\Lambda_{\text{CH}_3\text{COONa}} = 91.0 \text{ Scm}^2 / \text{equiv.}$$

$$\Lambda_{\text{HCl}}^{\circ} = 426.2 \text{ Scm}^2 / \text{equiv.}$$

What additional information\ quantity one needs to calculate Λ° of an aqueous solution of acetic acid? [2007]

Options:

- A. Λ of chloroacetic acid (ClCH_2COOH)
- B. Λ of NaCl
- C. Λ of CH_3COOK
- D. the limiting equivalent conductance of H^+ ($\lambda_{\text{H}^+}^{\circ}$).

Answer: B

Solution:

Note: According to Kohlrausch's law, equivalent conductivity of weak electrolyte, acetic acid (CH_3COOH) can be calculated as follows:

$$\Lambda_{\text{CH}_3\text{COOH}}^{\circ} = (\Lambda_{\text{CH}_3\text{COONa}}^{\circ} + \Lambda_{\text{HCl}}^{\circ}) - \Lambda_{\text{NaCl}}^{\circ}$$

$\therefore$ Value of $\Lambda_{\text{NaCl}}^{\circ}$ should also be known for calculating value of $\Lambda_{\text{CH}_3\text{COOH}}^{\circ}$

Question 185

The cell, $\text{Zn} | \text{Zn}^{2+}(1\text{M}) | \text{Cu}^{2+}(1\text{M}) | \text{Cu}$ ($E_{\text{cell}}^{\circ} = 1.10\text{V}$) was allowed to be completely discharged at 298K .

The relative concentration of Zn^{2+} to Cu^{2+} $\left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right)$ is

[2007]

Options:

- A. 9.65×10^4
- B. antilog (24.08)
- C. 37.3
- D. 10^{373} .

Answer: D

Solution:

$E_{\text{cell}} = 0$; when cell is completely discharged.

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right)$$

$$\text{or } 0 = 1.1 - \frac{0.059}{2} \log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right)$$

$$\log \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right) = \frac{2 \times 1.1}{0.059} = 37.3$$

$$\therefore \left(\frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} \right) = 10^{37.3}$$

Question 186

The molar conductivities $\Lambda_{\text{NaOAc}}^{\circ}$ and $\Lambda_{\text{HCl}}^{\circ}$ at infinite dilution in water at 25°C are 91.0 and $426.2\text{Scm}^2/\text{mol}$ respectively. To calculate $\Lambda_{\text{HOAc}}^{\circ}$, the additional value required is [2006]

Options:

A. $\Lambda_{\text{NaOH}}^{\circ}$

B. $\Lambda_{\text{NaCl}}^{\circ}$

C. $\Lambda_{\text{H}_2\text{O}}^{\circ}$

D. $\Lambda_{\text{KCl}}^{\circ}$

Answer: B

Solution:

$\Lambda_{\text{CH}_3\text{COOH}}^{\circ}$ is given by the following equation

$$\Lambda_{\text{CH}_3\text{COOH}}^{\circ} = (\Lambda_{\text{CH}_3\text{COONa}}^{\circ} + \Lambda_{\text{HCl}}^{\circ}) - (\Lambda_{\text{NaCl}}^{\circ})$$

Hence $\Lambda_{\text{NaCl}}^{\circ}$ is required.

Question 187

Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω . The conductivity of this solution is 1.29Sm^{-1} . Resistance of the same cell when filled with 0.2M of the same solution is 520Ω . The molar conductivity of 0.2M solution of electrolyte will be [2006]

Options:

A. $1.24 \times 10^{-4}\text{Sm}^2\text{mol}^{-1}$

B. $12.4 \times 10^{-4}\text{Sm}^2\text{mol}^{-1}$

C. $124 \times 10^{-4}\text{Sm}^2\text{mol}^{-1}$

D. $1240 \times 10^{-4}\text{Sm}^2\text{mol}^{-1}$

Answer: B

Solution:

$$R = 100\Omega, \kappa = \frac{1}{R} \left(\frac{l}{a} \right)$$

$$\frac{1}{a} \left(\text{cell constant} \right) = 1.29 \times 100\text{m}^{-1}$$

Given, $R = 520\Omega$

$$C = 0.2\text{M} = 0.2 \times 1000\text{mol m}^{-3}$$

Λ_m (molar conductivity) = ?

$\Lambda_m = \kappa \times V$ (κ can be calculated as cell constant is known.)

$$\kappa = \frac{1}{R} \left(\frac{l}{a} \right)$$

Hence,

$$\kappa = \frac{1.29\text{m}^{-1}}{520\Omega} = 0.248\text{Sm}^{-1}$$

$$\Lambda_m = \frac{\kappa}{C} = \frac{0.248\text{Sm}^{-1}}{0.2 \times 1000\text{mol m}^{-3}} = 12.4 \times 10^{-4}\text{Sm}^2\text{mol}^{-1}$$

Question188

The highest electrical conductivity of the following aqueous solutions is of [2005]

Options:

- A. 0.1M difluoroacetic acid
- B. 0.1M fluoroacetic acid
- C. 0.1M chloroacetic acid
- D. 0.1M acetic acid

Answer: A

Solution:

Difluoro acetic acid being strongest acid will furnish maximum number of ions showing highest electrical conductivity. The decreasing acidic strength of the carboxylic acids given is
difluoro acetic acid > fluoro acetic acid > chloro acetic acid > acetic acid.

Question189

Electrolyte:	KCl	KNO_3	HCl	$NaOAc$	$NaCl$
$\Lambda^\infty (Scm^2mol^{-1})$:	149.9	145	426.2	91	126.5

Calculate Λ_{HOAc}^∞ using appropriate molar conductance's of the electrolytes listed above at infinite dilution in H_2O at $25^\circ C$ [2005]

Options:

- A. 217.5
- B. 390.7
- C. 552.7
- D. 517.2

Answer: B

Solution:

$$\begin{aligned}\Lambda_{HCl}^\infty &= 426.2 \dots (i) \\ \Lambda_{ACONa}^\infty &= 91.0 \dots (ii) \\ \Lambda_{NaCl}^\infty &= 126.5 \dots (iii) \\ \Lambda_{AcOH}^\infty &= (i) + (ii) - (iii) \\ &= [426.2 + 91.0 - 126.5] = 390.7\end{aligned}$$

Question190

Aluminium oxide may be electrolysed at $1000^\circ C$ to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96,500 Coulombs). The cathode reaction is- $Al^{3+} + 3e^- \rightarrow Al^\circ$ Toprepare 5.12kg of

aluminium metal by this method we require
[2005]

Options:

- A. 5.49×10^1 C of electricity
- B. 5.49×10^4 C of electricity
- C. 1.83×10^7 C of electricity
- D. 5.49×10^7 C of electricity

Answer: D

Solution:

1 mole of $e^- = 1F = 96500C$
27g of Al is deposited by $3 \times 96500C$
5120g of Al will be deposited by
 $= \frac{3 \times 96500 \times 5120}{27} = 5.49 \times 10^7 C$

We know, $Q = \frac{mFZ}{M}$

$\therefore Q = \frac{5120 \times 96500 \times 3}{27} = 5.49 \times 10^7 C$

Question 191

For a spontaneous reaction the ΔG , equilibrium constant (K) and E_{cell}° will be respectively
[2005]

Options:

- A. -ve, >1 , -ve
- B. -ve, <1 , -ve
- C. +ve, >1 , -ve
- D. -ve, >1 , +ve

Answer: D

Solution:

Note: For spontaneous reaction ΔG should be negative. Equilibrium constant should be more than one ($\Delta G = -2.303RT \log K_c$, If $K_c = 1$ then $\Delta G = 0$; If $K_c > 1$ then $\Delta G = -ve$).
Again $\Delta G = -nF E_{\text{cell}}^\circ$
 E_{cell}° must be +ve to have $\Delta G -ve$.

Question 192

The limiting molar conductivities Λ° for NaCl, KBr and KCl are 126, 152 and $150 \text{ Scm}^2 \text{ mol}^{-1}$ respectively. The Λ for NaBr is
[2004]

Options:

- A. $278 \text{ Scm}^2 \text{ mol}^{-1}$

B. $176\text{Scm}^2\text{mol}^{-1}$

C. $128\text{Scm}^2\text{mol}^{-1}$

D. $302\text{Scm}^2\text{mol}^{-1}$

Answer: C

Solution:

$$\Lambda^\circ \text{NaCl} = \lambda^\circ \text{Na}^+ + \lambda^\circ \text{Cl}^- \dots \text{(i)}$$

$$\Lambda^\circ_{\text{KBr}} = \lambda^\circ_{\text{K}^+} + \lambda^\circ_{\text{Br}^-} \dots \text{(ii)}$$

$$\Lambda^\circ_{\text{KCl}} = \lambda^\circ_{\text{K}^+} + \lambda^\circ_{\text{Cl}^-} \dots \text{(iii)}$$

operating (i) + (ii) – (iii)

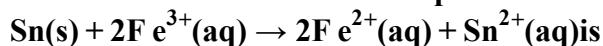
$$\begin{aligned} \Lambda^\circ_{\text{NaBr}} &= \lambda^\circ_{\text{Na}^+} + \lambda^\circ_{\text{Br}^-} \\ &= 126 + 152 - 150 = 128\text{Scm}^2\text{mol}^{-1} \end{aligned}$$

Question193

Consider the following

$$E^\circ \text{ values } E_{\text{Fe}^{3+}/\text{Fe}^{2+}} = +0.77\text{V}; E_{\text{Sn}^{2+}/\text{Sn}}^\circ = -0.14\text{V}$$

Under standard conditions the potential for the reaction



[2004]

Options:

A. 0.91V

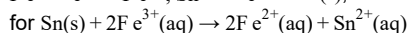
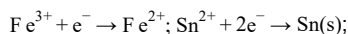
B. 1.40V

C. 1.68V

D. 0.63V

Answer: A

Solution:



$\therefore$ Standard potential for the given reaction or

$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ_{\text{Sn}/\text{Sn}^{2+}} + E_{\text{Fe}^{3+}/\text{Fe}^{2+}} \\ &= 0.14 + 0.77 = 0.91\text{V} \end{aligned}$$

Question194

The $E^\circ \text{M}^{3+}/\text{M}^{2+}$ values for Cr, Mn, Fe and Co are -0.41 , $+1.57$, $+0.77$ and $+1.97\text{V}$ respectively. For which one of these metals the change in oxidation state from +2 to +3 is easiest?

[2004]

Options:

A. Fe

B. Mn

C. Cr

D. Co

Answer: C

Solution:

The given values show that Cr has maximum oxidation potential, therefore its oxidation will be easiest. (Change the sign to get the oxidation values)

Question 195

The standard e.m.f. of a cell involving one electron change is found to be 0.591V at 25°C. The equilibrium constant of the reaction is

($F = 96,500 \text{ C mol}^{-1}$; $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

[2004]

Options:

A. 1.0×10^{10}

B. 1.0×10^5

C. 1.0×10^1

D. 1.0×10^{30}

Answer: A

Solution:

$$(a) E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log K_c$$

$$\text{or } 0 = 0.591 - \frac{0.0591}{1} \log K_c$$

$$\text{or } \log K_c = \frac{0.591}{0.0591} = 10 \text{ or } K_c = 1 \times 10^{10}$$

Question 196

In a cell that utilises the reaction

$\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ addition of H_2SO_4 to cathode compartment, will

[2004]

Options:

A. increase the E and shift equilibrium to the right

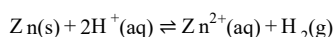
B. lower the E and shift equilibrium to the right

C. lower the E and shift equilibrium to the left

D. increase the E and shift equilibrium to the left

Answer: A

Solution:



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}][\text{H}_2]}{[\text{H}^+]^2}$$

Addition of H_2SO_4 will increase $[\text{H}^+]$ and E_{cell} will also increase and thus the equilibrium shifts towards RHS.

Question197

In a hydrogen-oxygen fuel cell, combustion of hydrogen occurs to [2004]

Options:

- A. produce high purity water
- B. create potential difference between two electrodes
- C. generate heat
- D. remove adsorbed oxygen from electrode surfaces

Answer: B

Solution:

In $H_2 - O_2$ fuel cell, the combustion of H_2 occurs to create potential difference between the two electrodes

Question198

Standard reduction electrode potentials of three metals A, B & C are respectively $+0.5V$, $-3.0V$ & $-1.2V$.

The reducing powers of these metals are [2003]

Options:

- A. $A > B > C$
- B. $C > B > A$
- C. $A > C > B$
- D. $B > C > A$

Answer: D

Solution:

Note: The higher the negative value of reduction potential, the more is the reducing power.
Hence $B > C > A$.

Question199

When during electrolysis of a solution of $AgNO_3$ 9650 coulombs of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be [2003]

Options:

- A. 10.8g
- B. 21.6g
- C. 108g
- D. 1.08g

Answer: A

Solution:

When 96500 coulomb of electricity is passed through the electroplating bath the amount of Ag deposited = 108g
∴ when 9650 coulomb of electricity is passed deposited Ag.

$$= \frac{108}{96500} \times 9650 = 10.8\text{g}$$

Question200

For a cell reaction involving a two-electron change, the standard e.m.f. of the cell is found to be 0.295V at 25°C. The equilibrium constant of the reaction at 25°C will be [2003]

Options:

A. 29.5×10^{-2}

B. 10

C. 1×10^{10}

D. 1×10^{-10}

Answer: C

Solution:

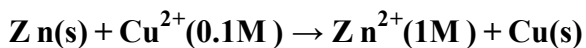
The equilibrium constant is related to the standard emf of cell by the expression

$$\log K = E_{\text{cell}}^{\circ} \times \frac{n}{0.059} = 0.295 \times \frac{2}{0.059}$$

$$\log K = \frac{590}{59} = 10 \text{ or } K = 1 \times 10^{10}$$

Question201

**For the redox reaction :
[2003]**



taking place in a cell, E_{cell}° is 1.10 volt. E_{cell} for the cell will be $\left(2.303 \frac{RT}{F} = 0.0591\right)$

Options:

A. 1.80 volt

B. 1.07 volt

C. 0.82 volt

D. 2.14 volt

Answer: B

Solution:

$$\begin{aligned}
 E_{\text{cell}} &= E_{\text{cell}}^{\circ} + \frac{0.059}{n} \log \frac{[\text{Cu}^{2+}]}{[\text{Zn}^{2+}]} \\
 &= 1.10 + \frac{0.059}{2} \log[0.1] \\
 &= 1.10 - 0.0295 = 1.07\text{V}
 \end{aligned}$$

Question202

Conductivity (unit Siemen's S) is directly proportional to area of the vessel and the concentration of the solution in it and is inversely proportional to the length of the vessel then the unit of the constant of proportionality is
[2002]

Options:

- A. Smmol^{-1}
- B. $\text{Sm}^2\text{mol}^{-1}$
- C. $\text{S}^{-2}\text{m}^2\text{mol}$
- D. $\text{S}^2\text{m}^2\text{mol}^{-2}$.

Answer: B

Solution:

$$\begin{aligned}
 \text{Given conductivity} &\propto \frac{\text{area} \times \text{conc.}}{\text{length}} \\
 &= \frac{\text{K m}^2\text{mol}}{\text{m} \times \text{m}^3} \\
 \therefore \text{K} &= \text{Sm}^2\text{mol}^{-1}
 \end{aligned}$$

Question203

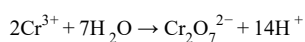
Which of the following reaction is possible at anode?
[2002]

Options:

- A. $2\text{Cr}^{3+} + 7\text{H}_2\text{O} \rightarrow \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+$ [2002]
- B. $\text{F}_2 \rightarrow 2\text{F}$
- C. $(1/2)\text{O}_2 + 2\text{H}^+ \rightarrow \text{H}_2\text{O}$
- D. none of these.

Answer: A

Solution:



O.S. of Cr changes from +3 to +6 by loss of electrons. At anode oxidation takes place.

Question204

When the sample of copper with zinc impurity is to be purified by electrolysis, the appropriate electrodes are Cathode Anode

[2002]

Options:

- A. Cathode - pure zinc
Anode - pure copper
- B. Cathode - impure sample
Anode - pure copper
- C. Cathode - impure zinc
Anode - impure sample
- D. Cathode - pure copper
Anode - impure sample.

Answer: D

Solution:

Pure metal always deposits at cathode.

Question205

EMF of a cell in terms of reduction potential of its left and right electrodes is [2002]

Options:

- A. $E = E_{\text{left}} - E_{\text{right}}$
- B. $E = E_{\text{left}} + E_{\text{right}}$
- C. $E = E_{\text{right}} - E_{\text{left}}$
- D. $E = -(E_{\text{right}} + E_{\text{left}})$.

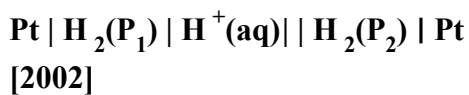
Answer: C

Solution:

$$\begin{aligned} E_{\text{cell}} &= \text{Reduction potential of cathode (right)} \\ &\quad - \text{reduction potential of anode (left)} \\ &= E_{\text{right}} - E_{\text{left}} \end{aligned}$$

Question 206

What will be the emf for the given cell



Options:

A. $\frac{RT}{F} \log_e \frac{P_1}{P_2}$

B. $\frac{RT}{2F} \log_e \frac{P_2}{P_1}$

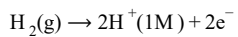
C. $\frac{RT}{F} \log_e \frac{P_2}{P_1}$

D. none of these.

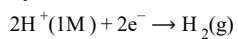
Answer: C

Solution:

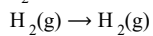
Oxidation half cell:-



P_1 Reduction half cell



P_2 The net cell reaction



P_t

$$E_{\text{cell}}^\circ = 0.00\text{V}; n = 2$$

$$\therefore E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \log_e K$$

$$= 0 - \frac{RT}{nF} \log_e \frac{P_2}{P_1}$$

$$\text{or } E_{\text{cell}} = \frac{RT}{2F} \log_e \frac{P_1}{P_2}$$
